SCALE-COVARIANT ATTENTION FOR TRAINING-FREE HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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Paper under double-blind review

ABSTRACT

Diffusion transformers trained near one megapixel fail at higher resolutions: attention meets unseen token counts and position offsets, and its cost grows quadratically. Each head's appetite for context, its balance between a local window and a coarse view of the whole image, can be measured at the training resolution and transfers to larger images. We introduce Scale-Covariant Attention (SCA), which calibrates once, fitting one weight per head, noise level, and image-to-window ratio in closed form against full attention. At the target, SCA opens with the global view alone, shrinks every window to native size over a four-step schedule, and mixes the two views with each head's weight for the current ratio. The schedule roughly halves repeated subjects: at 4096^2 a person or animal repeats in 26% of the 95 prompts that name one, against 57% without it and 4% for AReA. For FLUX.1-dev at 4096^2, SCA beats AReA on 12 of 15 metrics and runs $2.85\times$ faster; in HRDiT's progressive schedule it matches HRDiT's preference scores in under half the time. Code: github.com/shreshthsaini/scale-covariant-attention.

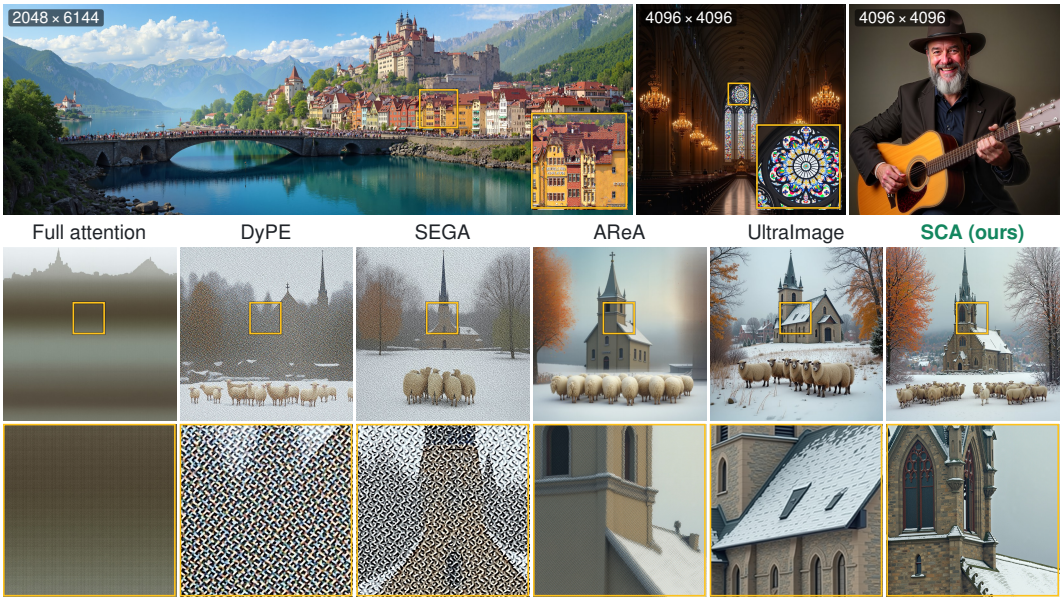

Figure 1: **SCA at 4K without training.** Top: curated SCA outputs of FLUX.1-dev at 2048×6144 and 4096^2, with insets at native pixels. Bottom: 1:1 crops of one curated 4096^2 prompt on which no baseline scores unusually low. Full attention gives a flat field, DyPE and SEGA a lattice; AReA, UltraImage, and SCA keep the church and flock. The shop signs in the panorama inset carry garbled lettering.

1 INTRODUCTION

Diffusion transformers (Peebles & Xie, 2022) such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained near one megapixel. At 4096×4096 the image has 16 times as many tokens, full attention costs about 256 times as much, and every layer meets token counts and position offsets it never saw in training. Full attention then returns stripes or a flat field

(Figure 1, bottom left). A native-size window per query keeps the offsets in range, but from step 1 it draws five Moai where the prompt asks for one (Figure 2a). We ask whether each head's balance between such a window and a coarse view of the whole image can be measured at the training resolution, and when in sampling each head should receive its window.

Training-free fixes either edit the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b) or restrict the keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). A native-size window cannot see an object larger than itself, so a window that dominates while a subject forms may settle that subject alone. AReA (Rajabi et al., 2026a), the most recent key-restricting method, attends to pooled keys for the first 8 of 28 steps, then to windows that shrink to native size, with one schedule for all heads. HRDiT and Ultra-Image give heads their own receptive fields (Xue et al., 2026; Zhao et al., 2025), but none measures a head's balance at the training resolution, where full attention is trustworthy (Section 5).

(a) mix from step 1 (b) global opening

Figure 2: **Windows repeat the subject.** A development prompt asking for one Moai at 4096^2. (a) Native-size windows mixed in from step 1 draw five; (b) a global-only opening keeps one large statue, plus a small one far off.

We answer by emulation. Shrinking a native-size neighbourhood and a downsampled view of the whole image by s mimics the image-to-window ratio of a target s times larger, while full attention stays in distribution as ground truth. A convex mix of the two views fits each head in closed form. Its weight differs strongly between heads and is stable across prompt splits (correlation 0.99). It also transfers: the emulated scale-2 table explains scale 4 with 2.4% more output error than the scale-4 table (Section 3). We call this measured transfer scale covariance; the weight depends mainly on the image-to-window ratio, so one table serves every window size.

We introduce Scale-Covariant Attention (SCA), which turns this measurement into an attention operator (Figure 3). Each head mixes a global branch over block-centre keys with a local branch, an exact sliding window. One schedule sets the window: the whole image, seen at native size by the global branch, for 8 of 28 steps, then 128, 112, 96, and 80 tokens, then the native 64. Each step reads every head's weight from the table at the current ratio. The local softmax also holds 1,024 bridge tokens sampled across the image, so a window sees subjects drawn outside it.

The schedule comes from a failure we measured. Our first operator switched every head from the global view to a native window in one step, at a weight near 0.7. Each window then redrew small subjects (Section 4.2). Shrinking the windows over four steps and adding the bridge cuts the share of prompts with a repeated person or animal from 57% to 26% (AReA: 4%) at 9% more time.

- We fit each head's mix of local and global attention in closed form at the training resolution, per noise level and image-to-window ratio, and show it transfers across scales (Section 3).
- We trace repeated subjects to a one-step switch from the global view to native windows, and halve them with a four-step schedule (Sections 2.4 and 4.2).
- We show that at 4096^2 on FLUX.1-dev, SCA beats AReA on 12 of 15 metrics with paired 95% intervals excluding zero, loses none, and runs $2.85\times$ faster (Section 4.1).

2 Scale-Covariant Attention

2.1 Setting and notation

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, where $[n] = \{0, \dots, n-1\}$. Each block applies joint attention (57 blocks of 24 heads in FLUX.1). It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. Rotary embeddings (Su et al., 2021) give token j a position $p_j \in \mathbb{Z}^2$ (FLUX.1 puts all text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query q placed at u, and a set $\mathcal{K}$ of (key, value, key position) triples,

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u') \in \mathcal{K}} \operatorname{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v, \tag{1}$$

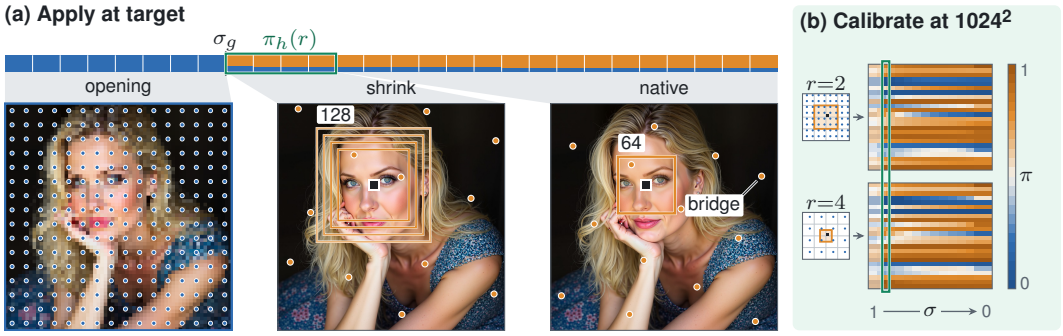

Figure 3: **SCA sampling and calibration.** (a) After the global opening, windows shrink from 128 to 64 tokens and their softmax adds 1,024 bridge tokens (dots); heads weigh the branches by π_h at $r = \max(H, W)/w$. (b) At 1024^2, π per head and noise bin is fit to full attention at emulated ratios 2 and 4 (zoom: Figure D.1).

where d is the head dimension. Full attention is $O_i^{\text{full}} = \mathcal{A}\big(q_i, p_i; \{(k_j, v_j, p_j)\}_{j \in \mathcal{T} \cup \mathcal{G}}\big)$. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer factors (s when equal), so full attention meets offsets s times the trained range. The latent at noise level σ is $x_\sigma = (1 - \sigma)x_0 + \sigma\epsilon$. Splitting the keys into near and far sets gives exactly $O_i^{\text{full}} = m_i O_i^{\text{near}} + (1 - m_i) O_i^{\text{far}}$ (Lemma 1), where each O attends to its own set and m_i is the near set's softmax mass. Every head thus mixes a near and a far field per query, which SCA approximates with two separately normalized operators and one weight per head.

2.2 Two operators

Global operator. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\ s_W b_2 + \lfloor s_W/2 \rfloor)$. The coarse map $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ sends a position to its block, so $c(\kappa(b)) = b$, and

$$\mathcal{K}^{\text{glob}} = \{(k_j, v_j, 0)\}_{j \in \mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b \in \mathcal{G}_0}, \qquad O_i^{\text{glob}} = \mathcal{A}\big(q_i,\ c(p_i);\ \mathcal{K}^{\text{glob}}\big), \tag{2}$$

with $c(p_i) = 0$ for text queries. This is AReA's first phase at pooling factor s, the image as one native canvas. It models content spanning a fixed fraction of the image: layout, object count, identity.

Local operator. Image queries form tiles of $t \times t$ tokens ($t = 8$); query i lies in tile $\tau = \lfloor p_i/t \rfloor$. The window $\mathcal{W}_w(i)$ of tile τ has side w, at most the grid side. It starts at $a(\tau) = \text{clip}\big(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor, 0, H - w\big)$ per axis, centred on the tile and clamped to the grid. Its keys are the text tokens, a bridge set $\mathcal{B}$ of image tokens, and the window, all at their original positions:

$$\mathcal{K}_i^{\text{loc}}(w) = \{(k_j, v_j, 0)\}_{j \in \mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{j \in \mathcal{B} \cup \mathcal{W}_w(i)}, \qquad O_i^{\text{loc}}(w) = \mathcal{A}\big(q_i,\ p_i;\ \mathcal{K}_i^{\text{loc}}(w)\big). \tag{3}$$

The bridge takes, in each cell c of an 8×8 partition of the grid, the 16 tokens whose pre-rotation keys lie farthest from the mean image key $\bar{k}^h$, summed over heads h:

$$\mathcal{B} = \bigcup_c \text{top}_{16}\big\{ \textstyle\sum_h \|k_j^h - \bar{k}^h\|^2 : j \in c \big\}. \tag{4}$$

It is recomputed at every block and step with AReA's selection rule (Rajabi et al., 2026a). Keys form a multiset; a bridge token inside the window enters twice. With $w = n_0$ it is an exact native-size sliding window, which models content spanning a fixed number of tokens: edges, texture, fine structure. Appendix A proves the two propositions below.

Proposition 1 (Trained position range)**.** *Assume exact integer positions and $t \le n_0 \le \min(H, W)$. Every query-key offset of the global operator, and every offset between an image query and a key of its window $\mathcal{W}_{n_0}(i)$, is at most $n_0 - 1$ per axis. Each global-operator query sees the native $T + n_0^2$ keys.*

Proposition 2 (Cost)**.** *Per head and layer, full attention scores $(N+T)^2$ query-key pairs, the global operator $(N + T)(T + N_0)$ with $N_0 = |\mathcal{G}_0| = n_0^2$, and the local operator $N(T + R + w^2)$ with $R = |\mathcal{B}|$. For fixed n_0, w, R, and T the total is $O\big(N(T + R + w^2 + N_0)\big)$, linear in N.*

Design choices. A block centre is a real token, so the global branch attends to keys and values the model produced; mean pooling does worse (Section 4.4). Native windows keep offsets in the trained

Algorithm 1: **Calibration**, once per model

```
# Input: prompts; full-attention sampler at 1024^2
# Output: pi*[s_c, l, h, beta]; full in App. E
def calibrate(q, k, v, l, sigma):  # every block
  O_full = attn(q, k, v)          # Eq. 1, all keys
  beta = min(int(14 * sigma), 13)    # noise bin
  for s_c in (2, 4):              # r = n / w = s_c
    f, w = s_c, n // s_c          # subsample, window
    t_c = round(t * w / n0)       # tile t n/(s_c n0)
    O_glob = global_branch(q, k, v, f)    # Eq. 2
    O_loc = local_branch(q, k, v, w, t_c) # Eq. 3
    dl, df = O_loc - O_glob, O_full - O_glob
    A[s_c, l, :, beta] += sum(dl * dl)    # Eq. 7
    B[s_c, l, :, beta] += sum(df * dl)  # over rows
    E[s_c, l, :, beta] += sum(df * df)
  return O_full                   # sampler unchanged
pi = clip(B / A, 0, 1); pi[A == 0] = 1  # Prop. 3
pi = fill_empty_bins(pi)  # read at r, Eq. 6
```

Algorithm 2: **Sampling**, one SCA attention call

```
# Input: q, k, v (T text, then H x W image), l, σ
# Output: O for all T + HW queries; App. E
fh, fw = H // n0, W // n0       # subsampling s
kb = k_img[fh//2::fh, fw//2::fw]  # κ(b); vb alike
Q = cat(rope(q_txt, 0), rope(q_img, c(p)))
K = cat(rope(k_txt, 0), rope(kb, b))
O_glob = sdpa(Q, K, cat(v_txt, vb))     # Eq. 2
if sigma > sigma_g:                  # 0.89, Eq. 5
  return O_glob       # opening phase: no local
w = window(sigma)   # 128, 112, 96, 80, n0; Eq. 6
B = bridge(k_img, cells=8, k=16)  # Eq. 4, 1,024
Kl = cat(k_txt, k_img[B], k_img)  # Vl alike
mask = flex_layout(T + len(B), t=8, w=min(w, H))
O_loc = flex_attention(rope(q_img, p),  # Eq. 3
    rope(Kl, p_Kl), Vl, block_mask=mask)  # App. D
pi = interp(sigma, max(H,W)/w, table[l])  # ratio r
O_img = pi*O_loc + (1-pi)*O_glob[T:]    # Eq. 5
return cat(O_glob[:T], O_img)     # text: O_glob
```

range, so SCA keeps FLUX.1's RoPE and softmax scale; text keys sit at position 0 in both branches (Remark 1). Bridge keys keep their true positions and windows wider than n_0 reach offsets up to $w-1$, so the local operator leaves the trained range for 1,024 keys and during the four wide-window steps. FLUX.1 also rounds positions in bf16 beyond 256 tokens per axis (Appendix D).

Worked example. At 4096^2 the latent grid is 256×256 tokens ($s = 4$, $N = 65{,}536$, $T = 512$). The global branch keeps one centre per 4×4 block, 4,096 keys on a 64×64 grid, so with the text a query sees 4,608 keys. In the native phase each 8×8 query tile attends to its 64×64 window, the 1,024 bridge tokens, and the text: 5,632 keys, against 66,048 keys at offsets up to 255 in full attention. Over 28 steps SCA scores 6.9 times fewer query-key pairs than full attention (Table D.1).

2.3 One schedule from the global view to the native window

Calibration (Section 2.5; Algorithm 1) gives each head h of block l a weight $\pi_{lh}(\sigma; r)$ for every noise level and emulated image-to-window ratio $r \in \{2, 4\}$. At the target (Algorithm 2), a window schedule $w(\sigma)$ sets the ratio $r(\sigma) = \max(H, W)/w(\sigma)$, and an image query takes

$$O_i \;=\; \begin{cases} O_i^{\text{glob}} & \sigma > \sigma_g, \\ \pi\, O_i^{\text{loc}}\big(w(\sigma)\big) + (1-\pi)\, O_i^{\text{glob}} & \text{otherwise,} \end{cases} \qquad \pi = \pi_{lh}\big(\sigma; r(\sigma)\big), \tag{5}$$

where σ_g is the gate and text queries take O_i^{glob}. Between calibrated ratios, π is linear in $\log r$:

$$\pi_{lh}(\sigma; r) = (1-\lambda)\,\pi_{lh}(\sigma; 2) + \lambda\,\pi_{lh}(\sigma; 4), \qquad \lambda = \text{clip}(\log_2 r - 1,\, 0,\, 1). \tag{6}$$

We use $\sigma_g = 0.89$, the first 8 of 28 steps at maximum shift as in AReA, and $w = 128$, 112, 96, and 80 tokens for σ in $(0.88, 0.89]$, $(0.86, 0.88]$, $(0.84, 0.86]$, and $(0.82, 0.84]$, then $w = n_0$. At 4096^2 these are the four steps after the opening, at ratios 2, 2.29, 2.67, and 3.2, and the last 16 steps run at ratio 4. Setting $w = n_0$ after the gate gives a one-step switch, the operator we call SCA without the schedule. The schedule's four sizes were set by hand; only the weights are calibrated.

2.4 Why a one-step switch repeats small subjects

Without the schedule, small copies of the subject form right after the gate (Figure 4c); three measurements locate the cause. **The calibrated weights are flat in the noise level.** At scale 4 the mean $\pi_h(\sigma)$ over the 1368 heads is 0.69 at $\sigma = 0.90$ and 0.77 at $\sigma = 0.11$, and 80% of heads exceed 0.5 at the gate. The local weight thus jumps from 0 to about 0.7 in one step. **A native window is a native canvas.** A 64×64 window with native offsets and all 512 text keys matches the model's 1024^2 training input. We hypothesize that the trained prior fills such a canvas with the prompt's subject while objects of 6 to 25 tokens are forming. **Duplication falls as the global view lasts longer.** On a fixed 46-prompt subset, people or animals repeat in 20, 15, and 7 of the 26 prompts naming them for gates at $\sigma = 0.93$, 0.89, and 0.85 (Figure 4a); with the global branch alone, only 1 does. Gate 0.85 costs quality, lowering the aesthetic score by 0.19 and MUSIQ_c by 2.0. Progressive SCA, whose 1024^2 stage fixes the layout before any window opens, repeats subjects in 6% of prompts.

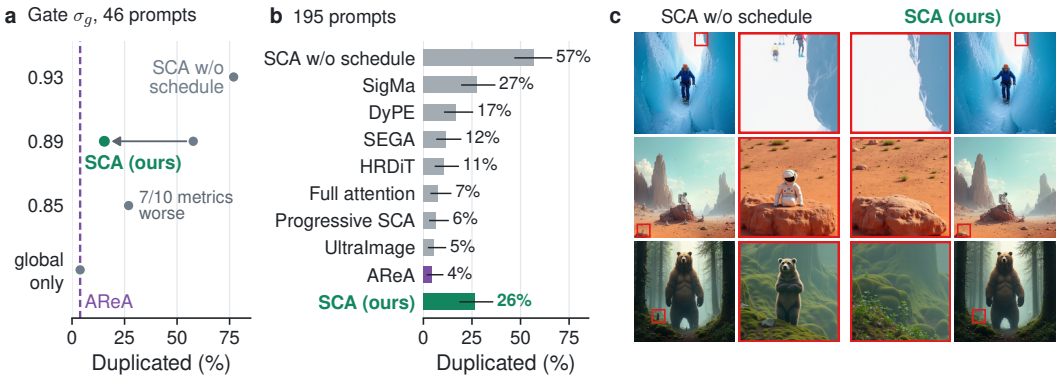

Figure 4: **A gradual handover cuts late duplication.** (a) Duplicated person or animal prompts (46-prompt subset) per gate; later gates cost quality, so ours keeps $\sigma_g = 0.89$ and adds the schedule. (b) All methods, 95 such prompts at 4096^2, with Wilson 95% intervals. (c) Uncurated, same seed, chosen as in Appendix P; one astronaut copy remains.

The schedule hands over in four steps instead of one, through windows that each hold more than a native canvas, and the bridge shows every window what is drawn elsewhere. Matched weights matter for quality. With scale-4 weights on 128-token windows, prompts that never had duplicates lose ImageReward, HPSv2, ARNIQA, and MUSIQ$_c$, and matched weights remove most of that loss (Appendix P). Argument 1 in Appendix A gives a frequency view of why windows fail at high noise.

2.5 Calibration by output matching

On an in-distribution $n \times n$ grid ($n = 64$ at 1024^2), emulated ratio r uses subsampling factor r, window side n/r, and tile side $\mathrm{round}(tn/(rn_0))$. The grid-to-window ratio is then r, as at a target of scale r. The sampler runs unmodified full attention (Algorithm 1), and at every block and step both emulated operators see the same queries, keys, and values. For block l, head h, noise bin β, and ratio r, sums over prompts, steps with $\sigma \in \beta$, image queries, and channels give

$$A = \sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}},\, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2, \tag{7}$$

where the emulated $O^{\mathrm{loc}}, O^{\mathrm{glob}}$ omit the bridge. Weight π has squared error $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 3 (Closed-form calibration and transfer cost)**.** *In each cell (l, h, β, r), J is minimized on $[0, 1]$ by $\pi^\star = \mathrm{clip}(B/A, 0, 1)$, with $\pi^\star = 1$ if $A = 0$. Any other weight $\pi' \in [0, 1]$ (fitted at another ratio or resolution, per layer, shared, or a hard switch) has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$.*

The same sums score every alternative rule without another forward pass, and one pass collects ratios 2 and 4. At inference, each weight is interpolated linearly in σ between the mean noise levels of occupied bins (14 of width $1/14$), constant beyond them, and in $\log r$ by Eq. 6. The native phase reads the ratio nearest $\max(s_H, s_W)$ in log space, here always 4. The local operator is one exact FlexAttention call with text and bridge keys as a shared prefix (Dong et al., 2024, Algorithm 2).

2.6 SCA inside a progressive schedule

Replacing only the attention, SCA fits inside a multi-stage sampler (Algorithm 5). HRDiT's released code (Xue et al., 2026) denoises at 1024^2 for 30 steps, then upsamples, re-noises, and repeats the last 17 and 10 steps of its schedule at 2048^2 and 4096^2. Progressive SCA keeps HRDiT's schedule, transitions, guidance, and seeds but replaces its four attention changes (NTK-scaled RoPE, a proportional attention scale, per-head key pruning, and position alignment) with SCA. Its stages re-enter at $\sigma = 0.81$ and 0.61, below the last wide window, so only the native window and bridge run.

3 What the heads want

We calibrate FLUX.1-dev on 32 purpose-written prompts at 1024^2 and 16 at 1280^2, disjoint from the evaluation sets. Errors are relative, $\sum J / \sum \|O^{\mathrm{full}}\|^2$, pooled over blocks, heads, and steps.

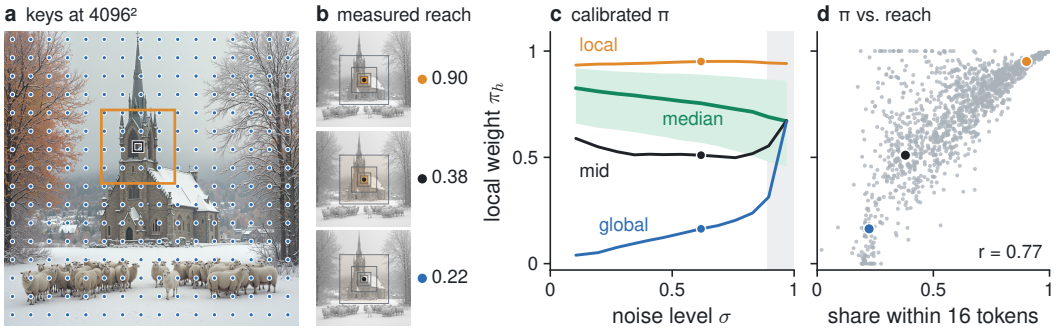

Figure 5: **Measured reach tracks calibrated π_h.** (a) Query tile, local window (amber), and every fourth block-centre key per axis (blue) on a 4096^2 SCA output; (b) full-attention mass of three heads at 1024^2, averaged over 32 calibration prompts. (c) Their scale-2 $\pi_h(\sigma)$ with median and interquartile range of all 1368 heads (grey: global only); (d) π_h against the mass share within 16 tokens, printed in (b), at $\sigma = 0.62$ ($r = 0.77$).

Table 1: **Main comparison at 4096^2.** FLUX.1-dev, 195 Aesthetic-4K prompts, seed label 0; HPSv2.1 $\times 100$, MUSIQ_c on 1024^2 crops, FID_p patch FID, Dup. duplicated share of the 95 person or animal prompts; **best**, second best, ours shaded; seconds per image on one GH200: clean batch-1 median or campaign mean on shared nodes (camp.); n/m: not measured; Table K.1 adds all metrics. Grey: published values without the schedule; their final-operator run is pending. Key result: among single-pass methods, SCA is best on every score and $2.85\times$ faster than AReA, but repeats subjects most often.

		Time (s)↓		Image↑	↑	Pick↑	↑	↑	↓	↓	↓
Method	Steps	clean	camp.	Reward	HPSv2.1	Score	CLIP	MUSIQ_c	FID	FID_p	Dup.
Single pass, shared sampler (28 steps, guidance 4.5)											
Full attention	28	186.5	196.2	−2.18	7.60	17.94	15.37	33.1	312.4	252.2	7
DyPE	28	n/m	195.9	−0.10	20.55	20.51	24.92	50.3	205.5	171.9	17
SEGA	28	n/m	190.4	0.56	24.34	21.64	26.95	50.3	164.9	125.8	12
AReA	28	202.4	264.4	0.92	27.16	22.36	27.69	50.4	145.8	102.9	**4**
SCA (ours)	28	71.1	**78.4**	1.25	29.73	23.12	**29.14**	**60.1**	140.2	**98.4**	26
Multi-stage, own recipe and budget											
UltraImage	30+30	n/m	308.5	1.28	30.77	23.33	28.84	41.7	142.0	109.5	5
HRDiT	30+17+10	108.6	110.4	**1.34**	**30.92**	**23.46**	29.12	53.5	**138.4**	103.0	11

Heads differ, and each head's weight depends on the ratio. At scale 2, $\pi_h(\sigma)$ has a standard deviation of 0.24 to 0.26 over the 1368 heads at every noise level. Half the (head, noise level) cells lie in $[0.2, 0.8]$ (Figures 5c, B.2, and B.1). At $\sigma = 0.62$ a head's scale-2 weight tracks its share of full-attention mass within 16 tokens ($r = 0.77$; Figure 5d). The mean weight is always lower at ratio 2 than at ratio 4 (0.65 against 0.69 at $\sigma = 0.90$), so a wider window takes less local weight.

The table is stable and transfers. Tables fitted on disjoint prompt halves correlate at 0.989 over all cells (0.986 at scale 4). Applied at scale 4, the scale-2 table gives 0.205 against 0.200 for the scale-4 table, below every shared rule fitted at scale 4; the 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585. The two resolutions emulate scale 2 with windows of 32 and 40 tokens, so the weights appear to depend mainly on the image-to-window ratio, as Eq. 6 assumes.

Per-head beats coarser rules; output error misses layout. Against the best hard switch, per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4. Per-layer tables reach 0.071 and 0.223, and one shared weight per noise level 0.077 and 0.230 (Figure B.3). A tokenwise error cannot see repeated subjects, so the opening and schedule override the fit at high noise (Section 2.4).

4 Experiments

Data and seeds. The main set (Table 1) is the Aesthetic-4K evaluation split with 4K references (Zhang et al., 2025a): all 195 prompts. Four of them (boats, Moai, woman and bear, underwater surfer) were used during development, which favours SCA slightly. A fixed 46-prompt

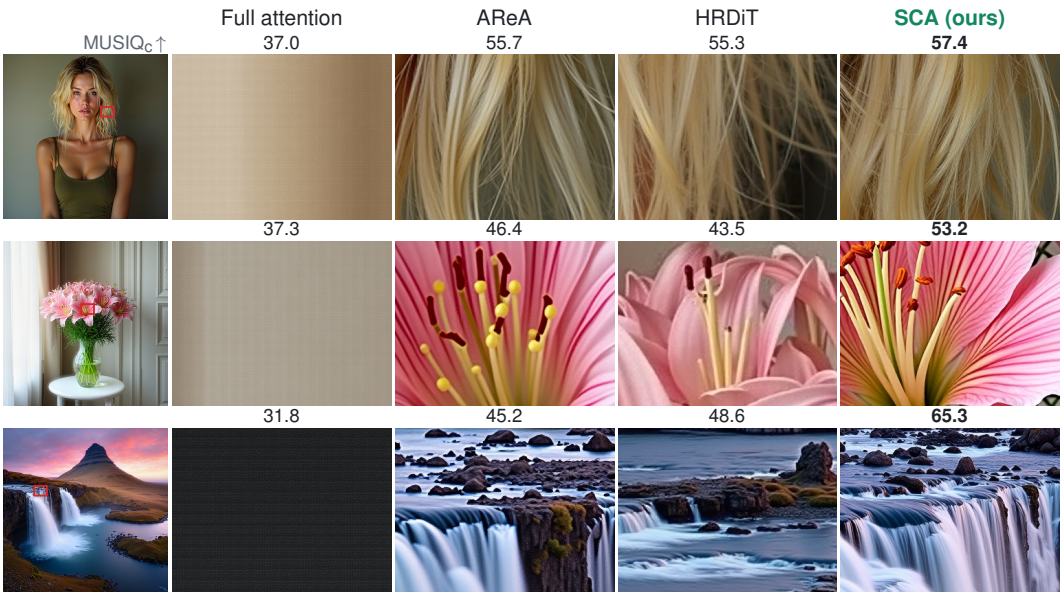

Figure 6: **SCA keeps native detail at 4096^2.** Three curated prompts (seed 0; curation favours SCA): SCA's image, then each method's red window at 1:1 under its MUSIQ_c (best bold). Full attention collapses to flat texture; Table 1 gives means and Figure T.2 uncurated samples. SCA's lily image streaks the table's back edge.

subset, chosen before any ablation ran, serves the ablations and the choice of the schedule. We also evaluate all 1000 HRGen-1K prompts (Rajabi et al., 2026a), which have no references. Sizes are 4096^2, 2048×4096, and 2048×6144 (height×width), plus 6144^2 and 8192^2 for the operator without the schedule. Seeds hash the prompt id and seed label 0 and are shared by all methods; labels 1 and 2 repeat all prompts.

Models and baselines. The main model is FLUX.1-dev (Black Forest Labs, 2024). FLUX.1-Krea-dev (195 prompts, against AReA) and FLUX.2-dev (50 prompts, against full attention, without the schedule) test other checkpoints, each calibrated on 16 prompts at 1024^2. SCA uses the window schedule of Section 2.3 and one calibrated table at every size (Table F.1). Full attention, AReA from its released code (Appendix H), DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), and SigMa (Zhao et al., 2026) share AReA's sampler: 28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$, bf16, and tiled VAE decoding. UltraImage (Zhao et al., 2025) runs its guided two-stage recipe (30 steps per stage) and HRDiT (Xue et al., 2026) its progressive recipe, each with its own budget (Appendix I). Each process uses one NVIDIA GH200 120GB; worker logs give at least 123.3 GH200-hours for the operator without the schedule (Appendix R).

Metrics. Preference and alignment scores are ImageReward, HPSv2, HPSv2.1, PickScore, CLIP score, and aesthetic score (Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023; Radford et al., 2021; Schuhmann, 2022). No-reference quality is MUSIQ on the full image and over four 1024^2 crops (MUSIQ_c; Figure 6), CLIP-IQA, and ARNIQA (Ke et al., 2021; Wang et al., 2022; Agnolucci et al., 2024). Distribution scores are FID and KID against all 195 official Aesthetic-4K references, Inception score at 299^2, and patch FID and KID on the same crops (Heusel et al., 2017; Salimans et al., 2016). Patch FID is our own definition and is never compared across papers. Intervals are 95% paired percentile bootstraps over prompt ids (1000 or 2000 draws), which also resample references for FID and KID (Figure 7; Appendix G).

Duplication. An open-vocabulary detector, Grounding DINO (Liu et al., 2024), counts each subject the prompt names. An image is duplicated when its person or animal count exceeds the prompt's; 95 of the 195 prompts imply one, with subject lists from the prompt text alone. The detector matches our visual review on all 20 labelled images of people or animals (Appendix P).

Table 2: **Multi-stage recipes and sampling budgets.** 4096^2, 195 prompts, campaign times, columns and bold as in Table 1; against SCA: $^+$ significantly better, $^\circ$ unresolved, * no interval, unmarked significantly worse; budget rows use SCA without the schedule. Grey: published values without the schedule; their final-operator run is pending. Key result: progressive SCA matches HRDiT's preference scores at 50.7 s against 110.4 s; Tables L.6, M.1, and M.2 add intervals.

		Time (s)↓		Image↑	↑	Pick↑	↑	↑	↓	↓	↓
Method	Steps	clean	camp.	Reward	HPSv2.1	Score	CLIP	MUSIQ_c	FID	FID_p	Dup.
Single pass, shared sampler (28 steps, guidance 4.5; other rows in Table 1)											
SigMa	28	n/m	192.5	−0.52	18.65	20.01	23.51	49.9	216.9*	184.6*	27
SCA (ours)	28	71.1	78.4	1.25	29.73	23.12	29.14	60.1	140.2	**98.4**	26
Multi-stage, own recipe and budget											
UltraImage	30+30	n/m	308.5	1.28°	30.77+	23.33+	28.84	41.7	142.0*	109.5*	**5**
HRDiT	30+17+10	108.6	110.4	**1.34**+	30.92+	**23.46**+	29.12°	53.5	**138.4***	103.0*	11
Prog. SCA	30+17+10	n/m	**50.7**	1.32+	**31.02**+	23.41+	**29.39°**	54.1	141.9*	107.7*	6
Without the schedule, other budgets											
56 steps	56	n/m	139.9	1.26+	30.55+	23.22+	29.14°	**64.1**+	139.8°	102.9	55
guidance 7	28	n/m	71.9	1.16	29.07	22.82	28.94°	61.4°	140.4°	**98.4°**	49
42 steps	42	n/m	106.8	1.26°	30.27+	23.17+	29.14°	63.1+	140.5	100.6°	57

4.1 Main results

SCA beats AReA on 12 of 15 metrics at 2.85× the speed. The gains include ImageReward (0.92 to 1.25, $+0.33$ $[0.25, 0.41]$) and FID (145.8 to 140.2, -5.6 $[-10.6, -0.4]$); three are unresolved (Tables 1 and K.2; Figures 1, 6, and 7). SCA takes 71.1 s per image and AReA 202.4 s. Our AReA run reproduces AReA's published FLUX.1-Krea-dev scores within 0.3 (Table H.1).

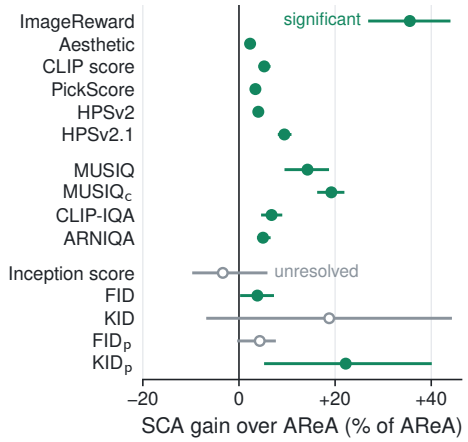

Figure 7: **Twelve of 15 metrics improve.** SCA minus AReA at 4096^2 as percent of AReA's mean, with 95% paired intervals. FID, KID, and their patch versions are negated so right is better; grey intervals span zero.

Every shared-sampler baseline trails SCA on all ten per-image metrics. This holds for DyPE, SEGA, and SigMa (Tables L.2 and L.6). UltraImage, at over 3× SCA's time, and HRDiT's three-stage recipe, at 1.4×, are significantly better on five and six per-image metrics, among them HPSv2.1 and PickScore, and worse on both MUSIQ scores (Table L.2).

Progressive SCA matches HRDiT's preference scores in less than half the time. HRDiT's lead thus comes from its schedule. Progressive SCA takes 50.7 s per image and HRDiT 110.4 s, as campaign means (Table 2; Figure M.2). ImageReward differs by -0.02 $[-0.09, 0.03]$, HPSv2.1 by $+0.10$ $[-0.14, 0.35]$, and PickScore by -0.05 $[-0.14, 0.04]$. Only CLIP-IQA is significantly worse (-0.9 $[-1.8, -0.1]$, $\times 100$), while HPSv2, MUSIQ, and ARNIQA are better; intervals that include zero do not establish equivalence (Table M.2). Against single-pass SCA, the parity costs fidelity: patch FID rises and MUSIQ_c falls (Appendix M).

4.2 Late duplication

Without the schedule, SCA repeats people or animals more often than any baseline. It does so in 57% of the 95 prompts that name them (50 to 57% across three seeds, 42 or 56 steps, and guidance 7), against 4% for AReA and 27% for SigMa (Figure 4b). **The schedule halves it but does not reach AReA.** With it, SCA repeats in 26% of the 95 prompts (Wilson 95% interval 18.5 to 36.0%; 26 to 28% across three seeds), more than every baseline except SigMa. At 2048×4096

Table 3: **Ablations.** 4096^2, 46-prompt subset; shaded default: absolute values; schedule rows: change against it; base-operator rows: against SCA without the schedule; orange: significantly worse, blue: better; IR, HPS, Pick, M_c: ImageReward, HPSv2.1 $\times 100$, PickScore, MUSIQ$_c$; Dup.: duplicated person or animal prompts of 26; Err.: relative output error, scale 4; Table K.5 adds all metrics and Table P.1 every schedule variant. Key result: the schedule cuts duplication, and changing its sizes moves the count by at most one prompt; every base-operator change lowers HPSv2.1 and PickScore.

Variant	Err.	IR	HPS	Pick	M_c	Dup.
SCA (ours)	0.200	1.17	30.07	23.02	60.7	4
Schedule						
without the schedule	0.200	−0.05	−0.09	−0.06	**+1.7**	15
bridge only	0.200	+0.01	+0.07	−0.04	**+0.5**	5
first window 160	0.200	−0.01	−0.05	−0.01	−0.2	4
2,048 bridge tokens	0.200	−0.02	−0.09	−0.04	−0.1	4
five-step schedule	0.200	−0.01	−0.05	−0.02	−0.2	3
Base operator (against SCA without the schedule)						
no opening (v0)	0.200	−1.33	−4.07	−2.18	+2.5	26
local only	0.261	−0.36	−2.86	−1.20	−0.3	25
global only	0.627	−0.40	−5.66	−1.44	−11.7	1
shared $\pi(\sigma)$	0.230	−0.09	−1.83	−0.38	−10.4	18
per-layer $\pi_\ell(\sigma)$	0.223	−0.05	−0.70	−0.16	−2.4	20
mean pooling	n/a	−1.52	−9.75	−2.60	−4.2	25

Table 4: **Generality and sensitivity.** Paired differences at 4096^2 against each group's reference over n prompts; orange: significantly worse; sensitivity rows vary SCA without the schedule (32 calibration prompts, $\sigma_g = 0.89$); Tables L.1, L.3, and L.4 add intervals. Grey: published values without the schedule; their final-operator run is pending. Key result: gains over AReA hold on HRGen-1K and with two more seeds.

Setting	n	ΔIR	ΔHPS	ΔM_c
Against AReA				
HRGen-1K	1000	+0.818	+4.93	+13.89
Seed 1	195	+0.306	+2.49	+10.27
Seed 2	195	+0.264	+2.53	+10.94
FLUX.1-Krea	195	−0.052	+0.63	+4.23
Without the schedule (published operator)				
4 prompts	46	+0.012	−0.02	+0.09
8 prompts	46	+0.010	+0.01	−0.03
16 prompts	46	−0.007	−0.01	−0.11
$\sigma_g = 0.85$	46	−0.015	−0.85	−2.04
$\sigma_g = 0.93$	46	−0.004	−0.15	+0.65
FLUX.2 vs. full	50	+0.456	+1.84	+21.44

and 2048×6144 it gives 24% and 33%, against 43% and 54% without it (AReA: 18% and 6%). On the 46-prompt subset used to choose the schedule it gave 15% (4 of 26; Table 3), so that subset understated the full-set rate. Its remaining copies are smaller and fewer (Figure 4c).

4.3 Other sizes and latency

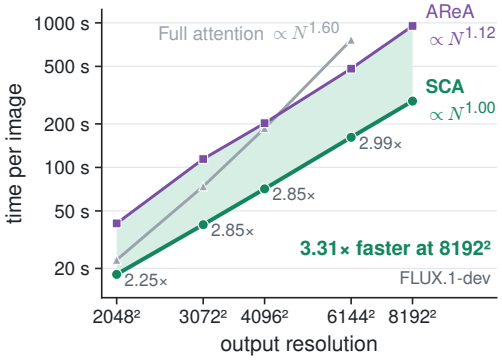

Figure 8: **Latency against size.** Batch 1, one GH200, median of three prompts; grey: speedup over AReA. Without the schedule; pending.

SCA is the fastest method we timed at every size; quality depends on the aspect ratio. At 2048×6144 SCA beats AReA on seven of 15 metrics and loses on the aesthetic score and patch FID; at 2048×4096 it wins three and loses three (Table K.3). Without the schedule it also beats AReA on 14 of 15 metrics at 6144^2 (ImageReward $+1.262$ $[1.051, 1.450]$). At 8192^2, 12 exploratory prompts favour it on preference and alignment scores, though grid artifacts remain (Appendix N).

SCA's time grows nearly linearly with pixel count. Fitting clean seconds $\propto N^b$ over the N image tokens gives $b = 1.00$ for SCA and 1.12 for AReA (Table D.2; Figure 8, five sizes from 2048^2 to 8192^2). Clean time is the median of three images after warm-up on an idle GPU, including text encoding and VAE decoding but not model loading. The schedule and the bridge add 9% to the time per image at 4096^2. At 8192^2 SCA takes 287.6 s against AReA's 952.0 s, $3.31\times$ faster.

4.4 Ablations

Table 3 changes one component at a time (examples in Figure T.7); Table 4 tests generality (Section 4.5). **Both parts of the schedule are needed.** The bridge alone leaves 5 of 26 prompts with a repeat, and wide windows with scale-4 weights soften clean prompts (Section 2.4). **The opening phase is essential.** Without it (v0), ImageReward falls by 1.33 $[1.06, 1.60]$ and all 26 prompts repeat subjects; CLIP-IQA and ARNIQA rise because each copy is sharp (Table K.5). **The per-head mix carries global context after the opening.** Local-only attention in the last 20 steps lowers ImageRe-

ward by 0.36 $[0.21, 0.52]$. **Per-head weights beat shared and per-layer weights, in the order output error predicts.** One shared weight per noise level lowers ImageReward by 0.09 $[0.01, 0.17]$ and MUSIQ$_c$ by 10.4, and per-layer weights still lose on HPSv2, HPSv2.1, PickScore, ARNIQA, and MUSIQ$_c$ (Section 3). **Neither branch works alone, and block centres beat block means.** Global only loses on every metric, and mean pooling blurs the global keys (ImageReward -1.52).

4.5 Generality and sensitivity

Gains hold across seeds, prompt sets, and checkpoints. Gains over AReA hold with seeds 1 and 2 and on 1000 HRGen-1K prompts (Tables 4 and L.3). On FLUX.1-Krea-dev (195 prompts), SCA wins six per-image metrics in a third of the time but trails AReA on ImageReward and patch FID (Table L.1). Without the schedule, calibrating on 4 to 16 prompts instead of 32 moves ImageReward by ≤ 0.012, and gates of 0.85 and 0.93 lose some metrics. On FLUX.2-dev, the same operator beats full attention on nine of ten per-image metrics in half the time (ImageReward 0.75 to 1.20).

5 Related work

Training-free high-resolution sampling. Methods alter convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023), fuse global and patch paths (Du et al., 2023; Yang et al., 2024), by frequency in FreeScale (Qiu et al., 2024), add native-resolution guidance (Du et al., 2024; Bu et al., 2025; Koh et al., 2025), or adapt RoPE (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2026; Rajabi et al., 2026b; Liu et al., 2026). Of the closest methods (Table J.1, Appendix J), AReA shares one schedule across heads and adds bridge tokens (Rajabi et al., 2026a), whose selection rule SCA borrows. ResDiT fuses position-scaled global and native-size patch attention between global-only and local-only phases (Ma et al., 2025).

Head- and layer-specific attention. HRDiT picks per-head scopes under a cost budget (Xue et al., 2026); UltraImage concentrates attention per head by entropy and noted stable local and global head roles before us (Zhao et al., 2025). Sparse attention varies windows, spans, or key scales per layer or head in diffusion transformers (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b), language models (Fu et al., 2024; Xiao et al., 2024), and vision transformers (Pan et al., 2022). Our local branch is neighbourhood attention (Hassani et al., 2022); ToDo attends to downsampled keys without rescaling coordinates (Smith et al., 2024). Head roles, two branches, a global opening, and shrinking windows have precedents; SCA adds a closed-form mixture per head, noise level, and image-to-window ratio, fitted by emulation at native resolution.

6 Limitations and conclusion

Limitations. Duplication is reduced, not solved. With the schedule a person or animal still repeats in 26% of the 95 prompts that name one at 4096^2, against 4% for AReA. We chose the schedule's sizes by hand on the 46-prompt subset, which is part of the evaluation set; there the rate was 15% (4 of 26), below the full-set rate. **Two no-reference metrics drop against SCA without the schedule.** On the 46 prompts, ARNIQA falls by 0.007 and MUSIQ$_c$ by 1.7. Sharp copies explain part of this: MUSIQ$_c$ falls by 1.4 where duplicates disappear. Part is real softening: on the 13 prompts clean under both operators, ImageReward falls by 0.06 and ARNIQA by 0.014. **Other limits.** Multi-stage recipes keep higher learned preference scores than single-pass SCA, and on FLUX.1-Krea-dev SCA trails AReA on two metrics (Sections 4.1 and 4.5). The 6144^2, 8192^2, FLUX.2-dev, and sensitivity results use SCA without the schedule. SigMa's dev-matched profile gives lattice-degraded images, possibly from our port. We did not run ResDiT (no verified official code) or UltraImage's direct-extrapolation variant. Metrics and the duplication detector are automatic, and intervals ignore multiple comparisons.

Conclusion. Each head's appetite for context can be measured once at the training resolution and reused at 4K for any window size. A global opening, four steps of shrinking windows, and weights read at the current ratio give the strongest and fastest single-pass training-free operator we tested for FLUX.1-dev at 4096^2. The schedule halves repeated subjects, from 57% to 26% of prompts, though AReA's 4% is lower. Code: github.com/shreshthsaini/scale-covariant-attention.

REPRODUCIBILITY STATEMENT

Section 4 states the protocol, including timing (Section 4.3), and Appendix G defines every metric and interval; Appendix P gives the duplication detector, its validation, and every schedule variant we ran. Appendix B gives the calibration geometry, statistics, and noise bins; Algorithms 1 and 2 give the procedure, and Appendix E gives it in full together with progressive SCA. Appendix F lists every hyperparameter, including the window schedule, and Appendix D the kernel layout, numerical checks, and position rounding. Appendices H and I give each baseline's code revision, recipe, and seeds, and Appendix R the hardware and logged compute. Code, calibration table, per-image metrics, detector outputs, and the paired-difference files behind every table will be released in the code repository (link in the abstract).

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APPENDIX CONTENTS

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A PROOFS

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into a near set $\mathcal{K}^{\text{near}}$ and a far set $\mathcal{K}^{\text{far}}$, with logits ℓ_j. Then $O^{\text{full}} = m\,O^{\text{near}} + (1-m)\,O^{\text{far}}$ with $m = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} / \sum_j e^{\ell_j}$ and $O^{\text{near}}, O^{\text{far}}$ the softmax outputs restricted to $\mathcal{K}^{\text{near}}$ and $\mathcal{K}^{\text{far}}$.*

Proof. Write $Z^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j}$, so that $Z^{\text{near}}O^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} v_j$, and likewise for the far set. Then $O^{\text{full}} = (Z^{\text{near}}O^{\text{near}} + Z^{\text{far}}O^{\text{far}})/(Z^{\text{near}} + Z^{\text{far}})$, and $Z^{\text{near}}/(Z^{\text{near}} + Z^{\text{far}}) = m$. □

Proof of Proposition 1. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \geq w$ and a tile $[t\tau,\, t\tau + t)$ with $t \leq w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \leq t\tau$ because $\lfloor t/2 \rfloor \leq \lfloor w/2 \rfloor$, and $a' + w \geq t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \geq \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \geq a' + w \geq t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \leq t\tau$ and $H \geq t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \leq n_0$ differ by at most $n_0 - 1$. The proposition makes no claim about the bridge keys, which keep their true positions. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proof of Proposition 2. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys, R bridge keys, and w^2 window keys. The sum is at most $(2N + T)(T + R + w^2 + N_0)$, which is $O(N(T + R + w^2 + N_0))$ for fixed n_0, w, R, and T. Value aggregation has the same pair counts. In the opening phase the local term is absent. The FlexAttention kernel visits whole blocks and masks the extra pairs: a query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 3. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\text{loc}} = O^{\text{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\text{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \le 0$, so $A\pi^\star - B \ge 0$ and $\pi' - \pi^\star \ge 0$. If $\pi^\star = 1$ then $B \ge A$, so $A\pi^\star - B \le 0$ and $\pi' - \pi^\star \le 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell, such as a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere, is a feasible π' in every cell, so its total error is at least the per-head table's, by $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$ or more. A table interpolated in σ between the mean sampled noise levels of occupied bins is not constant within a bin; the statement applies to the bin-constant table used to score rules in Section 3. □

Argument 1 (A frequency-based explanation for the opening phase). At noise level σ a spatial frequency band has signal-to-noise ratio $\big((1-\sigma)/\sigma\big)^2 S(f)$ for image spectrum S. If latent power falls with frequency, only low frequencies carry signal at high noise, and a window of w tokens spans less than one period of any frequency below $1/w$. Independent windows may then repeat local structures; a global branch supplies context across them. As hidden states also carry earlier layers' information, this is a hypothesis, not an impossibility result.

The frequency argument in more detail. Argument 1 rests on two facts and one assumption. The first fact is that the noisy latent is a sum of signal and independent white noise, so a band's signal-to-noise ratio scales as $((1-\sigma)/\sigma)^2 S(f)$; at $\sigma = 0.89$ the factor is 0.015, and at $\sigma = 0.5$ it is 1. The second is that a window of w tokens cannot distinguish a component of period longer than w from a smooth trend within the window, so windows cannot coordinate a layout at that scale except through what earlier layers mixed into their keys. The assumption is that natural-image latent spectra fall with frequency, which we have not measured for FLUX.1 latents. Together they predict that windowed attention fails first and worst at high noise and for objects larger than the window, which matches the duplication we observe (Figure A.1) and the effect of a 16-token window.

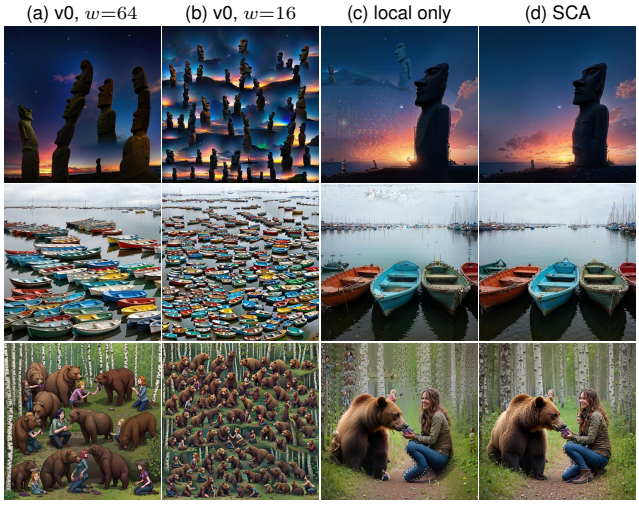

Figure A.1: **Windows at high noise duplicate subjects.** 4096^2, three development prompts. (a) v0, the calibrated mix from the first step: each 64-token window settles the layout on its own. (b) A 16-token window gives smaller, more numerous copies. (c) With the opening phase but local-only attention afterwards ($\pi = 1$), ghost copies and grid speckle remain. (d) SCA without the schedule keeps one main subject per row, though a small second Moai stands on the far shore and extra faces appear around the woman and the bear.

B Calibration details and statistics

This section expands Section 3 and Figures 3b and 5. Figure B.1 shows the calibrated tables, Figure B.2 every head's $\pi_h(\sigma)$ and the split-half agreement, Figures B.3 and B.4 the output errors per rule and per noise level, and Figure B.5 every head's weight per block.

Statistics. For every emulated scale, block, head, and noise bin, the pass stores A, B, E, the error of the local branch alone $\sum \|O^{\text{full}} - O^{\text{loc}}\|^2$, $\sum \|O^{\text{full}}\|^2$, and the query count, in float64, summed over image queries and channels (`sca/calib.py`). The sampler's trajectory is that of unmodified full attention; the emulated operators only observe. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the global branch only.

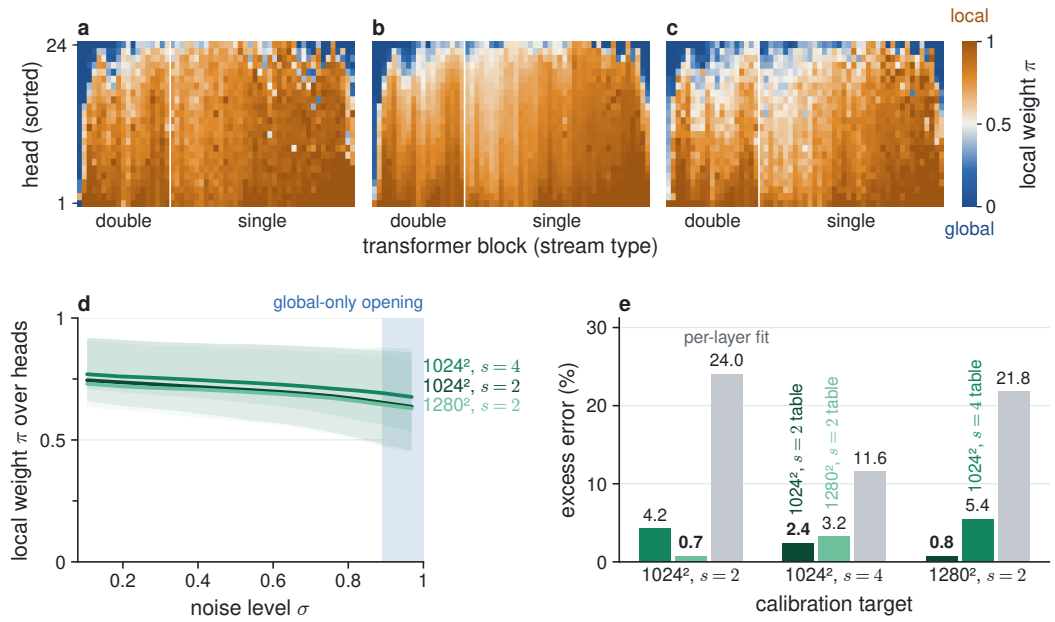

Figure B.1: **Heads differ widely, yet a calibrated table transfers across scale and resolution.** (a to c) $\pi_h(\sigma)$ at $\sigma = 0.10, 0.46, 0.83$ ($\sigma = 1$: pure noise); (d) mean and interquartile range across heads, shaded: global-only opening. (e) Excess error over each target's table: at most 5.4% transferred, 12 to 24% per-layer.

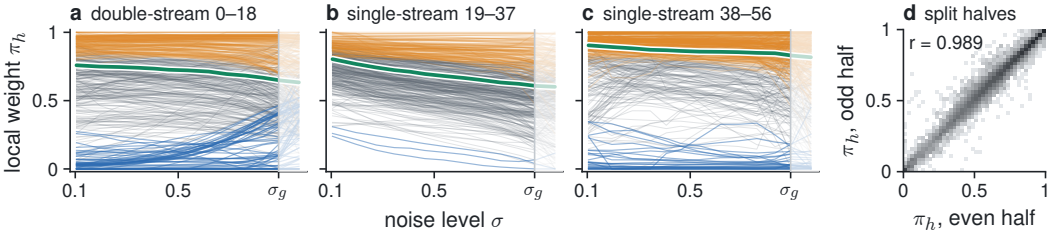

Figure B.2: **Heads want different mixes.** (a–c) $\pi_h(\sigma)$ of all 1368 heads at emulated scale 2; amber: mean above 0.8, blue: below 0.2, green: median; $\sigma > \sigma_g$ veiled. (d) Tables from disjoint prompt halves agree ($r = 0.989$).

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 subsampling onto a 32×32 grid, and 4×4 query tiles; emulated scale 4 uses a 16×16 window, 4×4 subsampling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window, 2×2 subsampling onto a 40×40 grid, and 5×5 tiles.

Noise bins. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest of the 14 bins is empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure B.4); within a block, the per-head weights span a median range of 0.66 to 0.84 at every noise level (Figure B.5). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261 to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. The generation results of Section 4.4 show the opposite preference for the opening steps, which is the gap between squared output error and layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass; image keys within distance

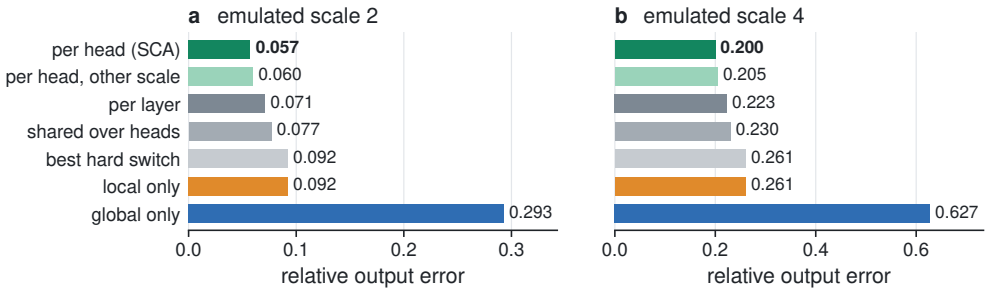

Figure B.3: **Per-head weights explain full attention best, and transfer across scales.** Relative output error per mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts; "other scale" swaps the scale-2 and scale-4 tables; the best hard switch ($\sigma^\star = 1$) equals local only. In-sample and held-out errors agree within 0.0003.

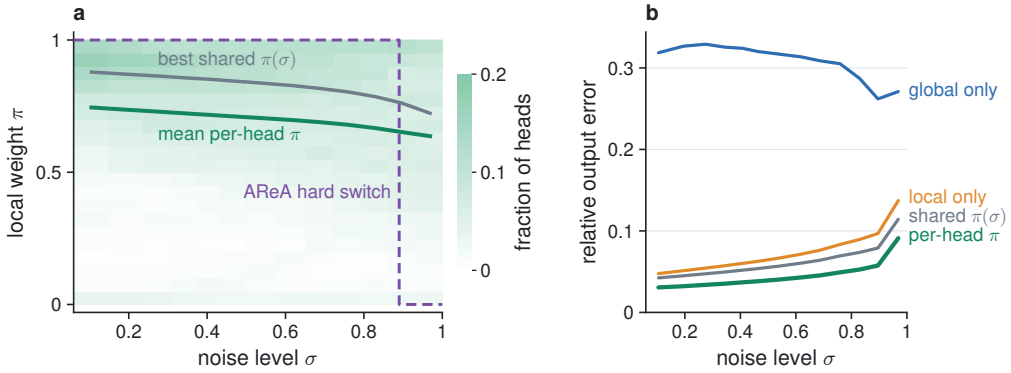

Figure B.4: **In output energy, no noise level prefers the global branch.** (a) Fraction of heads per π and noise bin, with mean $\pi_h(\sigma)$, best shared $\pi(\sigma)$ and AReA's switch; scale 2, 32 prompts, $\sigma = 1$: pure noise. (b) Error per noise bin: local beats global everywhere; per-head mixing beats both.

16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step, even at native resolution. Figure 5b and d use the per-head shares in the noise bin at $\sigma = 0.62$; panel b caps the opacity at 12 times the uniform density, which clips only the local head's core.

Calibration budget. We merge the stored sufficient statistics from the first 4, 8, or 16 calibration prompts and rerun the 46 ablation prompts with each resulting table. Table L.4 reports generation-level differences from the 32-prompt table. The test reuses calibration observations but uses separate prompts for generation evaluation; it is not a held-out output-error benchmark.

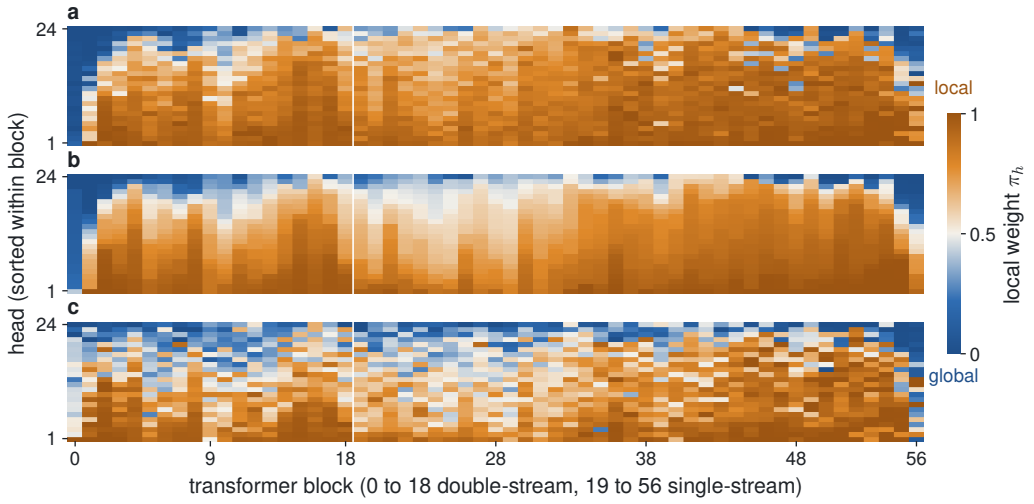

Figure B.5: **Heads in the same block differ widely at every noise level.** $\pi_h(\sigma)$ of the scale-2 table at $\sigma = 0.10$ (a), 0.54 (b), 0.97 (c), 57 blocks by 24 heads, 32 prompts, heads sorted per block by (b). Median within-block range 0.66 to 0.84; Figure B.1 shows the scale-4 inference table.

C CALIBRATION PROMPTS

This section lists the prompts behind the calibration of Section 3 and Algorithm 1 (`scripts/calib_prompts.jsonl`). The 1024^2 table uses prompts 1 to 32 and the 1280^2 table prompts 1 to 16. None of them is an evaluation prompt.

1. A candid street portrait of an elderly fisherman with a weathered face and a knitted wool cap, harbor in the background, overcast light
2. Two children laughing while running through a lawn sprinkler on a summer afternoon, backlit by the sun
3. A professional headshot of a young woman with curly red hair and freckles, soft studio lighting, neutral grey backdrop
4. A jazz trio performing in a dim basement club, saxophone player in the foreground, warm stage lights and haze
5. A chef plating a dessert in a busy restaurant kitchen, steam rising, stainless steel counters
6. A ballet dancer mid-leap in an empty rehearsal studio with tall windows and wooden floors
7. A red fox standing in fresh snow at the edge of a birch forest, early morning light
8. A macro photograph of a honeybee covered in pollen on a purple lavender flower
9. A herd of elephants crossing a dusty savanna river at sunset, long shadows
10. A tabby cat sleeping curled up on a stack of old books next to a window
11. A school of tropical fish swimming around a coral reef, sun rays through clear blue water
12. A bald eagle perched on a dead pine branch above a misty lake
13. A neon sign reading "OPEN 24 HOURS" in the window of a rainy diner at night, reflections on the wet sidewalk
14. A hand-painted wooden sign that says "Fresh Eggs" leaning against a farm fence, chickens nearby
15. A vintage movie theater marquee with the words "NOW SHOWING" lit by hundreds of bulbs
16. A chalkboard menu in a cafe listing coffee drinks and prices in neat handwriting
17. A highway road sign reading "Exit 42 Pine Valley" against a clear blue sky
18. A bookstore storefront with gold lettering "Second Chapter Books" on the glass, autumn leaves on the ground
19. A wide alpine valley with a winding river, snow-capped peaks and wildflower meadows, midsummer
20. Sand dunes in the desert at golden hour with sharp ridges and wind ripples
21. A rugged coastline with sea stacks and crashing waves under a stormy sky
22. A terraced rice field on a hillside in morning fog, farmers in conical hats
23. The northern lights over a frozen lake with a small wooden cabin, stars visible
24. A dense tropical rainforest with a waterfall and hanging vines, light filtering through the canopy
25. A cozy Scandinavian living room with a fireplace, sheepskin rug, and large windows showing snow outside
26. A modern open-plan kitchen with a marble island, pendant lights, and potted herbs

27. A cluttered artist studio with canvases, paint tubes, brushes in jars, and afternoon sunlight
28. A grand library reading room with long oak tables, green lamps, and arched ceilings
29. A small Japanese tatami room with a low table, sliding paper doors, and a garden view
30. A product photograph of a matte black wristwatch on a slate surface, dramatic side light
31. A pair of white running shoes floating against a pastel gradient background, studio product shot
32. A glass bottle of olive oil with a sprig of rosemary on a rustic wooden table, soft daylight
33. A flat lay of a leather notebook, fountain pen, glasses, and a cup of espresso on a desk
34. A ceramic teapot with a blue glaze and matching cups arranged on a linen tablecloth
35. A crowded night market in Taipei with food stalls, lanterns, and hundreds of people
36. A packed football stadium during a night match, floodlights and cheering fans
37. Commuters crossing a huge intersection in Tokyo during rush hour, seen from above
38. A music festival crowd at sunset with raised hands and confetti in the air
39. A busy train station concourse with travelers, departure boards, and a glass roof
40. A gothic cathedral facade with intricate stone carvings and a rose window, low angle
41. A row of colorful Victorian townhouses on a hilly street with a cable car
42. A brutalist concrete apartment block with repeating balconies under a pale sky
43. A glass skyscraper reflecting clouds, shot looking straight up from the street
44. A traditional Chinese pagoda beside a lake with cherry blossoms in full bloom
45. An old stone bridge over a canal in Venice with gondolas and laundry lines
46. An astronaut riding a horse across a red desert planet with two moons, cinematic
47. A detailed watercolor illustration of a lighthouse on a rocky island with seagulls
48. A steampunk airship floating above a Victorian city at dusk, intricate brass details

D IMPLEMENTATION DETAILS AND COST

This section expands Section 2.2 and Figure 3. Figure D.1 draws the two operators, the gate, and the mix at component level, together with the calibrated table for all 57 blocks. Table D.1 gives attention cost, and Table D.2 and Figure D.2 end-to-end latency; Appendix E writes out the corresponding algorithms.

FlexAttention window. Image tokens are reordered tile-major, so that a sparse block is a small spatial tile. Query blocks cover 8×16 tokens (128 queries) and key blocks 8×8 (64 keys), shifted so that unclamped windows start on a block edge. The T text keys and the 1,024 bridge keys come first, as a prefix of $T + 1{,}024$ keys shared by every query; the bridge keys keep their true positions. The wider windows of the four steps after the opening use the same tiling with more key blocks per query block. The mask function states window membership per query-key pair, so the result does not depend on the block layout, only the amount of skipped work does. For the operator without the schedule on GH200, 8×16 query blocks with 8×8 key blocks were fastest at 2048^2, 4096^2, and 2048×6144. RoPE, subsampling, both branches, and the mix compile into one graph per layer; the mix is in fp32.

Numerical checks. The 18 CPU tests in `tests/test_sca.py` use float32 to check full-mode agreement with stock FLUX attention, both branches against brute-force Equations 2 and 3 without the bridge, full-grid and factor-1 identities, calibration when the local branch is exact, the gate, and FlexAttention against an explicit gather reference. Tests in `tests/test_dedup.py` check the bridge selection (Equation 4) and the local branch with bridge keys against brute force, the window schedule, and that the operator reduces to the one without the schedule when these options are off. A separate GPU check in `scripts/bench_attn.py` compares the gather and FlexAttention implementations in bf16: the recorded maximum absolute difference is 0.015625, and both have the same maximum error against the fp32 reference on the tested grids. This is an observed absolute-error check; no per-element ULP bound was measured.

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256. Our frozen runs retain these rounded ids for RoPE. Passing the true grid from the sampler fixes window membership and subsampling geometry, but does not restore exact RoPE coordinates: the coarse-query positions are also computed from the rounded ids. At a 384-token axis, index 383 rounds to 384 and its scale-6 coarse coordinate becomes 64; at a 512-token axis, index 511 rounds to 512 and its scale-8 coarse coordinate also becomes 64. A local 64-token window can likewise have a rounded endpoint difference of 64 instead of the ideal bound 63. Proposition 1 therefore applies exactly to the mathematical operator, not to these finite-precision positions at 2048×6144, 6144^2, or 8192^2. The 4096^2 grid uses indices 0–255, which bf16 represents exactly. FLUX.2 uses integer position ids. Rebuilding FLUX.1 positions in higher precision would require a separate controlled experiment; the reported runs are unchanged.

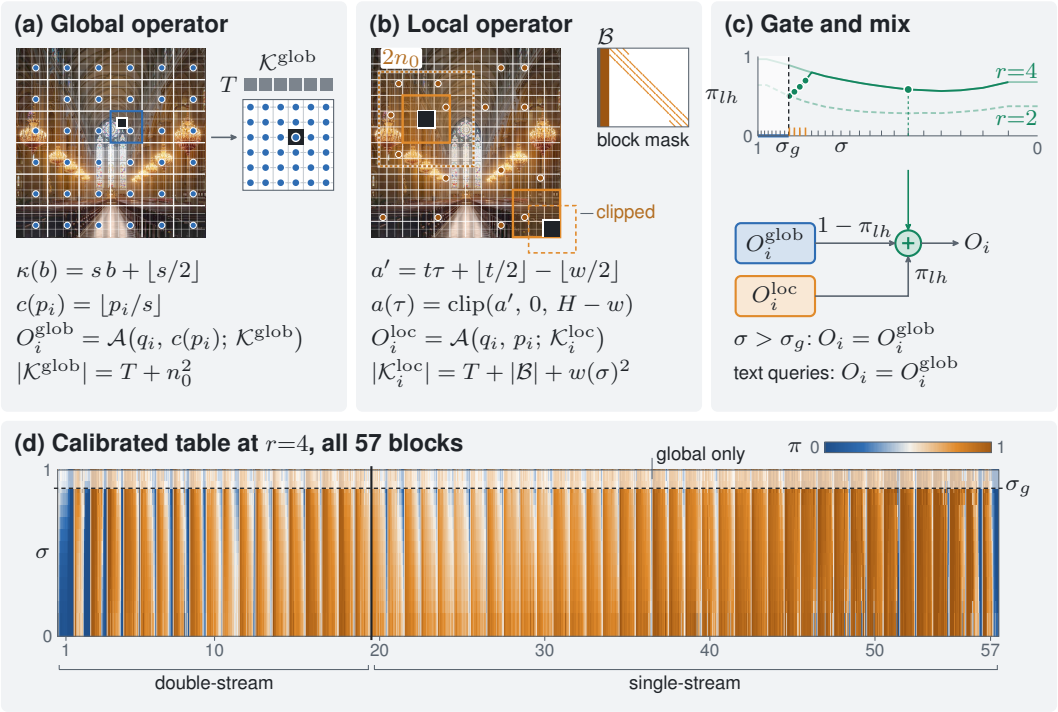

Figure D.1: **The operators at component level.** (a) The global operator keeps the centre token $\kappa(b)$ of each $s \times s$ block as a key at coarse position b and places query i at $c(p_i)$ (Equation 2). (b) Each $t \times t$ query tile attends in one softmax to the text keys, the bridge set $\mathcal{B}$ (dots; Equation 4) and a window at original positions, clipped at the border (dashed: unclipped a'), whose side shrinks from $2n_0$ (dotted) to n_0 after the gate (Equation 6); inset: block mask at $w = n_0$ (grey: text, dark amber: bridge; Equation 3). (c) Block 2, head 8 (chosen for a large gap; median 0.05) at $r = 2$ (dashed) and $r = 4$ (solid): window steps (amber ticks) read π_{lh} at $r = \max(H, W)/w$, native steps at $r = 4$; for $\sigma > \sigma_g$ every head returns O_i^{glob} (Equation 5). (d) The $r = 4$ table, all 57×24 heads. Toy grid: $H = 24$, $s = 4$, $n_0 = 6$, $t = 2$, one schematic bridge token per 6×6 cell.

Non-divisible grids. When H/n_0 is not an integer, the subsampling factor per axis is the divisor of H nearest to H/n_0, and the tile side the divisor nearest to t; neither case arises at the sizes we evaluate.

Table D.1: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$), with full attention over each row in parentheses. SCA runs the global branch alone for the 8 opening steps, local windows of side 128, 112, 96, and 80 for the next four, and the native window $n_0 = 64$ for the last 16, each with $T + 1{,}024$ text and bridge keys. Without the schedule the window is always n_0 and there is no bridge; "gated" skips the local branch in the first 8 steps. Bottom: measured time of one attention layer for SCA without the schedule (both branches and the mix) in bf16 on GH200, minimum over interleaved runs, with the speedup over full attention in parentheses. Source: `paper/figures/src/alg_drafts/cost_v2.txt`.

FLOPs per image grid, (s_H, s_W)	2048×4096 128×256, (2, 4)	4096×4096 256×256, (4, 4)	2048×6144 128×384, (2, 6)
Full attention	2.17×10^{16}	8.56×10^{16}	4.84×10^{16}
SCA without the schedule, no gate	5.97×10^{15}	1.19×10^{16}	8.93×10^{15}
SCA without the schedule, gated	5.12×10^{15} (4.2)	1.02×10^{16} (8.4)	7.66×10^{15} (6.3)
SCA (ours)	6.24×10^{15} (3.5)	1.24×10^{16} (6.9)	9.34×10^{15} (5.2)
One layer, bf16 (ms) token grid	2048^2 128×128	4096^2 256×256	2048×6144 128×384
Full attention (SDPA)	7.1	87.3	49.6
SCA without the schedule	5.2 (1.4)	18.7 (4.7)	14.0 (3.5)

End-to-end latency. Table D.2 and Figure D.2 come from one matched protocol on one GH200 (`results/latency_curve_batch1.csv`, which also lists the excluded warm-up images;

Table D.2: **Seconds per image**, batch 1, idle GH200, 28 steps with text encoding and VAE decoding, median of three timed images after an excluded warm-up; full attention not run at 8192^2. Grey: published values without the schedule; their final-operator run is pending.

Size (N tokens)	Full attention	AReA	SCA
2048^2 (16,384)	22.9	41.0	**18.2**
3072^2 (36,864)	73.9	114.3	**40.2**
4096^2 (65,536)	186.5	202.4	**71.1**
6144^2 (147,456)	762.7	482.3	**161.6**
8192^2 (262,144)	not run	952.0	**287.6**

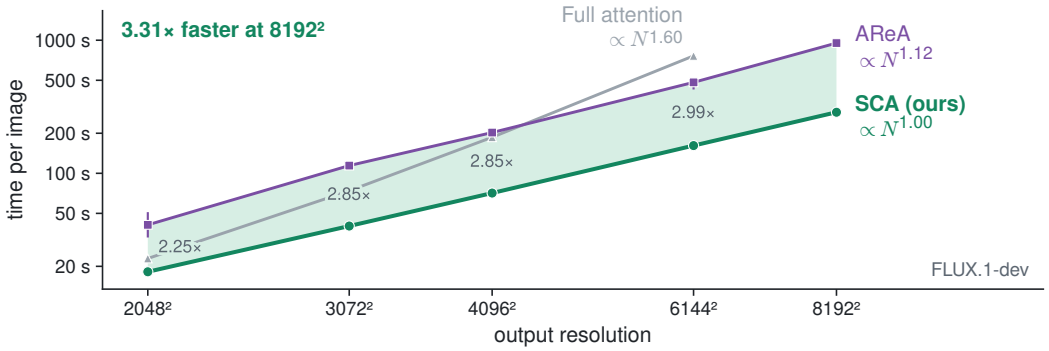

Figure D.2: **Latency at every size.** Figure 8 at full width: batch 1 on one GH200, median of three prompts, whiskers min to max; full attention was not run at 8192^2. Grey ratios: speedup over AReA per size; exponents fit seconds $\propto N^b$ (N image tokens). Without the schedule; pending.

`results/latency_curve_summary.csv` summarizes all cells but full attention at 6144^2). Every SCA timing in this paragraph is for SCA without the schedule; the schedule and the bridge add 9% at 4096^2 (Section 4.3). At each size, each method generates one excluded warm-up image and then three timed images in the same process, at batch 1 and 28 steps; we report the median. At 4096^2 the medians are 186.5 s for full attention (186.0, 186.6, 186.5), 202.4 s for AReA (192.7, 202.4, 204.3), and 71.1 s for SCA (71.0, 71.1, 71.1). At 2048^2 and 3072^2 they are 22.9 and 73.9 s for full attention, 41.0 and 114.3 s for AReA, and 18.2 and 40.2 s for SCA. At 6144^2 and 8192^2, AReA takes 482.3 and 952.0 s and SCA 161.6 and 287.6 s, so SCA reduces measured time by 66.5% and 69.8%. At 6144^2 full attention takes 762.7 s (763.5, 762.7, 762.5), $4.7\times$ SCA's time. An earlier 4096^2 benchmark, kept as a separate repeat (`results/latency_4096_batch1.csv`), gave 184.6 s for full attention (184.1, 184.7, 184.6), 212.6 s for AReA (207.3, 225.8, 212.6), 88.1 s for SCA + band guard (88.1, 98.7, 88.1; single-pass compiled band), and 71.0 s for SCA without the schedule or the guard (71.0, 83.1, 70.9). On the shared campaign nodes, in the 191-prompt run that also timed the band guard, per-image times at 4096^2 were 71 s for SCA without the guard, 120 s for SCA + band guard, and 262 s for AReA; the campaign code used an unfused band, so dropping it saved more there (49 s) than in the clean measurement (17 s). At the non-square sizes we have only campaign timings, with both methods on the same pool of shared nodes (mean over 195 prompts): SCA takes 38.1 s and AReA 112.4 s at 2048×4096, and 55.8 s and 150.5 s at 2048×6144.

E ALGORITHMS IN FULL

Algorithms 3 and 4 expand Algorithms 1 and 2 to every step the code performs, and Algorithm 5 gives progressive SCA (Section 2.6). Each step follows `sca/calib.py`, `sca/attn.py`, or `scripts/sample_progressive.py` in the code repository (link in the abstract), and each comment names the equation or setting that a line implements.

Algorithm 3: **Calibration**, in full

```
# Input: 32 prompts; FLUX.1-dev at 1024^2 (n = 64),
#   28 steps, guidance 4.5, full attention
# Output: pi[s_c, l, h, beta], centres[beta]
def bin(sigma): return min(int(14 * sigma), 13)
def on_step(sigma):              # once per step
  S[bin(sigma)] += sigma; C[bin(sigma)] += 1
def calibrate(q, k, v, l, sigma):  # every block
  O_full = attn(q, k, v)         # Eq. 1, all keys
  beta = bin(sigma)              # 14 bins, width 1/14
  for s_c in (2, 4):             # emulation, App. B
    f, w = s_c, n // s_c         # subsample, window
    t_c = round(t * w / n0)      # tile sides 4 and 2
    O_glob = global_branch(q, k, v, f)     # Eq. 2
    O_loc = local_branch(q, k, v, w, t_c) # Eq. 3
    dl, df = O_loc - O_glob, O_full - O_glob
    # Eq. 7: per head, over image rows, channels
    A[s_c, l, :, beta] += sum(dl * dl)
    B[s_c, l, :, beta] += sum(df * dl)
    E[s_c, l, :, beta] += sum(df * df)
    F[s_c, l, :, beta] += sum(O_full * O_full)
  return O_full                  # sampler unchanged
sample(prompts, on_step, calibrate)
pop = C > 0                      # 13 of 14 bins
centres = where(pop, S / C, (arange(14)+0.5)/14)
pi = clip(B / A, 0, 1); pi[A == 0] = 1  # Prop. 3
pi = fill_empty_bins(pi, pop)  # nearest bin
J = E - 2 * pi * B + pi**2 * A # error of the mix
rel_err = J.sum() / F.sum()    # Section 3
```

Algorithm 4: **Sampling**, in full

```
# Input: q, k, v before RoPE (T text, then H x W
#   image), block l, noise level sigma
# Output: O for all T + HW queries; sca/attn.py
# Per image: s_H, s_W = H // n0, W // n0; pi_s is
#   the table entry nearest max(s_H, s_W) in log
# Per step, all 57 x 24 heads, ends held:
w = n0                           # Eq. 6
for sig_k, w_k in ((.88, 128), (.86, 112),
                   (.84, 96), (.82, 80)):
  if sigma > sig_k: w = w_k; break
if w == n0: pi = interp(sigma, centres, pi_s)
else:                 # ratio r, log-linear
  r = max(H, W) / w; u = clip(log2(r / 2), 0, 1)
  pi = (1-u) * interp(sigma, centres, table[2]) \
     + u * interp(sigma, centres, table[4])
kb = k_img[s_H//2::s_H, s_W//2::s_W]  # κ(b)
vb = v_img[s_H//2::s_H, s_W//2::s_W]
cp = p // (s_H, s_W)             # c(p_i); text at 0
Q = cat(rope(q_txt, 0), rope(q_img, cp))
K = cat(rope(k_txt, 0), rope(kb, b))
O_glob = sdpa(Q, K, cat(v_txt, vb))   # Eq. 2
if sigma > sigma_g:                   # 0.89, Eq. 5
  return O_glob                # opening phase
# bridge (Eq. 4), per block and step:
d = ((k_img - k_img.mean(0))**2).sum(heads, ch)
B = top_k(d, 16) per cell, 8 x 8 cells  # 1,024
kr = rope(k_img, p)              # true positions
kl = cat(rope(k_txt, 0), kr[B], kr)
vl = cat(v_txt, v_img[B], v_img)  # no dedup
P = T + len(B)                   # prefix, seen by all
w = (min(w, H), min(w, W))     # window sides
a = clip(t*(p//t) + t//2 - w//2, 0, (H, W) - w)
keep(i, j) = j < P or a_i <= p_j < a_i + w  # Eq. 3
perm = tile_major(H, W)          # 8 x 16 query blocks
mask = block_mask(keep, perm) # 8 x 8 key blocks
O_loc = flex_attention(rope(q_img, p)[perm],
    kl, vl, mask)[inverse(perm)]
O_img = pi[l] * O_loc + (1 - pi[l]) * O_glob[T:]
return cat(O_glob[:T], O_img) # fp32 mix, Eq. 5
```

Algorithm 5: **Progressive SCA**, HRDiT's schedule with SCA attention (Section 2.6)

```
# Input: prompt, seed; FLUX.1-dev in bf16; SCA tables at scales 2 and 4
# Output: 4096^2 image; scripts/sample_progressive.py
sig = shift(linspace(1, 1/30, 30), mu=1.15)     # one 30-step schedule serves all stages, as in HRDiT
x = noise(1024^2, seed)                         # stage 1: stock dense attention, native RoPE
for i in range(30):                             # Euler; ref keeps each step's x0 estimate
  v = flux(x, sig[i], g=3.5); ref[i] = x - sig[i] * v; x = x + (sig[i+1] - sig[i]) * v
for res, k, g, s_c, a in ((2048, 13, 4.5, 2, 1.0), (4096, 20, 6.0, 4, 0.25)):
  img = sharpen(bicubic(decode(x), res))        # 2 img - blur(img), blur width ~ U(0.1, 2.0)
  x = sig[k] * noise + (1 - sig[k]) * encode(img)   # re-enter at sigma 0.81, then 0.61
  use_sca(table[s_c], n0=64, t=8, sigma_g=0.89, bridge=(8, 16))  # replaces NTK RoPE, attention scale,
                                                # pruning, alignment; w = n0, as sig[k] < 0.82 (Eq. 6)
  N = 30 - k                                    # 17, then 10 steps; sig[k] < sigma_g: no opening phase
  for n, i in enumerate(range(k, 30)):
    v = flux(x, sig[i], g); x0 = x - sig[i] * v; r, ref[i] = ref[i], x0    # SCA in every block
    x0 = x0 + a * (N - n) / N * (lowpass(up(r)) - lowpass(x0))  # Butterworth, order 4, cut-off 0.2
    u = momentum((x - x0) / sig[i], 0.5 * (N - n) / N)          # HRDiT's velocity momentum
    x = x + (sig[i+1] - sig[i]) * u
return decode(x)
```

F HYPERPARAMETERS

Table F.1 expands the Models and hyperparameters paragraph of Section 4 and lists every setting of the main runs.

Table F.1: **All hyperparameters** of the main runs (code commit `7bd64c0`), produced by `scripts/sample.py --method sca --pi-table calib/k1_1024.npz --gate-sigma 0.89 --bridge-cells 8 --bridge-k 16 --win-sched 0.88:128,0.86:112,0.84:96,0.82:80 --win-pi-match --win-pi-interp`. Appendix I gives the settings that differ for UltraImage, HRDiT, SigMa, and FLUX.2-dev.

Group	Setting	Value
Model	checkpoint, precision	FLUX.1-dev, bf16 weights and activations
	blocks, heads, head dim	19 double-stream + 38 single-stream, 24, 128
	native grid n_0, text tokens T	64 (1024^2 pixels, 16 pixels per token), 512
Sampler	steps, solver	28, flow-matching Euler
	guidance, shift	4.5 (distilled guidance), $\mu = 1.15$ at every size
	seed, decoding	seed 0 per prompt, shared across methods; tiled VAE decoding
Local branch	tile t	8
	window $w(\sigma)$	128, 112, 96, 80 for σ in $(0.88, 0.89]$, $(0.86, 0.88]$, $(0.84, 0.86]$, $(0.82, 0.84]$, then 64; clamped at borders; original positions
	bridge	16 tokens per cell of an 8×8 grid (1,024), farthest from the mean key, recomputed per block and step, true positions
	kernel	FlexAttention; query blocks 8×16, key blocks 8×8, tile-major order
Global branch	subsampling	block centre, factor (s_H, s_W) = target grid / n_0
	positions	queries at $c(p)$, keys at block index, text at 0
Mixing	table	per (block, head, σ bin, ratio), $57 \times 24 \times 14 \times 2$, ratios 2 and 4
	ratio	$r = \max(H, W)/w(\sigma)$, weight linear in $\log_2 r$ between ratios 2 and 4, nearest outside
	interpolation	linear in σ between the mean sampled noise levels of occupied bins, constant outside
	gate	$\sigma_g = 0.89$ (8 of 28 steps global only, then 4 wide-window steps)
Calibration	prompts, resolution	32 at 1024^2 (16 more at 1280^2 for transfer)
	emulated scales	2 and 4, one pass; 14 bins of width $1/14$
	hardware	one GH200; complete calibration runtime was not retained
Band guard	(not in final method)	average-pool $p = 16$, bilinear upsampling

G EVALUATION PROTOCOL AND METRICS

This section expands the Data and Metrics paragraphs of Section 4: Table G.1 summarizes the protocol, and the paragraphs below define each metric and interval.

Prompts and references. The Aesthetic-4K evaluation split (Zhang et al., 2025a) has 195 prompts with 4K reference images. Four of them (boats, Moai, woman and bear, underwater surfer) were used to choose the opening phase and window; they stay in the evaluation set for every method, which favours SCA slightly. The 195 reference rows hold 168 distinct images (27 duplicate rows, each with its own caption); distribution metrics keep the official rows and weights, as Diffusion-4K's statistics do. The 46-prompt ablation subset was fixed before any ablation was run. HRGen-1K (Rajabi et al., 2026a) has 1000 prompts and no references; we use all of them. The 48 calibration prompts were written for this purpose and share no prompt with Aesthetic-4K (the largest word-set Jaccard overlap with any evaluation prompt is 0.26).

Implementations. torch-fidelity 0.3.0, pyiqa 0.1.16 (MUSIQ) and 0.1.15 (CLIP-IQA, ARNIQA), hpsv2 1.2.0, and image-reward 1.5.

Table G.1: **Evaluation protocol.** The FLUX.1-dev attention comparisons share sampler settings and prompt seeds. UltraImage and HRDiT use their own multi-stage protocols; FLUX.2-dev uses its own native-resolution schedule (Appendix I). Prompt counts include the four development prompts. Table F.1 lists the hyperparameters; the paragraphs below define the metrics.

Models	FLUX.1-dev (Black Forest Labs, 2024) (main results); FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) (195 prompts); FLUX.2-dev (Black Forest Labs, 2025) (50 prompts); bf16.
Sampler	28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, seed label 0 per prompt, tiled VAE decoding; UltraImage, HRDiT, and FLUX.2-dev exceptions are detailed in Appendix I.
Sizes	4096^2, 2048×4096, 2048×6144 (195 prompts each); 6144^2 (100-prompt subset); 8192^2 (12 exploratory paired prompts, metrics and qualitative examples).
Prompts	Aesthetic-4K evaluation set with 4K references (Zhang et al., 2025a), 195 prompts, fixed 46-prompt ablation subset; HRGen-1K (Rajabi et al., 2026a), all 1000 prompts.
Baselines	Full attention; AReA, released code (Rajabi et al., 2026a); DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (two-stage defaults; Zhao et al., 2025), HRDiT (released progressive protocol; Xue et al., 2026); SigMa (Zhao et al., 2026) on 195 prompts.
Metrics	ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; FID, KID, IS, patch FID/KID; MUSIQ (full, 1024^2 crops), CLIP-IQA, ARNIQA; 95% paired bootstrap intervals over prompts.
Hardware	One NVIDIA GH200 per process; verified timing logs identify the GH200 120 GB model. Clean latency uses batch 1; campaign timings are marked separately.

Preference and alignment metrics. ImageReward v1.0 (Xu et al., 2023) (bicubic resize to 224, centre crop). HPSv2 and HPSv2.1 (Wu et al., 2023): cosine of normalized image and text features with the official validation transform (longest side 224, padded). PickScore (Kirstain et al., 2023): $\exp(\text{logit scale})$ times the cosine. CLIP score (Radford et al., 2021): $100 \max(\cos, 0)$ with ViT-L/14. Aesthetic: the LAION improved aesthetic predictor (sac+logos+ava1, linear MSE) on the L2-normalized CLIP ViT-L/14 image embedding (Schuhmann, 2022).

No-reference quality. MUSIQ (Ke et al., 2021) (pyiqa defaults) on the full image at native resolution, and MUSIQ_c, its mean over four fixed random 1024^2 crops per image; crop positions depend only on the prompt id, so they are identical across methods. CLIP-IQA (Wang et al., 2022) and ARNIQA (Agnolucci et al., 2024) (pyiqa defaults) on the image downscaled with antialiased bicubic resampling to a long side of 1024 pixels, the scale both were trained on.

Distribution metrics and reference correction. FID (Heusel et al., 2017) and KID compare the generated images (195 on the main set, 195 on the panel) with all 195 official reference rows, using Inception-v3 features (torch-fidelity) after bilinear resizing to 299^2. The reference rows contain 168 distinct image contents; repeated rows retain their official weights. Patch FID and KID are our own definition: the same Inception features on four fixed random 1024^2 crops per generated image and per reference row (780 generated and 780 reference crops on the main set). AReA and SEGA report a patch FID without defining its crops, so patch FID is never compared across papers. An audit found that the original feature-bank assembly keyed reference crops by resolved image path, so repeated images reused the last row's crop features. We corrected the lookup to retain each row's intended cached crops. Whole-image FID/KID are unchanged; all patch point estimates reported here use the corrected assembly. Original CSVs remain archived for provenance. Inception score uses full images at 299^2, with 10 splits.

Intervals. Per-image intervals are 95% paired percentile bootstraps over prompt ids: original summaries use 1000 draws; separately recomputed comparisons use 2000 draws with seed 0. For the main SCA–AReA FID/KID and patch-FID/KID comparison, 1000 draws resample generated prompt ids jointly across methods and resample the 168 distinct reference-image identities, keeping all rows and crops from each sampled identity together. This retains the 195-row weighted point estimate while accounting for repeated reference images. The resulting FID difference is -7.74 $[-13.11, -2.61]$; patch FID is -3.79 $[-7.46, 1.29]$ and patch KID ($\times 10^3$) is -2.29 $[-4.60, 0.05]$. An earlier version of this analysis reported a significant patch-KID improvement; under the corrected evaluation its interval includes zero. The whole-image and patch FID/KID comparisons at both non-square sizes, against every added baseline, and for the sampling-budget and progressive

runs also use reference-image clusters. Other whole-image FID/KID intervals are labelled historical reference-row bootstraps; they have not been recomputed with image-identity clusters. Historical patch intervals are not attached to corrected estimates and are omitted where recomputation is incomplete. FID's finite-sample bias can place point estimates outside percentile intervals. Intervals are unadjusted for multiple comparisons and measure prompt/reference sampling uncertainty, not variability over independent model training runs.

H REPRODUCING AREA

This section expands the AReA comparison of Section 4.1: our protocol for AReA and our runs beside its published numbers at 4096^2 (Table H.1) and on HRGen-1K (Table H.2). We run AReA's released code with its FLUX.1 defaults (block-centre pooling, first phase for 8 of 28 steps, halo width 8, bridge ratio 0.5), the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's published numbers match FLUX.1-Krea-dev, the default of its released code: rerunning that code on Krea-dev reproduces its published ImageReward, aesthetic score, and PickScore within 0.1 and its CLIP, HPS, and MUSIQ within 0.3 (Table H.1), while on FLUX.1-dev the same code gives ImageReward 0.92 against the published 1.32. Our direct SCA–AReA comparison holds the FLUX.1-dev settings fixed. UltraImage and FLUX.2-dev use separate protocols (Appendix I). AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; the matched sweep measures 202.4 s and 186.5 s on one GH200 (medians of three images). We use commit `9e38ec2` of the released repository, which includes a cost cap on halo and bridge tokens and a jagged halo edge; neither appears in AReA's paper, which does not say whether its Table 1 used them. Recomputed from the code with 512 text tokens, the first-phase pooling factors are 4, 4, 4, 4, 2, 2, 2, 2 at 4096×4096 and 2048×4096, and 8, 8, 8, 4, 4, 4, 2, 2 at 2048×6144. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted, so the first window falls back to 192 tokens and the cap drops the bridge and halo on the first three second-phase steps. Other differences from the release defaults are the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts. On FLUX.1-Krea-dev, AReA's default checkpoint, we rerun AReA and recalibrate SCA (16 prompts at 1024^2, scale-4 table, other settings unchanged) with the same 28-step, guidance-4.5 sampler on all 195 prompts; DyPE and SEGA run the Krea transformer through the frozen baseline harness on the 100 prompts of rows 4 to 103. There SCA wins six metrics and loses patch FID and, at the edge of significance, ImageReward, at a third of AReA's time (Table L.1). Table H.2 expands the HRGen-1K block of Table H.1 to every row of AReA's Table 2.

Table H.1: **Published numbers at 4096^2 next to our runs.** Aesthetic-4K, 195 prompts: AReA's Table 1 (HPS and MUSIQ from its Table 6; "FLUX.1"), our runs with FLUX.1-Krea-dev, the checkpoint AReA's released code uses (28 steps, guidance 4.5), and our FLUX.1-dev protocol. Last block: HRGen-1K (all rows in Table H.2). HPS: HPSv2 $\times 100$; MUSIQ on the full image; FID at 299^2 against the 195 reference rows; patch FID is omitted because AReA does not state its crops. †Fewer than all prompts; not ranked. **Bold** / underline: best / second best among ranked rows (HRGen-1K: bold only); ours shaded. Key result: Our run of AReA's code on FLUX.1-Krea-dev reproduces its published ImageReward, Aesthetic, and PickScore within 0.1 and its CLIP, HPS, and MUSIQ within 0.28; against that run SCA is higher on HPS and MUSIQ, within 0.1 on ImageReward, Aesthetic, CLIP, and PickScore, and worse on FID; on HRGen-1K, where we did not rerun AReA on FLUX.1-Krea-dev, SCA on that checkpoint exceeds every published AReA score. Grey: published values without the schedule; their final-operator run is pending.

Method	n	Image-Reward↑	Aes-thetic↑	CLIP↑	Pick-Score↑	HPS↑	MUSIQ↑	FID↓
Published by AReA, single pass (FLUX.1)								
Full attention	195	−0.72	5.80	25.72	20.34	–	–	179.8
UltraImage	195	−0.14	6.11	26.77	20.85	–	–	169.4
DyPE	195	1.11	6.32	28.98	22.79	–	–	155.2
SigMa	195	1.01	6.16	28.36	22.47	–	–	156.8
SEGA	195	1.27	6.31	29.16	23.16	–	–	150.6
AReA	195	**1.32**	6.37	**29.71**	**23.31**	29.69	45.2	148.5
Our runs, FLUX.1-Krea-dev, same prompts and metrics								
AReA	195	1.28	6.32	29.49	23.21	29.41	45.0	**142.6**
DyPE	100†	0.96	6.41	28.73	22.50	28.39	42.7	171.6
SEGA	100†	1.13	6.31	29.19	23.01	29.15	46.0	164.2
SCA (ours)	195	1.23	**6.39**	29.41	23.20	**29.70**	**46.3**	144.1
Our runs, FLUX.1-dev, shared sampler								
AReA	195	0.92	6.27	27.69	22.36	28.62	39.4	145.8
SCA (ours)	195	1.25	6.42	29.14	23.12	29.77	45.0	140.2
HRGen-1K, 1000 prompts without references: AReA's Table 2 and our FLUX.1-Krea-dev run								
AReA, published (FLUX.1)	1000	0.60	5.87	29.89	21.44	27.50	–	–
SCA (ours), FLUX.1-Krea-dev	1000	**0.78**	**6.03**	**30.10**	**21.64**	**28.06**	43.7	–

Table H.2: **HRGen-1K: AReA's published Table 2 next to our runs.** All 1000 prompts at 4096^2, no reference images. Top: AReA's Table 2 ("FLUX.1", which its released code runs as FLUX.1-Krea-dev). Bottom: SCA with a Krea-calibrated table on FLUX.1-Krea-dev (28 steps, guidance 4.5), and our paired FLUX.1-dev runs of AReA and SCA. HPS: HPSv2 $\times 100$; IS: Inception score; MUSIQ on the full image; †fewer than 1000 prompts. **Bold** / underline: best / second best among the published rows and our FLUX.1-Krea-dev row. Table H.1 holds the Aesthetic-4K comparison. Key result: SCA on FLUX.1-Krea-dev exceeds AReA's published ImageReward, Aesthetic, CLIP, PickScore, HPS, and IS. Grey: published values without the schedule; their final-operator run is pending.

Method	n	Image-Reward↑	Aes-thetic↑	CLIP↑	Pick-Score↑	HPS↑	IS↑	MUSIQ↑
Published by AReA (FLUX.1)								
Full attention	1000	−1.46	5.12	23.36	19.04	23.20	7.2	–
UltraImage	1000	−0.03	5.77	28.26	20.46	26.90	10.2	–
DyPE	1000	0.33	5.83	29.35	20.94	26.60	10.9	–
SigMa	1000	0.06	5.63	27.91	20.54	25.80	10.7	–
SEGA	1000	0.44	5.82	29.63	21.18	27.00	12.0	–
AReA	1000	0.60	5.87	29.89	21.44	27.50	13.9	–
Our runs								
SCA (ours), FLUX.1-Krea-dev	1000	**0.78**	**6.03**	**30.10**	**21.64**	**28.06**	**15.7**	43.7
AReA, our run, FLUX.1-dev	1000	−0.03	5.92	26.08	20.43	26.16	12.7	37.4
SCA (ours), FLUX.1-dev	1000	0.79	6.25	28.73	21.61	28.11	15.3	45.1

I Other baselines and FLUX.2-dev

DyPE and SEGA. We use the existing baseline harness frozen at commit `c05ebdc`, with FLUX.1-dev, 28 steps, guidance 4.5, maximum shift, tiled VAE decoding, and the same prompt-

derived seeds as SCA. These settings isolate the attention/position treatment on the checkpoint used in our main comparison. They do not reproduce the methods' reported FLUX.1-Krea-dev results; on Krea weights both trail AReA on ImageReward (Table L.1). Their low FLUX.1-dev scores should therefore not be read as a ranking under each method's preferred checkpoint.

UltraImage. We run a guided two-stage configuration from the FLUX branch of the released implementation through `scripts/run_ultraimage.py`. It uses a 1024^2 first stage and a 4096^2 second stage, 30 steps per stage, guidance 3.5 then 7, the `super_imax` guidance schedule, shifted-window patchification, proportional attention, RIFLEx positions, and attention decay. Each prompt also needs a 2+2-step block-average preparation pass. The reported campaign time includes preparation when not cached and the two-stage call; main-stage and preparation times are retained separately in sidecars. Each call reseeds both the generator and the global RNG, including the VAE sampling path, from the prompt id and seed label. Thus seed labels are paired, but the two-stage method does not share an identical latent trajectory with the one-stage methods. The wrapper preserves an upstream scheduler behaviour: assigning `scheduler.config.shift` does not update the scheduler's internal shift in the installed diffusers version, so both stages retain shift 3.0. This is a reproduction caveat, not a newly tuned setting. We report this system-level comparison separately from the common-sampler attention comparisons.

HRDiT. We run the released progressive protocol of the official repository (commit `f3be935`) on FLUX.1-dev in fp16: 30 steps at 1024^2 with guidance 3.5, then 17 steps at 2048^2 with guidance 4.5 and 10 steps at 4096^2 with guidance 6, 57 steps in all, with latent upsampling and the shipped per-head scope plan and attention settings. Its sampling budget therefore differs from the shared protocol, and we report it with the multi-stage recipes. HRDiT's global Torch random state is reset per prompt because its Gaussian blur draws a random width. Its clean time, 108.6 s, is the median of three batch-1 runs on the latency prompts (the first, 144.4 s, includes warm-up). The 195-prompt HRDiT evaluation used the original crop assembly, under which patch FID is 109.9 for HRDiT and 105.9 for SCA without the schedule. Recomputed from the cached crop features with the corrected assembly of Table 1 (`repro/evaluate_corrected.py`), HRDiT's patch FID is 103.0, against 99.1 for SCA without the schedule and 102.9 for AReA.

Progressive SCA. `scripts/sample_progressive.py` reimplements HRDiT's released schedule with the stock diffusers FLUX components and changes only the attention (Section 2.6). One 30-step schedule with dynamic shift $\mu = 1.15$ (from the 1024^2 sequence length) serves all stages, and the recorded HRDiT noise levels are reproduced to 10^{-6}. The 2048^2 stage repeats steps 14 to 30 and the 4096^2 stage steps 21 to 30. Each transition decodes the latent, upsamples it bicubically, applies HRDiT's unsharp mask with a blur width drawn from the per-prompt global seed, re-encodes it, and re-noises it with the per-prompt generator; guidance is 3.5, 4.5, and 6.0, and HRDiT's low-band structure guidance and velocity momentum are kept. At 1024^2 both methods use dense attention with native RoPE. At 2048^2 and 4096^2 we remove NTK RoPE, the proportional scale, head pruning, and position alignment, and use SCA with the FLUX.1-dev table at scales 2 and 4, $n_0 = 64$, $t = 8$, and $\sigma_g = 0.89$, which no high-resolution step exceeds. Deliberate differences from HRDiT's runner are bf16 instead of fp16 and a finite-value check at the end of each stage instead of after every step; with bf16 and stock components the 1024^2 images, and hence the layouts, differ from HRDiT's for the same seed (Figure M.2). Reported time is CUDA-synchronized and covers text encoding, all stages, transitions, and the final decode, excluding model loading; per-stage means are 5.1, 12.3, and 26.1 s, peak memory 37.8 GiB. The gate and every other setting were fixed before generation; two development prompts served as a smoke test.

SigMa. The dev-matched profile runs the released code (commit `fbe3a86`) on FLUX.1-dev under the shared sampler: 28 steps, guidance 4.5, bf16, YaRN positions, and the shift capped at its maximum from the target sequence length. SigMa's paper states FLUX.1-dev with guidance 3.5, while its released command line hardcodes FLUX.1-Krea-dev with guidance 4.5; we ran only the profile that matches our protocol. The outputs are finite, fully decoded 4096^2 images, but the images we inspected show a regular lattice of dots, as DyPE's do at 4096^2 on FLUX.1-dev. This may reflect our port or profile rather than the method, so the SigMa row should not be read as the method's best result.

Adapter for HRDiT and SigMa. SigMa runs on 195 prompts and HRDiT on all 195, both through one adapter with campaign per-prompt CPU generators, resident models, and tiled VAE decoding. These differ from the upstream command-line seed, offload, and decode choices, so the runs are controlled adaptations, not reproductions of the authors' tables. Campaign time includes text encoding, decoding, and finite-value checks after every step, which add synchronization overhead; it excludes model loading and is not a clean latency benchmark.

FLUX.2-dev. The model has 56 attention blocks and 48 heads per block. We reuse the two branches and calibration rule with a native grid of 64^2 tokens, local window 64, and query tile 8. The four-axis positions preserve the time and text axes while coarsening the two spatial axes. Calibration uses 16 prompts at 1024^2, 14 noise bins, and emulated scales 2 and 4; sampling uses the scale-4 table. Both full attention and SCA use 28 steps, guidance 4.0, seed label 0, and a shift fixed to the native sequence length, $\mu = 2.1514$. The pipeline's default extrapolated shift at 4096^2 would be about 11.55; we therefore do not interpret this comparison as a default-pipeline benchmark. The gate is 0.959 so that the first 8 steps use only global attention. The 50 evaluation prompts are Aesthetic-4K rows 4–53, a 50-prompt subset. Prompt embeddings are cached for both methods; reported times include denoising and tiled VAE decoding, and exclude text encoding and model loading. At emulated scale 4, fitted per-head relative output error is 0.1790, versus 0.1889 for per-layer weights and 0.1924 for local-only attention; split-half correlation is 0.950 and cross-half error 0.1794. These are calibration diagnostics, not generation ablations of the individual components on FLUX.2.

J Mechanisms of related methods

This section expands Section 5: Table J.1 compares how the closest training-free methods change attention.

Table J.1: **How training-free methods change attention.** Entries follow each paper's text or, where marked *, its mechanism; offsets: whether image query-key offsets stay in the trained range (text keys aside, Remark 1); cost: order in the image tokens N at a fixed native size. Key result: among single-pass methods, only SCA fits per-head weights in advance at the training resolution, and it runs at linear cost.

Method	Local view	Global view	Per-head reach	Offsets	Cost	Passes
FreeScale	overlapping crops	full attention	×	n/a	$O(N^2)^*$	cascade
DyPE	×	full, scaled RoPE	×	×*	$O(N^2)^*$	single
SEGA	×	full, NTK, logit scale	×	×	$O(N^2)$	single
UltraImage	×	full, edited RoPE	entropy at run time	×*	$O(N^2)^*$	one or two
HRDiT	per-head scopes	×	budget, fitted per size	✓*	$O(N^2)^*$	cascade
ResDiT	native patches	full, interpolated	×	✓	$O(N^2)^*$	single
AReA	shrinking windows	pooled keys	×	×	$O(N^2)^*$	single
SCA (ours)	shrinking windows	block-centre keys	fitted at 1024^2	window	$O(N)$	single

K Extended results with intervals

This section expands the main tables to all metrics and sizes. Table K.1 expands Table 1, Table K.2 gives its paired intervals against AReA, Table K.3 covers the non-square sizes, and Table K.4 checks the calibration table there. Table K.5 expands Table 3; Tables K.6 and K.7 cover the band guard.

Appendix L pairs the added baselines against SCA. All these tables are populated from archived evaluation CSV files. Per-image means were checked against the per-image records; distribution-level estimates use cached features and the corrected crop assembly (Appendix G). Set-level metrics (FID, KID, their patch versions, Inception score) use the whole set. FID/KID and patch-FID/KID intervals on the 195-prompt sets resample reference-image clusters; other FID/KID intervals retain the historical reference-row bootstrap unless a corrected image-cluster interval exists, and Inception-score intervals resample only generated images.

Table K.1: **At 4096^2, SCA beats AReA on every per-image metric at about a third of its latency.** FLUX.1-dev, 195 prompts (100 at 6144^2). HPS, CLIP-IQA, and ARNIQA are $\times 100$; higher is better except latency and FID. Bold: best per size. * / †: significantly better / worse than AReA (95% paired interval; none for full attention's FID and FID_p). SCA band w/o sched.: the band-guard run without the schedule, on the 191 prompts other than the development prompts. At 6144^2 the row is SCA without the schedule. Latency at 4096^2: batch-1 seconds per image on an idle GH200, median of three. ‡Campaign mean on shared nodes, not clean latency. $^\circ$UltraImage uses its two-stage defaults. Grey: published values without the schedule; their final-operator run is pending.

Method	Lat. (s)↓	Image Reward	Aesth.	CLIP	Pick Score	HPSv2	HPSv2.1	FID↓	FID_p ↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096×4096, scale factors (4, 4)													
Full attention	186.5	$-2.18^\dagger$	$4.50^\dagger$	$15.37^\dagger$	$17.94^\dagger$	$20.75^\dagger$	$7.60^\dagger$	312.4	252.2	$22.9^\dagger$	$33.1^\dagger$	$49.0^\dagger$	$54.4^\dagger$
DyPE	$195.9^\ddagger$	-0.10	5.72	24.92	20.51	25.53	20.55	205.5	171.9	36.3	50.3	59.6	60.2
SEGA	$190.4^\ddagger$	0.56	6.02	26.95	21.64	27.13	24.34	164.9	125.8	37.3	50.3	66.7	64.8
UltraImage$^\circ$	$308.5^\ddagger$	**1.28**	**6.50**	28.84	**23.33**	**29.96**	**30.77**	142.0	109.5	33.6	41.7	**72.8**	**72.3**
AReA (released code)	202.4	0.92	6.27	27.69	22.36	28.62	27.16	145.8	102.9	39.4	50.4	67.7	67.4
SCA band w/o sched.	88.1	1.20^*	6.46^*	28.87^*	22.90^*	29.40^*	28.95^*	142.3	98.8	41.4^*	**61.0***	70.9^*	70.8^*
SCA (final)	71.1	1.25^*	6.42^*	**29.14***	23.12^*	29.77^*	29.73^*	**140.2***	**98.4**	**45.0***	60.1^*	72.3^*	70.7^*
2048×4096, scale factors (2, 4)													
AReA (released code)	$112.4^\ddagger$	**1.13**	**6.41**	**28.59**	**22.66**	28.36	**27.64**	144.6	**111.7**	51.9	61.6	**74.9**	**73.0**
SCA (final)	**41.1**‡	1.11	$6.17^\dagger$	$28.09^\dagger$	22.61	**28.57***	27.49	**142.5**	114.9	**56.2***	**65.3***	$72.5^\dagger$	72.6
2048×6144, scale factors (2, 6)													
AReA (released code)	$150.5^\ddagger$	0.39	**6.14**	26.52	21.51	26.41	23.36	163.0	**105.7**	42.8	53.0	67.5	69.3
SCA (final)	**60.5**‡	**0.64***	$5.88^\dagger$	**26.78**	**21.66**	**26.59***	**23.60**	**142.8***	$109.7^\dagger$	**47.7***	**59.4***	**67.8**	**69.5**
6144×6144, scale factors (6, 6), 100 prompts													
AReA (released code)	$507.9^\ddagger$	-0.42	5.80	23.84	20.30	24.95	18.78	216.7	188.6	28.3	44.5	53.8	47.8
SCA w/o schedule	**164.7**‡	**0.84***	**6.24***	**27.12***	**21.89***	**27.48***	**25.16***	**169.9***	**122.4***	**37.5***	**48.3***	**60.1***	**62.6***

Table K.2: **All metrics at 4096^2**, 195 prompts, paired against AReA. Δ is method minus AReA with its 95% paired bootstrap interval; **blue** / *orange*: significantly better / worse than AReA. Intervals use 1000 prompt draws for SCA final (FID, KID, and patch versions also resample reference-image clusters) and 2000 for full attention's per-image metrics; full attention's set-level differences and campaign time are points without marks. SCA + band: the band-guard run without the schedule, on the 191 prompts other than the development prompts, paired on those prompts; its corrected patch values have no intervals. ‡Campaign mean on shared nodes (clean latency: Table D.2). Sources: `results/a4k195_main_sca_v2/` (SCA final), `results/a4k195_main/`, `final4k_*`.

Metric	AReA	SCA final	Δ [95% CI]	SCA + band	Δ [95% CI]	Full attention	Δ [95% CI]
ImageReward	0.921	1.248	**+0.327** [+0.247, +0.405]	1.198	**+0.281** [+0.208, +0.358]	−2.180	−3.101 [−3.218, −2.984]
Aesthetic	6.27	6.42	**+0.15** [+0.10, +0.20]	6.46	**+0.18** [+0.12, +0.24]	4.50	−1.77 [−1.87, −1.67]
CLIP score	27.69	29.14	**+1.45** [+1.13, +1.82]	28.87	**+1.20** [+0.79, +1.61]	15.37	−12.32 [−13.12, −11.54]
PickScore	22.36	23.12	**+0.76** [+0.64, +0.88]	22.90	**+0.54** [+0.40, +0.71]	17.94	−4.42 [−4.63, −4.22]
HPSv2 (×100)	28.62	29.77	**+1.15** [+0.96, +1.34]	29.40	**+0.75** [+0.53, +0.99]	20.75	−7.87 [−8.24, −7.50]
HPSv2.1 (×100)	27.16	29.73	**+2.56** [+2.20, +2.97]	28.95	**+1.75** [+1.31, +2.22]	7.60	−19.56 [−20.41, −18.72]
MUSIQ	39.4	45.0	**+5.6** [+3.7, +7.4]	41.4	**+2.0** [+0.0, +3.9]	22.9	−16.5 [−18.0, −14.9]
MUSIQ_c	50.4	60.1	**+9.7** [+8.2, +11.1]	61.0	**+10.6** [+9.2, +12.3]	33.1	−17.3 [−19.1, −15.5]
CLIP-IQA (×100)	67.7	72.3	**+4.6** [+3.1, +6.1]	70.9	**+3.3** [+1.8, +4.9]	49.0	−18.7 [−20.5, −16.9]
ARNIQA (×100)	67.4	70.7	**+3.4** [+2.5, +4.4]	70.8	**+3.5** [+2.4, +4.6]	54.4	−12.9 [−14.5, −11.3]
Inception score	6.66	6.44	−0.22 [−0.65, +0.39]	6.10	−0.33 [−0.88, +0.18]	3.82	−2.84
FID↓	145.8	140.2	**−5.6** [−10.6, −0.4]	142.3	−3.7 [−9.6, +1.0]	312.4	+166.6
KID↓ ($\times 10^3$)	7.17	5.82	−1.35 [−3.17, +0.49]	5.48	−1.63 [−3.84, +0.51]	164.15	+156.98
FID_p ↓	102.9	98.4	−4.4 [−7.9, +0.4]	98.8	−4.7	252.2	+149.3
KID_p ↓ ($\times 10^3$)	11.40	8.87	**−2.53** [−4.57, −0.60]	9.67	−1.77	171.01	+159.61
Seconds per image‡ ↓	264.4	78.4	**−186.0** [−199.6, −173.4]	120.0	**−141.7** [−155.8, −127.1]	196.2	−68.1

Table K.3: **All metrics at the non-square sizes**, 195 prompts each, SCA final paired against AReA (released code). Conventions as in Table K.2: evaluator intervals over 1000 prompt draws, and for FID, KID, and their patch versions 1000 draws that also resample reference-image clusters. Sources: `results/a4k195_2048x4096_sca_v2/` and `a4k195_2048x6144_sca_v2/` (SCA final), `results/a4k195_2048x4096/` and `a4k195_2048x6144/`.

	2048 × 4096, factors (2, 4)			2048 × 6144, factors (2, 6)		
Metric	AReA	SCA	Δ [95% CI]	AReA	SCA	Δ [95% CI]
ImageReward	1.132	1.106	−0.026 [−0.107, +0.047]	0.391	0.638	**+0.247** [+0.125, +0.376]
Aesthetic	6.41	6.17	−0.23 [−0.29, −0.18]	6.14	5.88	−0.26 [−0.37, −0.15]
CLIP score	28.59	28.09	−0.49 [−0.89, −0.15]	26.52	26.78	+0.26 [−0.30, +0.79]
PickScore	22.66	22.61	−0.06 [−0.17, +0.06]	21.51	21.66	+0.15 [−0.03, +0.32]
HPSv2 (×100)	28.36	28.57	**+0.21** [+0.09, +0.32]	26.41	26.59	**+0.17** [+0.06, +0.31]
HPSv2.1 (×100)	27.64	27.49	−0.16 [−0.45, +0.12]	23.36	23.60	+0.23 [−0.14, +0.63]
MUSIQ	51.9	56.2	**+4.3** [+2.9, +5.8]	42.8	47.7	**+4.9** [+3.1, +6.8]
MUSIQ_c	61.6	65.3	**+3.7** [+2.5, +4.8]	53.0	59.4	**+6.4** [+4.9, +8.0]
CLIP-IQA (×100)	74.9	72.5	−2.4 [−3.4, −1.5]	67.5	67.8	+0.3 [−1.1, +1.8]
ARNIQA (×100)	73.0	72.6	−0.4 [−1.1, +0.1]	69.3	69.5	+0.2 [−0.6, +1.0]
Inception score	6.27	6.82	+0.55 [−0.12, +1.04]	5.48	6.41	**+0.93** [+0.28, +1.39]
FID↓	144.6	142.5	−2.1 [−6.9, +3.3]	163.0	142.8	**−20.2** [−24.7, −11.9]
KID↓ ($\times 10^3$)	7.81	7.02	−0.79 [−3.25, +1.95]	17.92	7.01	**−10.91** [−14.53, −7.26]
FID_p ↓	111.7	114.9	+3.2 [−0.5, +7.1]	105.7	109.7	+4.0 [+1.0, +10.8]
KID_p ↓ ($\times 10^3$)	14.26	15.20	+0.95 [−0.46, +2.59]	13.56	15.09	+1.54 [−2.01, +5.42]
Seconds per image‡ ↓	112.4	41.1	**−71.3** [−73.3, −69.2]	150.5	60.5	**−90.0** [−92.2, −87.8]

Table K.4: **Calibration table at 2048 × 4096**, 46 ablation prompts, SCA without the schedule throughout, paired against the same operator with the scale-4 table; **blue** / *orange*: significantly better / worse than this reference. Sources: `results/ps2_*`.

Metric	SCA, scale-2 table	AReA (released code)
ImageReward	+0.012 [−0.025, +0.050]	+0.027 [−0.141, +0.192]
Aesthetic	−0.00 [−0.03, +0.02]	**+0.29** [+0.16, +0.43]
CLIP score	−0.02 [−0.22, +0.17]	+0.19 [−0.53, +0.91]
PickScore	**+0.05** [+0.00, +0.09]	+0.14 [−0.09, +0.39]
HPSv2 (×100)	**+0.09** [+0.04, +0.13]	−0.19 [−0.44, +0.06]
HPSv2.1 (×100)	**+0.17** [+0.08, +0.26]	+0.39 [−0.21, +1.01]
MUSIQ	+0.0 [−0.5, +0.5]	−3.1 [−7.5, +1.2]
MUSIQ_c	−0.0 [−0.4, +0.2]	−4.1 [−6.7, −1.3]
CLIP-IQA (×100)	−0.6 [−1.2, +0.0]	+1.3 [−0.8, +3.4]
ARNIQA (×100)	−0.2 [−0.6, +0.1]	+0.4 [−1.5, +2.8]

Table K.5: **Ablation intervals for the base-operator rows of Table 3**, fixed 46-prompt ablation subset at 4096^2, paired against SCA without the schedule (2000 bootstrap draws), whose column gives absolute values; every variant also runs without the schedule. **Blue** / *orange*: significantly better / worse. Panels split the variants by component; each panel's last row gives patch FID (set-level, no paired interval). Sources: `results/ablnb_*`.

Weight rule after the opening phase

Metric	SCA w/o sched.	$\pi = 1$	shared $\pi(\sigma)$	per-layer
ImageReward	1.124	-0.357 $[-0.524, -0.207]$	-0.090 $[-0.173, -0.012]$	-0.054 $[-0.159, +0.030]$
Aesthetic	6.45	-0.21 $[-0.35, -0.07]$	-0.09 $[-0.18, -0.02]$	$+0.01$ $[-0.06, +0.07]$
CLIP score	29.41	-1.93 $[-2.68, -1.20]$	-0.38 $[-0.78, +0.08]$	-0.13 $[-0.54, +0.28]$
PickScore	22.96	-1.20 $[-1.44, -0.99]$	-0.38 $[-0.51, -0.26]$	-0.16 $[-0.28, -0.05]$
HPSv2 ($\times 100$)	29.71	-1.78 $[-2.07, -1.50]$	-0.96 $[-1.19, -0.73]$	-0.50 $[-0.63, -0.37]$
HPSv2.1 ($\times 100$)	29.98	-2.86 $[-3.35, -2.36]$	-1.83 $[-2.24, -1.43]$	-0.70 $[-0.98, -0.41]$
MUSIQ	46.1	-3.5 $[-6.5, -0.7]$	-4.9 $[-7.9, -1.9]$	$+0.7$ $[-1.6, +3.0]$
MUSIQ_c	62.5	-0.3 $[-2.5, +1.7]$	-10.4 $[-12.8, -8.3]$	-2.4 $[-3.6, -1.1]$
CLIP-IQA ($\times 100$)	71.3	-4.6 $[-6.3, -3.0]$	-7.7 $[-9.4, -6.0]$	-0.9 $[-2.0, +0.1]$
ARNIQA ($\times 100$)	70.5	-1.0 $[-2.0, +0.0]$	-6.5 $[-8.5, -4.8]$	-1.5 $[-2.3, -0.9]$
$\text{FID}_p \downarrow$	168.0	192.2	166.7	167.7

Global branch and opening phase

Metric	SCA w/o sched.	$\pi = 0$	mean pool	v0, no gate
ImageReward	1.124	-0.395 $[-0.507, -0.295]$	-1.518 $[-1.847, -1.193]$	-1.328 $[-1.595, -1.064]$
Aesthetic	6.45	-0.76 $[-0.87, -0.64]$	-0.61 $[-0.77, -0.44]$	-0.08 $[-0.17, +0.01]$
CLIP score	29.41	-3.48 $[-4.43, -2.59]$	-3.59 $[-4.76, -2.49]$	-1.65 $[-2.58, -0.84]$
PickScore	22.96	-1.44 $[-1.66, -1.22]$	-2.60 $[-3.00, -2.17]$	-2.18 $[-2.51, -1.85]$
HPSv2 ($\times 100$)	29.71	-3.32 $[-3.72, -2.94]$	-4.04 $[-4.66, -3.43]$	-1.89 $[-2.34, -1.45]$
HPSv2.1 ($\times 100$)	29.98	-5.66 $[-6.35, -4.98]$	-9.75 $[-11.30, -8.17]$	-4.07 $[-5.11, -3.04]$
MUSIQ	46.1	-3.5 $[-6.7, -0.2]$	-1.8 $[-5.3, +1.9]$	-5.6 $[-9.0, -2.1]$
MUSIQ_c	62.5	-11.7 $[-13.6, -9.8]$	-4.2 $[-8.4, -0.5]$	$+2.5$ $[-0.4, +5.5]$
CLIP-IQA ($\times 100$)	71.3	-27.0 $[-30.2, -23.8]$	-10.1 $[-13.0, -7.1]$	$\mathbf{+2.9}$ $[+1.0, +4.9]$
ARNIQA ($\times 100$)	70.5	-45.1 $[-47.6, -42.5]$	-4.8 $[-6.7, -2.6]$	$\mathbf{+4.3}$ $[+2.7, +6.1]$
$\text{FID}_p \downarrow$	168.0	237.0	184.6	202.6

Table K.6: **Intervals for the band-guard ablations** (Table Q.1 and further variants), 46 ablation prompts at 4096^2, paired against SCA + band guard, whose absolute values form the first column; **blue** / *orange*: significantly better / worse than this reference. Corrected patch FID is shown without intervals. All rows use SCA without the schedule. Sources: `results/abl_*`, `route_*`.

Metric	Reference	no band	no gate	shared $\pi(\sigma)$	per-layer	$\pi = 1$
ImageReward	1.165	−0.041 [−0.109, +0.021]	−0.357 [−0.511, −0.222]	−0.032 [−0.084, +0.015]	−0.024 [−0.055, +0.004]	−0.015 [−0.075, +0.040]
Aesthetic	6.43	+0.01 [−0.05, +0.08]	−0.23 [−0.35, −0.11]	−0.06 [−0.13, +0.00]	+0.03 [−0.01, +0.06]	+0.03 [−0.02, +0.10]
CLIP score	29.01	+0.40 [−0.04, +0.92]	−1.15 [−1.82, −0.47]	−0.09 [−0.53, +0.31]	−0.03 [−0.28, +0.23]	+0.16 [−0.19, +0.53]
PickScore	22.71	**+0.25** [+0.09, +0.41]	−0.86 [−1.13, −0.59]	+0.02 [−0.10, +0.14]	+0.01 [−0.06, +0.08]	**+0.12** [+0.03, +0.22]
HPSv2 (×100)	29.18	**+0.54** [+0.31, +0.75]	−0.49 [−0.87, −0.14]	−0.20 [−0.42, +0.01]	−0.08 [−0.19, +0.04]	**+0.16** [+0.04, +0.29]
HPSv2.1 (×100)	28.81	**+1.17** [+0.71, +1.64]	−1.39 [−2.18, −0.63]	−0.33 [−0.68, +0.01]	−0.01 [−0.19, +0.19]	**+0.55** [+0.30, +0.79]
MUSIQ	43.1	**+2.9** [+0.7, +5.7]	−0.7 [−3.6, +2.4]	+0.5 [−2.5, +4.0]	−0.3 [−1.9, +1.8]	+1.0 [−0.3, +2.6]
MUSIQ_c	62.1	+0.4 [−0.9, +1.6]	−1.0 [−3.4, +1.7]	−5.7 [−7.2, −4.5]	−1.5 [−2.4, −0.7]	**+0.9** [+0.3, +1.4]
CLIP-IQA (×100)	69.6	**+1.7** [+0.3, +3.2]	−0.6 [−2.7, +1.5]	−6.5 [−8.4, −4.7]	−1.2 [−2.1, −0.5]	+0.5 [−0.9, +1.9]
ARNIQA (×100)	69.6	+0.9 [−0.6, +3.1]	+1.6 [−0.4, +4.5]	−4.6 [−7.1, −2.9]	−1.2 [−2.0, −0.6]	+0.6 [−0.1, +1.4]
$\text{FID}_p \downarrow$	166.7	+1.3	+8.9	−4.6	−1.3	+1.6

Table K.7: **Band-guard ablations, continued.** SCA without the schedule; conventions as in Table K.6.

Metric	Reference	route 0.5	route 0.3	mean pool	$\pi = 0$	v0
ImageReward	1.165	−0.051 [−0.095, −0.009]	−0.026 [−0.082, +0.024]	−0.848 [−1.083, −0.615]	−0.436 [−0.537, −0.336]	−1.369 [−1.633, −1.113]
Aesthetic	6.43	−0.02 [−0.08, +0.04]	+0.02 [−0.03, +0.07]	−0.39 [−0.54, −0.25]	−0.75 [−0.86, −0.63]	−0.07 [−0.15, +0.03]
CLIP score	29.01	−0.00 [−0.31, +0.28]	+0.13 [−0.23, +0.51]	−2.96 [−4.18, −1.67]	−3.08 [−3.98, −2.16]	−1.25 [−2.02, −0.40]
PickScore	22.71	+0.04 [−0.03, +0.12]	**+0.09** [+0.01, +0.17]	−1.80 [−2.10, −1.49]	−1.19 [−1.41, −0.96]	−1.94 [−2.23, −1.65]
HPSv2 (×100)	29.18	+0.02 [−0.11, +0.15]	+0.11 [−0.01, +0.24]	−2.35 [−2.81, −1.94]	−2.79 [−3.18, −2.45]	−1.35 [−1.74, −0.95]
HPSv2.1 (×100)	28.81	+0.18 [−0.04, +0.40]	**+0.34** [+0.12, +0.58]	−5.81 [−6.96, −4.74]	−4.48 [−5.17, −3.80]	−2.90 [−3.82, −2.01]
MUSIQ	43.1	+1.2 [−0.2, +3.0]	+0.8 [−0.4, +2.5]	−1.8 [−5.5, +1.7]	−0.6 [−4.4, +3.4]	−2.7 [−6.1, +0.7]
MUSIQ_c	62.1	+0.5 [−0.0, +1.1]	**+0.8** [+0.2, +1.4]	−0.8 [−5.0, +3.6]	−11.3 [−13.3, −9.6]	+2.9 [−0.3, +6.2]
CLIP-IQA (×100)	69.6	+0.3 [−1.3, +1.5]	+0.6 [−0.7, +1.7]	−5.4 [−9.3, −1.6]	−25.2 [−29.1, −21.4]	**+4.6** [+2.1, +7.3]
ARNIQA (×100)	69.6	+0.1 [−0.6, +0.8]	+0.6 [−0.1, +1.3]	−1.1 [−3.6, +1.9]	−44.3 [−47.4, −40.7]	**+5.1** [+2.8, +8.1]
$\text{FID}_p \downarrow$	166.7	+1.5	+1.1	+18.7	+70.3	+35.9

L ADDITIONAL PAIRED INTERVALS

This section expands Table 4 and the baseline rows of Tables 1 and 2 with per-metric intervals. Tables L.2 and L.5 cover the added baselines, Table L.3 the prompt-set, seed, and model checks, Table L.4 the calibration-budget and gate checks, Table L.6 every baseline of Table 2, and Table L.1 the 6144^2 and FLUX.1-Krea-dev runs. Figure L.1 plots four of these metrics for five of the seven baselines run on all 195 prompts (HRDiT and SigMa are in Table L.6), signed as SCA without the schedule minus baseline. Every difference in the tables is method minus the named reference. Positive values favour the method for the ten per-image metrics, and **blue** / *orange* mark intervals that exclude zero in favour of / against the method. These comparisons resample matched prompt ids, with 2000 draws and seed 0; they do not treat multiple crops as independent images. The HRGen-1K sample has 1000 prompts, the two extra seeds 195, the sensitivity experiments 46, FLUX.2-dev 50, and the main baseline comparison 195. Tables use the separately recomputed per-image CSVs; these intervals differ slightly from the original 1000-draw evaluator summaries. No multiplicity correction is applied.

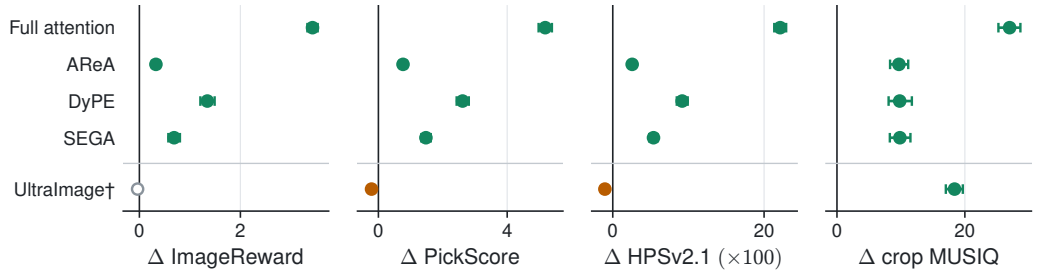

Figure L.1: **SCA beats every shared-protocol baseline on all four metrics; against UltraImage the result is mixed.** SCA minus baseline, mean over the same 195 prompts at 4096^2, whiskers 95% prompt-bootstrap intervals (2000 draws, one seed); open markers: interval contains zero; orange: significant loss. †Own two-stage recipe, not an equal-cost comparison.

6144^2 and FLUX.1-Krea-dev. Table L.1 gives every metric with its paired interval for the 100-prompt 6144^2 comparison and for FLUX.1-Krea-dev, where DyPE and SEGA also trail AReA on ImageReward. The Krea table follows the FLUX.1-dev calibration procedure with 16 prompts at 1024^2 (Appendix H).

Table L.1: **At 6144^2 SCA without the schedule beats AReA on 14 of 15 metrics; on FLUX.1-Krea-dev SCA wins six metrics and loses ImageReward and patch FID, in a third of the time.** Method minus AReA with 95% paired intervals over 100 prompts at 6144^2 and for DyPE and SEGA, which ran only those, and all 195 for SCA on FLUX.1-Krea-dev; first column FLUX.1-dev at 6144^2, others FLUX.1-Krea-dev at 4096^2 with a Krea-calibrated SCA table. Per-image intervals use 1000 prompt draws; FID, KID, and their patch versions resample prompts and reference-image clusters. Lower is better for FID, KID, their patch versions, and time. **Blue** / *orange*: significantly better / worse than AReA. Time is the campaign mean on shared GH200 nodes. Grey: published values without the schedule; their final-operator run is pending.

	6144^2	FLUX.1-Krea-dev, 4096^2		
Metric	SCA w/o sched.	SCA	DyPE	SEGA
ImageReward	**+1.262** [+1.051, +1.450]	−0.052 [−0.103, −0.001]	*−0.282* [−0.424, −0.173]	*−0.110* [−0.189, −0.030]
Aesthetic	**+0.44** [+0.31, +0.55]	**+0.07** [+0.04, +0.10]	+0.05 [−0.02, +0.12]	−0.05 [−0.11, +0.02]
CLIP score	**+3.28** [+2.32, +4.18]	−0.08 [−0.33, +0.15]	*−0.95* [−1.47, −0.49]	*−0.49* [−0.91, −0.07]
PickScore	**+1.58** [+1.31, +1.88]	−0.01 [−0.10, +0.07]	*−0.73* [−0.93, −0.55]	*−0.23* [−0.36, −0.10]
HPSv2 (×100)	**+2.53** [+2.12, +2.94]	**+0.29** [+0.17, +0.42]	*−1.02* [−1.24, −0.80]	*−0.25* [−0.43, −0.09]
HPSv2.1 (×100)	**+6.37** [+5.39, +7.38]	**+0.63** [+0.37, +0.91]	*−1.24* [−1.76, −0.76]	−0.25 [−0.61, +0.09]
MUSIQ	**+9.2** [+6.9, +11.7]	**+1.3** [+0.3, +2.3]	*−2.6* [−4.6, −0.4]	+0.7 [−1.2, +2.8]
MUSIQ$_c$	**+3.8** [+1.4, +6.2]	**+4.2** [+3.4, +5.0]	**+3.2** [+1.8, +4.6]	+1.3 [−0.1, +2.6]
CLIP-IQA (×100)	**+6.3** [+4.3, +8.4]	+0.2 [−0.6, +1.2]	+0.9 [−0.7, +2.3]	**+1.8** [+0.4, +3.2]
ARNIQA (×100)	**+14.9** [+13.0, +17.0]	**+1.4** [+1.0, +1.8]	*−1.6* [−2.6, −0.7]	−0.4 [−1.3, +0.4]
FID↓	**−46.8** [−58.6, −35.3]	+1.4 [−3.0, +7.0]	+5.0 [−2.8, +16.3]	−2.3 [−10.0, +7.4]
KID↓ (×10^3)	**−23.20** [−31.77, −16.84]	−0.12 [−2.15, +1.79]	−0.81 [−4.63, +3.26]	−0.85 [−4.00, +1.53]
FID$_p$ ↓	**−66.2** [−70.2, −48.9]	+8.1 [+5.5, +12.8]	*+7.7* [+5.0, +18.5]	+0.1 [−3.1, +6.7]
KID$_p$ ↓ (×10^3)	**−56.88** [−68.46, −47.50]	+1.79 [−0.26, +4.00]	+1.32 [−1.44, +4.94]	−0.81 [−2.48, +0.90]
Inception score	+0.45 [−0.06, +0.84]	+0.32 [−0.31, +0.77]	+0.25 [−0.22, +0.69]	+0.09 [−0.43, +0.56]
Time (s)↓	**−343.1** [−349.3, −337.4]	**−165.8** [−177.9, −154.4]	**−24.3** [−28.9, −19.6]	**−20.1** [−25.0, −15.5]

Table L.2: **DyPE and SEGA trail SCA on every per-image metric; UltraImage and HRDiT are mixed.** Baseline minus SCA at 4096^2, 195 prompts; **blue** / *orange*: baseline significantly better / worse than SCA. UltraImage and HRDiT use their own multi-stage recipes; HRDiT's set-level metrics are in Table 1.

Metric	DyPE	SEGA	UltraImage	HRDiT
ImageReward	−1.345 [−1.490, −1.207]	−0.688 [−0.803, −0.573]	+0.035 [−0.031, +0.102]	**+0.094** [+0.032, +0.157]
Aesthetic	−0.69 [−0.79, −0.60]	−0.40 [−0.48, −0.33]	**+0.08** [+0.04, +0.12]	**+0.11** [+0.07, +0.16]
CLIP score	−4.22 [−4.83, −3.63]	−2.19 [−2.62, −1.79]	−0.31 [−0.60, −0.01]	−0.02 [−0.29, +0.27]
PickScore	−2.61 [−2.81, −2.43]	−1.48 [−1.63, −1.33]	**+0.21** [+0.12, +0.31]	**+0.34** [+0.25, +0.43]
HPSv2 (×100)	−4.25 [−4.55, −3.97]	−2.64 [−2.88, −2.40]	**+0.19** [+0.03, +0.35]	+0.11 [−0.05, +0.27]
HPSv2.1 (×100)	−9.18 [−9.91, −8.46]	−5.39 [−5.91, −4.82]	**+1.04** [+0.75, +1.33]	**+1.19** [+0.92, +1.47]
MUSIQ	−8.7 [−10.8, −6.6]	−7.7 [−9.5, −5.9]	−11.4 [−13.3, −9.5]	−4.6 [−6.2, −2.9]
MUSIQ_c	−9.8 [−11.7, −8.1]	−9.8 [−11.5, −8.3]	−18.4 [−19.7, −17.0]	−6.6 [−7.9, −5.3]
CLIP-IQA (×100)	−12.8 [−14.6, −11.0]	−5.7 [−7.3, −4.2]	+0.5 [−0.6, +1.6]	**+3.5** [+2.6, +4.4]
ARNIQA (×100)	−10.5 [−11.9, −9.1]	−6.0 [−7.2, −4.8]	**+1.6** [+0.9, +2.3]	**+0.9** [+0.2, +1.5]

Table L.3: **The gains hold on HRGen-1K, two more seeds, and FLUX.2-dev.** SCA minus AReA for HRGen-1K and seeds 1 and 2; SCA without the schedule minus full attention for FLUX.2-dev. All at 4096^2; **blue**: significantly better than the reference. Grey: published values without the schedule; their final-operator run is pending.

Metric	HRGen-1K	Seed 1	Seed 2	FLUX.2-dev w/o sched.
ImageReward	**+0.818** [+0.768, +0.870]	**+0.306** [+0.242, +0.378]	**+0.264** [+0.197, +0.338]	**+0.456** [+0.320, +0.598]
Aesthetic	**+0.33** [+0.30, +0.36]	**+0.11** [+0.06, +0.16]	**+0.09** [+0.04, +0.14]	**+0.30** [+0.18, +0.42]
CLIP score	**+2.65** [+2.40, +2.89]	**+1.22** [+0.89, +1.54]	**+1.37** [+1.08, +1.66]	+0.38 [−0.38, +1.12]
PickScore	**+1.19** [+1.13, +1.24]	**+0.77** [+0.67, +0.88]	**+0.74** [+0.62, +0.86]	**+0.56** [+0.29, +0.86]
HPSv2 (×100)	**+1.95** [+1.86, +2.05]	**+1.17** [+1.01, +1.33]	**+1.18** [+0.99, +1.38]	**+0.85** [+0.50, +1.21]
HPSv2.1 (×100)	**+4.93** [+4.70, +5.17]	**+2.49** [+2.16, +2.83]	**+2.53** [+2.13, +2.90]	**+1.84** [+1.10, +2.55]
MUSIQ	**+7.7** [+6.8, +8.5]	**+5.5** [+3.8, +7.2]	**+6.3** [+4.4, +8.1]	**+16.0** [+11.9, +19.9]
MUSIQ_c	**+13.9** [+13.2, +14.6]	**+10.3** [+8.9, +11.6]	**+10.9** [+9.3, +12.6]	**+21.4** [+18.2, +24.8]
CLIP-IQA (×100)	**+9.0** [+8.3, +9.8]	**+4.6** [+3.2, +6.1]	**+4.2** [+3.0, +5.4]	**+6.6** [+4.7, +8.6]
ARNIQA (×100)	**+7.2** [+6.7, +7.7]	**+3.4** [+2.5, +4.3]	**+3.5** [+2.6, +4.4]	**+7.1** [+5.4, +8.8]

Table L.4: **The calibration budget barely matters; a gate at $\sigma_g = 0.85$ lowers seven metrics.** Variant minus default SCA, both without the schedule, 46 ablation prompts; **blue** / *orange*: variant significantly better / worse than default SCA. Budgets use the first 4, 8, or 16 of the 32 calibration prompts; the default gate is $\sigma_g = 0.89$.

Metric	4 prompts	8 prompts	16 prompts	$\sigma_g = 0.85$	$\sigma_g = 0.93$
ImageReward	+0.012 [−0.007, +0.029]	+0.010 [−0.004, +0.024]	−0.007 [−0.020, +0.005]	−0.015 [−0.098, +0.064]	−0.004 [−0.097, +0.069]
Aesthetic	+0.01 [−0.00, +0.02]	+0.01 [−0.00, +0.02]	+0.00 [−0.01, +0.01]	−0.19 [−0.26, −0.12]	+0.00 [−0.05, +0.06]
CLIP score	+0.08 [−0.03, +0.19]	+0.05 [−0.04, +0.15]	+0.04 [−0.04, +0.13]	−0.27 [−0.82, +0.26]	−0.64 [−1.03, −0.29]
PickScore	−0.01 [−0.03, +0.01]	+0.00 [−0.01, +0.02]	+0.01 [−0.01, +0.03]	−0.12 [−0.27, +0.02]	−0.22 [−0.38, −0.07]
HPSv2 (×100)	+0.02 [−0.02, +0.05]	+0.02 [−0.00, +0.05]	+0.00 [−0.02, +0.03]	−0.48 [−0.69, −0.28]	+0.02 [−0.16, +0.20]
HPSv2.1 (×100)	−0.02 [−0.07, +0.04]	+0.01 [−0.03, +0.05]	−0.01 [−0.05, +0.04]	−0.85 [−1.22, −0.48]	−0.15 [−0.56, +0.24]
MUSIQ	−0.1 [−0.4, +0.3]	−0.1 [−0.5, +0.2]	**+0.1** [+0.0, +0.3]	−1.9 [−3.5, −0.2]	+0.4 [−1.0, +2.0]
MUSIQ_c	+0.1 [−0.1, +0.3]	−0.0 [−0.2, +0.1]	−0.1 [−0.2, +0.0]	−2.0 [−3.0, −1.1]	+0.6 [−0.4, +1.9]
CLIP-IQA (×100)	+0.2 [−0.1, +0.4]	+0.2 [−0.1, +0.4]	−0.1 [−0.4, +0.1]	−3.2 [−4.8, −1.7]	+0.0 [−1.2, +1.1]
ARNIQA (×100)	+0.1 [−0.1, +0.3]	+0.0 [−0.1, +0.2]	−0.0 [−0.1, +0.1]	−4.0 [−5.9, −2.3]	**+1.8** [+0.7, +3.1]

Table L.5: **All three added baselines have significantly higher FID, patch FID, and patch KID than SCA.** Method minus SCA without the schedule on 195 prompts (negative favours the method); *orange*: significantly worse than SCA. Patch estimates use corrected crops; all four intervals use 1000 draws that resample generated prompts and reference-image clusters.

Metric	DyPE	SEGA	UltraImage
FID↓	+67.4 [+56.2, +80.3]	+26.9 [+21.1, +39.1]	+4.0 [+1.1, +10.2]
KID↓ ($\times 10^3$)	+40.39 [+30.63, +52.87]	+4.70 [+0.14, +10.48]	+0.19 [−1.83, +2.18]
FID_p ↓	+72.8 [+58.7, +81.6]	+26.7 [+18.8, +35.3]	+10.4 [+6.2, +13.9]
KID_p ↓ ($\times 10^3$)	+49.34 [+40.09, +59.95]	+13.97 [+9.33, +20.00]	+7.31 [+5.10, +9.47]

195-prompt panel. Table L.6 pairs each baseline of Table 2 with SCA on the same 195 prompts and seeds. Per-image intervals use 2000 prompt draws with seed 0; FID, KID, and their patch versions use 1000 draws that resample generated prompts and reference-image clusters. Table M.1 gives the matching intervals for the 56-step and guidance-7 SCA runs.

Table L.6: **On the 195-prompt panel, every shared-protocol baseline trails SCA on all ten per-image metrics, while HRDiT and UltraImage lead on several preference metrics.** Baseline minus SCA (28 steps) at 4096^2 with 95% paired intervals; HPSv2, HPSv2.1, CLIP-IQA, and ARNIQA are $\times 100$, KID and KID_p $\times 10^3$. For FID, KID, their patch versions, and campaign time (seconds per image), lower is better, so negative favours the baseline. **Blue** / *orange*: baseline significantly better / worse than SCA. SigMa uses its dev-matched profile; UltraImage and HRDiT use their own recipes (Appendix I).

	Single pass, shared sampler				
Metric	Full attn.	DyPE	SigMa	SEGA	AReA
ImageReward	−3.428 [−3.530, −3.327]	−1.345 [−1.490, −1.207]	−1.771 [−1.920, −1.624]	−0.688 [−0.803, −0.573]	−0.327 [−0.405, −0.250]
Aesthetic	−1.92 [−2.00, −1.83]	−0.69 [−0.79, −0.60]	−0.82 [−0.92, −0.72]	−0.40 [−0.48, −0.33]	−0.15 [−0.20, −0.09]
CLIP score	−13.77 [−14.54, −13.01]	−4.22 [−4.83, −3.63]	−5.64 [−6.28, −4.98]	−2.19 [−2.62, −1.79]	−1.45 [−1.80, −1.12]
PickScore	−5.18 [−5.39, −4.98]	−2.61 [−2.81, −2.43]	−3.11 [−3.32, −2.90]	−1.48 [−1.63, −1.33]	−0.76 [−0.88, −0.66]
HPSv2	−9.02 [−9.34, −8.69]	−4.25 [−4.55, −3.97]	−4.99 [−5.31, −4.71]	−2.64 [−2.88, −2.40]	−1.15 [−1.34, −0.96]
HPSv2.1	−22.13 [−22.92, −21.35]	−9.18 [−9.91, −8.46]	−11.08 [−11.83, −10.35]	−5.39 [−5.91, −4.82]	−2.56 [−2.94, −2.18]
MUSIQ	−22.1 [−23.7, −20.5]	−8.7 [−10.8, −6.6]	−9.5 [−11.5, −7.4]	−7.7 [−9.5, −5.9]	−5.6 [−7.4, −3.9]
MUSIQ_c	−27.0 [−28.7, −25.2]	−9.8 [−11.7, −8.1]	−10.2 [−12.1, −8.5]	−9.8 [−11.5, −8.3]	−9.7 [−11.1, −8.3]
CLIP-IQA	−23.3 [−24.8, −21.9]	−12.8 [−14.6, −11.0]	−14.3 [−16.3, −12.4]	−5.7 [−7.3, −4.2]	−4.6 [−6.1, −3.2]
ARNIQA	−16.3 [−17.7, −14.9]	−10.5 [−11.9, −9.1]	−10.9 [−12.3, −9.5]	−6.0 [−7.2, −4.8]	−3.4 [−4.3, −2.5]
Baseline minus SCA w/o schedule					
FID↓	+174.4 [+150.2, +179.4]	+67.4 [+56.2, +80.3]	+78.9 [+66.7, +91.0]	+26.9 [+21.1, +39.1]	+7.7 [+2.6, +13.1]
KID↓	+158.0 [+140.3, +176.0]	+40.4 [+30.6, +52.9]	+50.4 [+39.1, +64.0]	+4.7 [+0.1, +10.5]	+1.0 [−1.5, +3.2]
FID_p ↓	+153.1 [+122.3, +149.6]	+72.8 [+58.7, +81.6]	+85.5 [+70.2, +93.7]	+26.7 [+18.8, +35.3]	+3.8 [−1.3, +7.5]
KID_p ↓	+161.9 [+146.6, +178.2]	+49.3 [+40.1, +60.0]	+62.4 [+52.0, +74.7]	+14.0 [+9.3, +20.0]	+2.3 [−0.1, +4.6]
Time (s)↓	+124.7 [+120.8, +129.4]	+124.4 [+120.5, +128.6]	+121.0 [+119.0, +123.0]	+118.9 [+118.0, +119.7]	+192.9 [+180.3, +206.4]

	Multi-stage, own recipe	
Metric	UltraImage	HRDiT
ImageReward	+0.035 [−0.031, +0.102]	**+0.094** [+0.032, +0.157]
Aesthetic	**+0.08** [+0.04, +0.12]	**+0.11** [+0.07, +0.16]
CLIP score	−0.31 [−0.60, −0.01]	−0.02 [−0.29, +0.27]
PickScore	**+0.21** [+0.12, +0.31]	**+0.34** [+0.25, +0.43]
HPSv2	**+0.19** [+0.03, +0.35]	+0.11 [−0.05, +0.27]
HPSv2.1	**+1.04** [+0.75, +1.33]	**+1.19** [+0.92, +1.47]
MUSIQ	−11.4 [−13.3, −9.5]	−4.6 [−6.2, −2.9]
MUSIQ_c	−18.4 [−19.7, −17.0]	−6.6 [−7.9, −5.3]
CLIP-IQA	+0.5 [−0.6, +1.6]	**+3.5** [+2.6, +4.4]
ARNIQA	**+1.6** [+0.9, +2.3]	**+0.9** [+0.2, +1.5]
Baseline minus SCA w/o schedule		
FID↓	+4.0 [+1.1, +10.2]	+0.4 [−2.3, +6.4]
KID↓	+0.2 [−1.8, +2.2]	−1.6 [−3.6, +0.3]
FID_p ↓	+10.4 [+6.2, +13.9]	+3.9 [+0.7, +8.0]
KID_p ↓	+7.3 [+5.1, +9.5]	+2.8 [+0.9, +4.8]
Time (s)↓	+237.0 [+233.5, +241.7]	+38.8 [+37.2, +40.5]

M SAMPLING-BUDGET, GUIDANCE, AND SCHEDULE CHECKS

This section expands the budget and schedule rows of Table 2: Figure M.1 plots quality against time, Table M.1 gives the budget and guidance checks, Table M.2 the progressive schedule against HRDiT, and Figure M.2 two random prompts under each schedule.

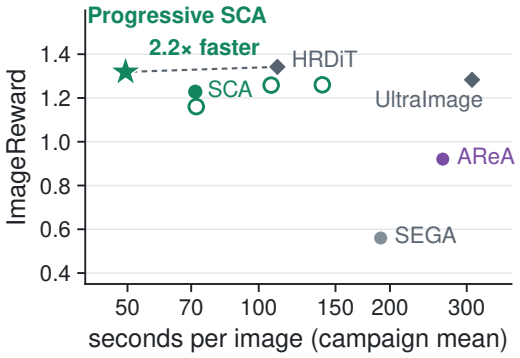

Figure M.1: **Quality against time.** 195 prompts, 4096^2. Open circles: SCA at guidance 7, 42 and 56 steps. Every SCA point, progressive included, is without the schedule.

On the same 195 evaluation prompts, we compare the default 28-step run of SCA without the schedule with 56 steps at guidance 4.5 and with 28 steps at guidance 7 (Table M.1). The 56-step run keeps the noise threshold at 0.89; it therefore changes the discretization of both the opening and mixed phases. It improves ImageReward by 0.032 $[0.0002, 0.062]$, HPSv2.1 by 0.88, PickScore by 0.13, and MUSIQ_c by 2.90, but increases corrected patch FID by 3.85 $[1.51, 6.33]$ with reference-image cluster resampling. Increasing guidance to 7 alone lowers seven per-image metrics, ImageReward by 0.068 $[0.025, 0.115]$ among them, and raises only MUSIQ. Thus the default schedule does not maximize every metric, and extra sampling can change the preference-quality balance.

Against UltraImage on these 195 prompts, 56-step SCA has ImageReward difference -0.023 $[-0.089, 0.040]$, HPSv2.1 difference -0.22 $[-0.51, 0.05]$, and PickScore difference -0.11 $[-0.21, -0.01]$. The first two intervals span zero and do not establish equivalence; PickScore favours UltraImage. Its MUSIQ advantage is 11.61 $[9.54, 13.60]$ and its MUSIQ_c advantage 22.42 $[21.20, 23.59]$. Campaign means are 71.5 s for 28-step SCA without the schedule, 139.9 s for 56-step SCA, and 308.5 s for UltraImage on these 195 prompts. This is a sampling-budget sensitivity check, not an equal-FLOP or equal-wall-clock experiment: UltraImage has two resolutions, different guidance, and per-prompt preparation. Against HRDiT on the same prompts (Table 2), the 56-step run stays lower on ImageReward (1.26 against 1.34), HPSv2.1 (30.55 against 30.92), and PickScore (23.22 against 23.46) while taking longer (139.9 against 110.4 s); these are point estimates without paired intervals. Table M.2 adds a 42-step run sized to HRDiT's campaign time (106.8 against 110.4 s) and progressive SCA (Section 2.6); the 42-step run is without the schedule, progressive SCA uses it. The extra steps raise PickScore, HPSv2, HPSv2.1, ARNIQA, and MUSIQ_c over 28 steps but leave ImageReward unresolved, and HRDiT stays significantly ahead of the 42-step run on ImageReward, PickScore, HPSv2.1, aesthetic score, and CLIP-IQA while trailing it on both MUSIQ scores and ARNIQA. Progressive SCA is significantly worse than HRDiT only on CLIP-IQA and better on HPSv2, MUSIQ, and ARNIQA; against single-pass SCA it gains ImageReward, PickScore, aesthetic score, both HPS scores, CLIP-IQA, and ARNIQA, and loses both MUSIQ scores. Figure M.1 places every setting of the panel by ImageReward and time, and Figure M.2 shows HRDiT, progressive SCA, and single-pass SCA on two random prompts, both SCA variants with the schedule.

Distribution metrics of the progressive and 42-step runs. All runs here are without the schedule, with 195 generated images, 168 reference identities, and 1000 bootstrap draws (`results/a4k195_panel/corrected_fidp_ci/paired_deltas.csv`). Against single-pass SCA at 28 steps, progressive SCA has significantly higher patch FID ($+7.2$ $[3.2, 11.0]$) and patch KID ($+0.005$ $[0.003, 0.007]$), while its FID ($+3.0$ $[-0.2, 8.3]$) and KID differences span zero. For 42-step SCA only FID is significantly higher ($+2.4$ $[0.1, 5.7]$); KID, patch FID ($+1.5$ $[-0.5, 3.6]$), and patch KID span zero.

Table M.1: **Doubling the steps raises eight metrics; guidance 7 lowers seven and raises MUSIQ.** On 195 prompts, all with SCA without the schedule, the first two columns compare with default SCA; the last compares 56-step SCA with UltraImage. Per-image differences (first-named setting minus second) and 95% paired intervals, 2000 draws; **blue** / *orange*: the interval excludes zero in favour of / against the first-named setting.

Metric	56 vs. 28 steps	Guidance 7 vs. 4.5	56 steps vs. UltraImage
ImageReward	**+0.032** [+0.000, +0.062]	−0.068 [−0.115, −0.025]	−0.023 [−0.089, +0.040]
Aesthetic	**+0.03** [+0.01, +0.06]	−0.10 [−0.14, −0.07]	−0.03 [−0.07, +0.02]
CLIP score	−0.01 [−0.23, +0.22]	−0.21 [−0.45, +0.03]	+0.30 [−0.01, +0.62]
PickScore	**+0.13** [+0.07, +0.18]	−0.27 [−0.34, −0.20]	−0.11 [−0.21, −0.01]
HPSv2 (×100)	**+0.32** [+0.23, +0.40]	−0.25 [−0.35, −0.16]	+0.11 [−0.03, +0.26]
HPSv2.1 (×100)	**+0.88** [+0.71, +1.04]	−0.60 [−0.80, −0.41]	−0.22 [−0.51, +0.05]
MUSIQ	−0.2 [−1.5, +1.0]	**+1.4** [+0.3, +2.6]	**+11.6** [+9.5, +13.6]
MUSIQ_c	**+2.9** [+2.2, +3.6]	+0.2 [−0.4, +0.8]	**+22.4** [+21.2, +23.6]
CLIP-IQA (×100)	**+0.9** [+0.4, +1.4]	−1.0 [−1.7, −0.4]	**+1.1** [+0.0, +2.1]
ARNIQA (×100)	**+1.5** [+1.0, +1.9]	−1.2 [−1.5, −0.8]	+0.4 [−0.4, +1.2]

Table M.2: **Against HRDiT, progressive SCA is significantly worse only on CLIP-IQA.** 4096^2, 195 prompts, seed label 0. Per-image differences (first-named setting minus second) with 95% paired intervals over 2000 prompt draws; **blue** / *orange*: the interval excludes zero in favour of / against the first-named setting. Progressive: HRDiT's schedule with SCA attention (Section 2.6); 42 steps: single-pass SCA without the schedule, sized to HRDiT's campaign time. Intervals that include zero do not establish equivalence.

			SCA w/o schedule	
Metric	Progressive vs. HRDiT	Progressive vs. SCA	42 steps vs. SCA	HRDiT vs. 42 steps
ImageReward	−0.022 [−0.088, +0.034]	**+0.072** [+0.010, +0.134]	+0.030 [−0.001, +0.059]	**+0.083** [+0.023, +0.150]
Aesthetic	−0.027 [−0.070, +0.017]	**+0.087** [+0.038, +0.130]	+0.020 [−0.007, +0.047]	**+0.071** [+0.022, +0.119]
CLIP score	+0.27 [−0.01, +0.54]	+0.25 [−0.02, +0.52]	−0.01 [−0.19, +0.20]	−0.02 [−0.31, +0.27]
PickScore	−0.05 [−0.14, +0.04]	**+0.29** [+0.20, +0.38]	**+0.08** [+0.03, +0.13]	**+0.29** [+0.19, +0.39]
HPSv2 (×100)	**+0.38** [+0.24, +0.51]	**+0.48** [+0.32, +0.64]	**+0.23** [+0.15, +0.31]	−0.10 [−0.26, +0.07]
HPSv2.1 (×100)	+0.10 [−0.14, +0.35]	**+1.29** [+0.99, +1.59]	**+0.60** [+0.44, +0.75]	**+0.65** [+0.38, +0.93]
MUSIQ	**+1.3** [+0.2, +2.4]	−3.4 [−5.1, −1.6]	−0.4 [−1.5, +0.7]	−4.7 [−6.3, −2.9]
MUSIQ_c	+0.6 [−0.7, +1.8]	−6.0 [−7.3, −4.6]	**+1.9** [+1.3, +2.5]	−9.6 [−10.9, −8.4]
CLIP-IQA (×100)	−0.9 [−1.8, −0.1]	**+2.6** [+1.6, +3.5]	+0.4 [−0.1, +0.8]	**+2.4** [+1.5, +3.3]
ARNIQA (×100)	**+2.9** [+2.3, +3.4]	**+3.7** [+3.2, +4.3]	**+1.1** [+0.7, +1.4]	−0.7 [−1.4, −0.1]

Figure M.2: **Progressive SCA keeps one large subject, as HRDiT does; single-pass SCA can shrink or repeat it.** Two prompts drawn at random from the original 46-prompt panel; same 1:1 512 × 512 crop per row; times in Table 2.

N EXPLORATORY 8192^2 COMPARISON

Table N.1 reports all 12 completed prompt pairs at 8192^2 under the shared FLUX.1-dev protocol. SCA has higher ImageReward, CLIP, PickScore, both HPS variants, and ARNIQA, with paired intervals excluding zero. Aesthetic score and the three remaining quality metrics have intervals spanning zero. In particular, MUSIQ, MUSIQ_c, and CLIP-IQA have point estimates favouring AReA. These mixed quality scores and the grid and stripe failures in Figure T.11 limit the evidence for reliable 8K generation. With only 12 prompts, this comparison is exploratory; we do not report distributional metrics for this set.

All 8K scores were computed on CPU. HPS uses float32 here, whereas the main GPU evaluation uses mixed precision, so absolute scores across resolutions should not be interpreted as a controlled scaling comparison. Both methods within this table use the same evaluator settings. For native-resolution MUSIQ, attention queries are processed in chunks of 256 while retaining every key and native image patch. This bounds intermediate memory without changing the attention context; pre-trained equivalence checks at three smaller image sizes reproduced the unchunked scores exactly.

Table N.1: **Exploratory 8192^2 comparison: SCA is significantly better than AReA on six of ten metrics.** FLUX.1-dev, 12 matched prompts, seed 0. All metrics are higher-is-better. Intervals are paired prompt-bootstrap 95% percentile intervals (2000 draws, bootstrap seed 0); **blue**: significantly better than AReA.

Metric	AReA	SCA	SCA − AReA [95% CI]
ImageReward	−0.376	0.675	**+1.051** [+0.386, +1.754]
Aesthetic	5.64	5.72	+0.08 [−0.26, +0.44]
CLIP score	22.38	25.52	**+3.14** [+0.63, +6.11]
PickScore	20.35	21.27	**+0.92** [+0.24, +1.75]
HPSv2 (×100)	24.60	26.13	**+1.53** [+0.56, +2.66]
HPSv2.1 (×100)	16.59	20.51	**+3.92** [+1.31, +6.85]
MUSIQ	28.6	25.7	−2.9 [−6.8, +0.8]
MUSIQ_c	44.0	40.1	−3.8 [−10.4, +1.6]
CLIP-IQA (×100)	57.1	52.0	−5.1 [−10.5, +0.6]
ARNIQA (×100)	37.3	54.6	**+17.3** [+11.9, +22.6]

O CALIBRATION TABLE AT NON-SQUARE SIZES

At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and SCA uses the table of the emulated scale nearest the larger factor, scale 4. On the 46 ablation prompts at 2048×4096, the scale-2 table changes the paired metrics negligibly (Table K.4): HPSv2 +0.09, HPSv2.1 +0.17 (both ×100), and PickScore +0.05 are significant but small, and ImageReward ($+0.01$ $[-0.03, 0.05]$) and the other metrics are within noise. These small changes do not resolve the mixed SCA–AReA results at this size, and we keep the default rule.

P LATE DUPLICATION: DETECTOR AND VARIANTS

This section expands Sections 2.4 and 4.2 and Figure 4: how duplicates are counted, how well the count agrees with visual review, and every variant of the schedule we ran on the 46-prompt subset.

Detector. Grounding DINO (Liu et al., 2024) (the released tiny checkpoint) runs on each image at two scales, longest side 800 and 1333 pixels, with all subjects of the prompt joined into one text query; a box counts when its score is at least 0.35, the detector's default, fixed before any image was scored. Boxes from both scales are merged by non-maximum suppression at IoU 0.5 and by containment (at least 90% of a box inside another of at least 30% of its area), and boxes covering over 90% of the image are dropped. Subject lists and implied counts were written from the prompt text alone, before any image was scored. A prompt is eligible when it implies a count for a person or an animal: 95 of the 195 Aesthetic-4K prompts and 26 of the 46-prompt subset. An image is duplicated when any such count exceeds the implied count. Rates carry Wilson 95% intervals, and paired comparisons use an exact McNemar test.

Validation. We compared the detector with visual labels of 30 images of SCA without the schedule. On the 20 images whose prompt names a person or an animal, it agrees with the original labels on 18; inspecting the two disagreements at full resolution showed small copies the first review had missed (three climbers along the top edge of the ice-climber image, a second portrait of the man in the artist's corner), so it agrees with the reviewed labels on all 20. Over any counted subject it agrees on 26 of 30, with two false positives (a brimmed hat read as a second umbrella, a two-channel waterfall) and two misses. Thresholds from 0.3 to 0.4 give the same rates.

Variants. Table P.1 lists every variant we ran on the 46 prompts, with its duplication count, time per image, and the per-image metrics on which it is significantly worse than SCA without the schedule. Losses are split between prompts whose duplicates the variant removes and prompts clean under both operators; a loss on the clean prompts is real softening. Every variant that alters the whole trajectory (a held low band, wide windows with the scale-4 weight, a larger bridge) softens clean prompts. The bridge alone is nearly free but leaves 5 of 26 prompts duplicated. The schedule with ratio-matched weights is the best trade-off we found. It leaves 4 of the 26 subset prompts duplicated (15%) but 25 of all 95 eligible prompts (26%), so the subset understated the full-set rate.

Examples. The pairs in Figure 4c follow a fixed rule. The examples in panel (c) are not curated: starting from the ice-climber prompt (Aesthetic-4K id `997293625843621481`), we took the next two prompts in id order that the detector flags for SCA without the schedule (ids `minus1071233499893339092` and `minus244748176062666l988`). Each box is the 512×512 pixel square holding the most extra detections, and the rule was fixed before any image of the final operator was viewed.

Table P.1: **Every schedule variant on the 46-prompt subset.** Each row is one run at 4096^2, seed 0: Dup. is the number of the 26 people and animal prompts with a duplicated subject (open-vocabulary detector), s/img is the median time per image over the 50 generated images, Worse lists the metrics (of 12) whose paired 95% interval against published SCA lies on the losing side, and Softens repeats that test for the ten per-image metrics on prompts with no duplicated subject in either run. Key result: the schedule cuts duplication from 15/26 to 4/26 at 9% more time per image and loses only MUSIQ_c and ARNIQA; no run reaches AReA's 1/26 without softening clean prompts.

Variant	Dup. ↓	s/img ↓	Worse than published SCA	Softens clean prompts
Without the schedule (published)	15/26	70.8	reference	reference
AReA	1/26	n/a	all 10 per-image	n/a
Global bridge (8×8 cells; 16 keys per cell unless noted)				
Bridge, 16 keys per cell	5/26	73.6	MUSIQ_c	MUSIQ_c
Bridge, 32 keys per cell	5/26	75.9	MUSIQ_c	none
Bridge, 64 keys while $\sigma > 0.6$	4/26	79.1	MUSIQ_c	IR, HPSv2, ARNIQA
Bridge only while $\sigma > 0.5$	5/26	72.7	MUSIQ_c	MUSIQ_c
Softer opening after the gate				
Linear π ramp to $\sigma = 0.7$	5/26	70.8	Aes., Pick, HPSv2, HPSv2.1, MUSIQ, MUSIQ_c, CLIP-IQA, ARNIQA, KID_p	IR, Aes., CLIP, Pick, HPSv2, HPSv2.1, MUSIQ, MUSIQ_c, CLIP-IQA, ARNIQA
Linear π ramp, bridge	3/26	73.6	Aes., CLIP, Pick, HPSv2, HPSv2.1, MUSIQ, MUSIQ_c, CLIP-IQA, ARNIQA, KID_p	Aes., CLIP, Pick, HPSv2, HPSv2.1, MUSIQ, MUSIQ_c, CLIP-IQA, ARNIQA
Per-head opening by $\sigma = 0.4$	5/26	70.8	Aes., HPSv2, HPSv2.1, MUSIQ, CLIP-IQA	Pick, HPSv2, HPSv2.1
Per-head opening by 0.5, bridge	4/26	73.6	Aes., HPSv2, HPSv2.1, MUSIQ_c, CLIP-IQA	Aes., CLIP, Pick, HPSv2, HPSv2.1, ARNIQA
Low-band anchor from the opening				
Anchor, linear fade to $\sigma = 0.3$	10/26	71.0	Pick, HPSv2, HPSv2.1, MUSIQ, MUSIQ_c, CLIP-IQA, ARNIQA, KID_p	HPSv2, HPSv2.1, MUSIQ, ARNIQA
Anchor, cosine fade to $\sigma = 0.3$	9/26	71.5	Pick, HPSv2, HPSv2.1, MUSIQ, MUSIQ_c, CLIP-IQA, ARNIQA, KID_p	HPSv2, HPSv2.1, ARNIQA
Linear anchor, refreshed every 4 steps	10/26	79.3	HPSv2, MUSIQ	HPSv2, MUSIQ
Linear anchor, bridge	2/26	73.6	Pick, HPSv2, HPSv2.1, MUSIQ_c, CLIP-IQA, ARNIQA, KID_p	HPSv2, HPSv2.1, MUSIQ_c, ARNIQA
Linear anchor, bridge, 128 windows	3/26	80.6	Pick, HPSv2, HPSv2.1, MUSIQ_c, CLIP-IQA, ARNIQA, KID_p	IR, Pick, HPSv2, HPSv2.1, MUSIQ_c, CLIP-IQA, ARNIQA
Wider windows after the gate, all with the bridge				
96 windows ($\sigma > 0.8$), scale-4 π	4/26	76.7	MUSIQ_c, ARNIQA	IR, HPSv2, ARNIQA
128 windows ($\sigma > 0.8$), scale-4 π	1/26	83.0	HPSv2.1, MUSIQ_c, ARNIQA	IR, CLIP, HPSv2, HPSv2.1, MUSIQ_c, ARNIQA
128 windows ($\sigma > 0.85$), scale-4 π	3/26	77.9	MUSIQ_c, ARNIQA	IR, ARNIQA
128 windows ($\sigma > 0.8$), matched π	3/26	80.9	MUSIQ_c, ARNIQA	IR, HPSv2, ARNIQA
128 windows ($\sigma > 0.85$), matched π	5/26	77.8	MUSIQ_c, ARNIQA	IR, ARNIQA
128 windows for heads $\pi < 0.5$	5/26	75.1	MUSIQ_c	none
128 windows for heads $\pi < 0.7$	4/26	77.1	MUSIQ_c	HPSv2, MUSIQ, MUSIQ_c, ARNIQA
Stride-2 dilated windows ($\sigma > 0.8$)	6/26	73.9	HPSv2, MUSIQ, MUSIQ_c, CLIP-IQA, ARNIQA	Aes., CLIP, HPSv2, MUSIQ, MUSIQ_c, CLIP-IQA
Shrinking window schedule with matched π and the bridge				
SCA (ours), windows 128/112/96/80/64	4/26	77.1	MUSIQ_c, ARNIQA	IR, ARNIQA
Windows 160/112/96/80/64	4/26	77.9	MUSIQ_c, ARNIQA	IR, HPSv2, ARNIQA
Windows 128/112/96/80/64, 32 bridge keys	4/26	79.0	MUSIQ_c, CLIP-IQA, ARNIQA	IR, HPSv2, ARNIQA
Windows 128/120/112/96/80/64	3/26	77.8	MUSIQ_c, ARNIQA	IR, HPSv2, ARNIQA

Q THE LOW-BAND DUPLICATION GUARD

The band guard was our first fix for late duplication. It acts on the operator without the schedule and is not part of SCA; the window schedule of Section 2.3 replaced it (Appendix P).

Construction. Let LP_p average-pool a head's image output over $p \times p$ tokens and upsample it bilinearly, and $\mathrm{HP}_p = \mathrm{I} - \mathrm{LP}_p$. The guard replaces the low band of the mix M by that of the global branch:

$$O \;=\; \mathrm{LP}_p\big(O^{\mathrm{glob}}\big) + \mathrm{HP}_p\big(M\big) \;=\; O^{\mathrm{glob}} + \pi_h(\sigma)\,\mathrm{HP}_p\big(O^{\mathrm{loc}} - O^{\mathrm{glob}}\big). \tag{8}$$

The local branch then contributes only its difference from the global branch above the cutoff. The cutoff should lie below $1/(2s)$, where the global branch aliases, and above $1/w$, below which the window cannot resolve; with $w = 64$ and $s = 4$ this leaves periods of 8 to 64 tokens. We used $p = 16$; $p = 8$ and $p = 32$ were worse on four development prompts, and a noise-dependent cutoff behaved like $p = 16$. FreeScale (Qiu et al., 2024) also fuses two self-attention outputs by frequency band, in the opposite direction: it keeps the blurred low band of attention over overlapping crops and takes the high band from full self-attention at the target size. ResDiT also combines global low frequencies and local high frequencies, using patch-wise FFT masks (Ma et al., 2025). Our optional guard uses pooled global attention and preserves the fitted per-head weight on the local-minus-global high-band residual. Its global-low/local-high construction is therefore not itself novel.

Table Q.1: **Ablations around the band-guard configuration at 4096^2**, 46 ablation prompts. The reference row gives absolute values for SCA + band guard (band 16), without the schedule like every row; every other row changes one thing, keeps the guard unless stated, and reports its paired difference to the reference. Marks and units as in Table 3; corrected patch FID is shown without intervals.

Variant	Mix err.	Image Reward	Aesth.	HPSv2	HPSv2.1	Pick Score	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ_c	$\mathrm{FID}_p \downarrow$
SCA + band guard (reference)	0.200	1.17	6.43	29.18	28.81	22.71	29.01	69.6	69.6	43.1	62.1	166.7
no band (**= SCA w/o sched.**)	0.200	−0.04	+0.01	**+0.54**	**+1.17**	**+0.25**	+0.40	**+1.7**	+0.9	**+2.9**	+0.4	+1.3
no gate	0.200	−0.36	−0.23	−0.49	−1.39	−0.86	−1.15	−0.6	+1.6	−0.7	−1.0	+8.9
one shared $\pi(\sigma)$	0.230	−0.03	−0.06	−0.20	−0.33	+0.02	−0.09	−6.5	−4.6	+0.5	−5.7	−4.6
per-layer $\pi_\ell(\sigma)$	0.223	−0.02	+0.03	−0.08	−0.01	+0.01	−0.03	−1.2	−1.2	−0.3	−1.5	−1.3
local only ($\pi = 1$)	0.261	−0.01	+0.03	**+0.16**	**+0.55**	**+0.12**	+0.16	+0.5	+0.6	+1.0	**+0.9**	+1.6
hard routing, threshold 0.5	n/a	−0.05	−0.02	+0.02	+0.18	+0.04	−0.00	+0.3	+0.1	+1.2	+0.5	+1.5
hard routing, threshold 0.3	n/a	−0.03	+0.02	+0.11	**+0.34**	**+0.09**	+0.13	+0.6	+0.6	+0.8	**+0.8**	+1.1
mean pooling	n/a	−0.85	−0.39	−2.35	−5.81	−1.80	−2.96	−5.4	−1.1	−1.8	−0.8	+18.7

History: what it fixes and what it costs. On the development prompts, the band alone (no opening phase) fixed the boats but left extra statues and faces on the Moai and woman-and-bear prompts; the opening phase alone got the main subject right, but small extra figures appeared late on some prompts; both together gave one subject on the boats, Moai, and woman-and-bear prompts. This was our first full configuration. On the 46 ablation prompts, however, removing the guard improves HPSv2 (+0.54), HPSv2.1 (+1.17), PickScore (+0.25), CLIP-IQA (+1.7), and MUSIQ (+2.9), all significant, leaves the other metrics within noise, and saves 17 s per image on an idle GPU. The guard also hid what the per-head mix contributes. With it (Table Q.1), local only matched the per-head mix within 2% on preference metrics, because the guard replaces the low band of every head's output with that of the global branch whatever the head's weight, so it injects global low frequencies even at $\pi = 1$; without the guard local only loses on nearly every metric (Section 4.4). Likewise, with the guard one shared weight and per-layer weights matched the per-head table on preference metrics and lost only on no-reference quality, and hard per-head routing (10 to 17% of heads global at threshold 0.5, about 6% at 0.3) stayed within 2% of the mix. The guard is an option for prompts where late duplication is visible, not part of the default method.

R COMPUTE AND REPRODUCIBILITY

Compute. Generation and calibration used single NVIDIA GH200 GPUs on a shared cluster; the verified timing logs identify the GH200 120 GB model. Some metric evaluations ran on CPUs. The surviving logs do not establish the complete calibration runtime. Generation at 4096^2 takes about 72 s per image for SCA without the schedule, 264 s for AReA, and 196 s for full attention

on shared nodes (campaign means, Table 1); the main comparisons, ablations, and non-square sizes comprise more than 2000 generated images. Surviving worker logs establish a lower bound of 123.3 GH200 worker-hours through 2026-10-17 06:59 UTC, including model loading, calibration, evaluation, and failed attempts. The bound is the union of 581 closed task intervals per physical node, with timestamp precision treated conservatively; it measures occupied worker time, not GPU kernel activity. Missing historical logs, unmatched starts, and ongoing work prevent a complete campaign total. Overlapping scheduler allocation time and image-sidecar durations are not added to this bound.

Reproducibility. The code (attention processor, calibration, sampling, evaluation wrappers, AReA wrapper, and tests), the calibration table `k1_1024.npz`, the per-image metric files, and the per-comparison summary and paired-delta CSV files behind every table will be released. Main-text metric values and paired intervals are derived from those CSV files. Each SCA generation task ran from a frozen copy of the code named in the task. Per-image sidecar files record the sampling settings and, where available, a code fingerprint; the inherited Krea baseline records do not establish one source revision for every image.

S BROADER IMPACT

SCA shortens generation at 4096^2 relative to full attention under our measured protocol, which helps applications such as print, design, and scientific visualization. Energy use remains unmeasured. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

T QUALITATIVE GALLERIES

Figure T.1 shows curated SCA outputs. The other galleries, Figures T.2 to T.11, are uncurated unless stated: prompts are drawn at random with a fixed seed from pools fixed without looking at images, which exclude the development prompts (Figures T.2 to T.5 also exclude the 16 prompts then chosen for the curated and failure-case figures), and the selection seed, source hashes and crop coordinates of every gallery are archived in its `provenance.json`. Unless a caption says otherwise, images are FLUX.1-dev outputs at seed 0 with 28 steps and guidance 4.5. Every crop of the uncurated galleries and of Figures 6 and M.2 shows native pixels of the outlined window, which is placed automatically at the highest edge energy averaged over the methods shown, never by hand. Figure T.6 uses FLUX.2-dev with guidance 4.0 and a FLUX.2 calibration table. Figure T.10 uses the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE and a Krea-calibrated SCA table; DyPE and SEGA come from the recorded Krea campaign, in which seven SEGA outputs were regenerated from an archived runtime because their original source snapshots could not be recovered, and the inherited baselines span several runtime revisions. In Figure T.8, UltraImage runs its two-stage defaults (30+30 steps, guidance 3.5 then 7), so that comparison is not equal-cost.

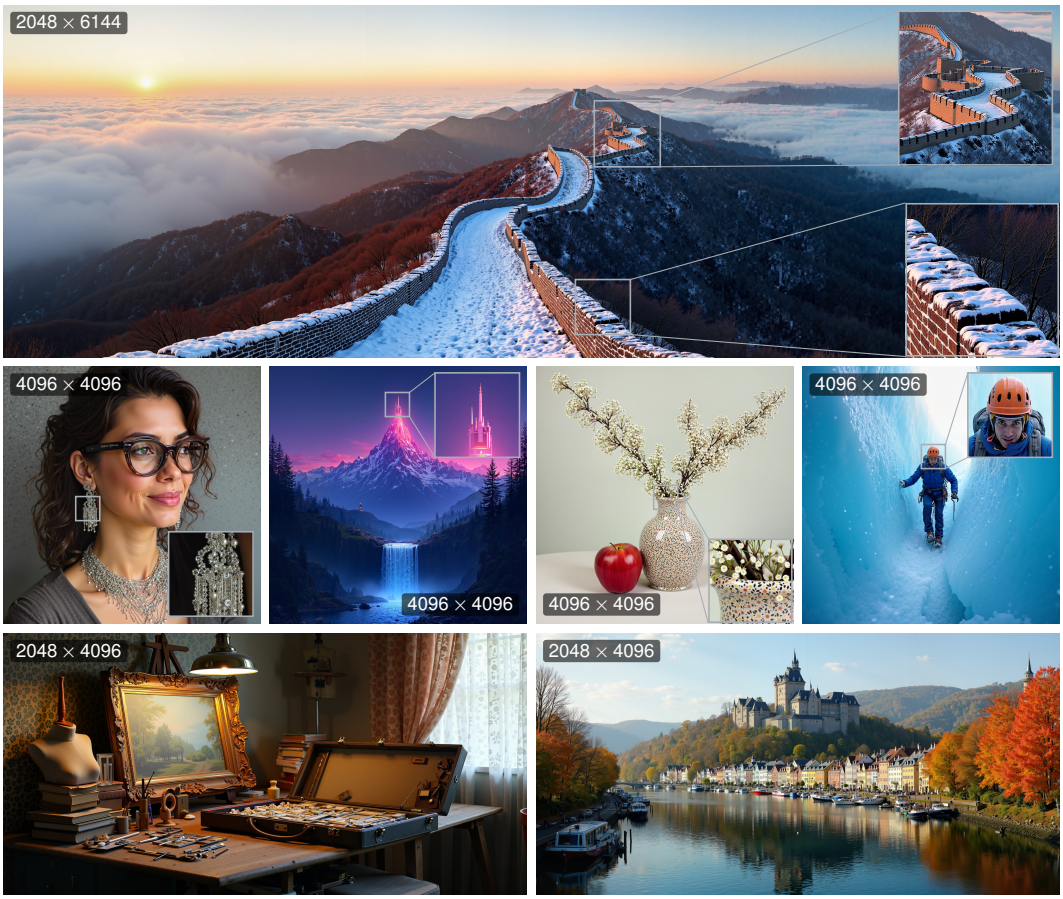

Figure T.1: **SCA keeps layout and fine detail across three aspect ratios without training.** Curated SCA outputs at 4096^2, 2048×4096 and 2048×6144, picked by eye for variety; insets are 1:1 crops of the outlined windows.

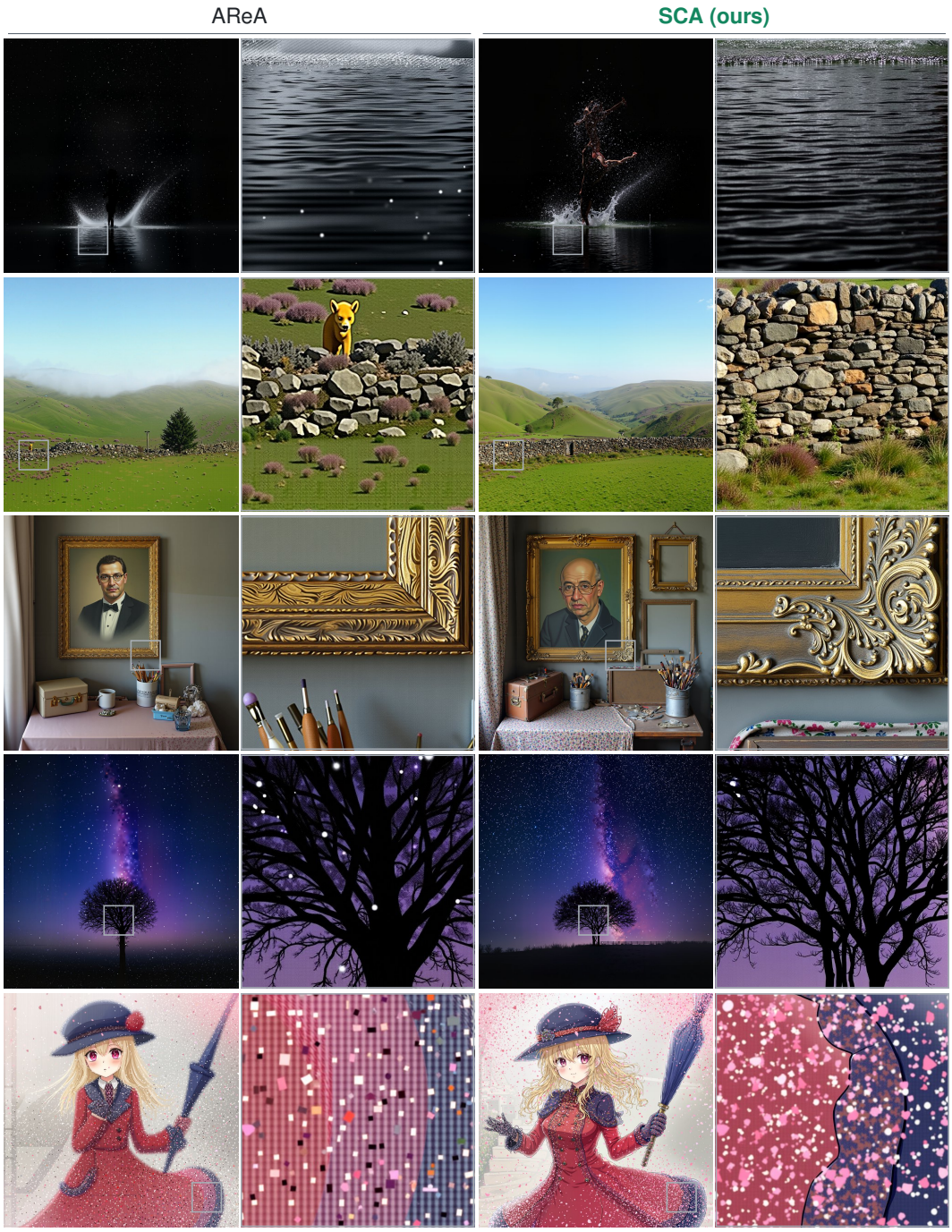

Figure T.2: **Uncurated, 4096^2: SCA keeps texture that AReA blurs or grids, but adds a faint second dancer in row one.** Prompts one to five of 32 drawn at random (seed 0); SCA with the window schedule; after each image, a 1:1 crop of its outlined window.

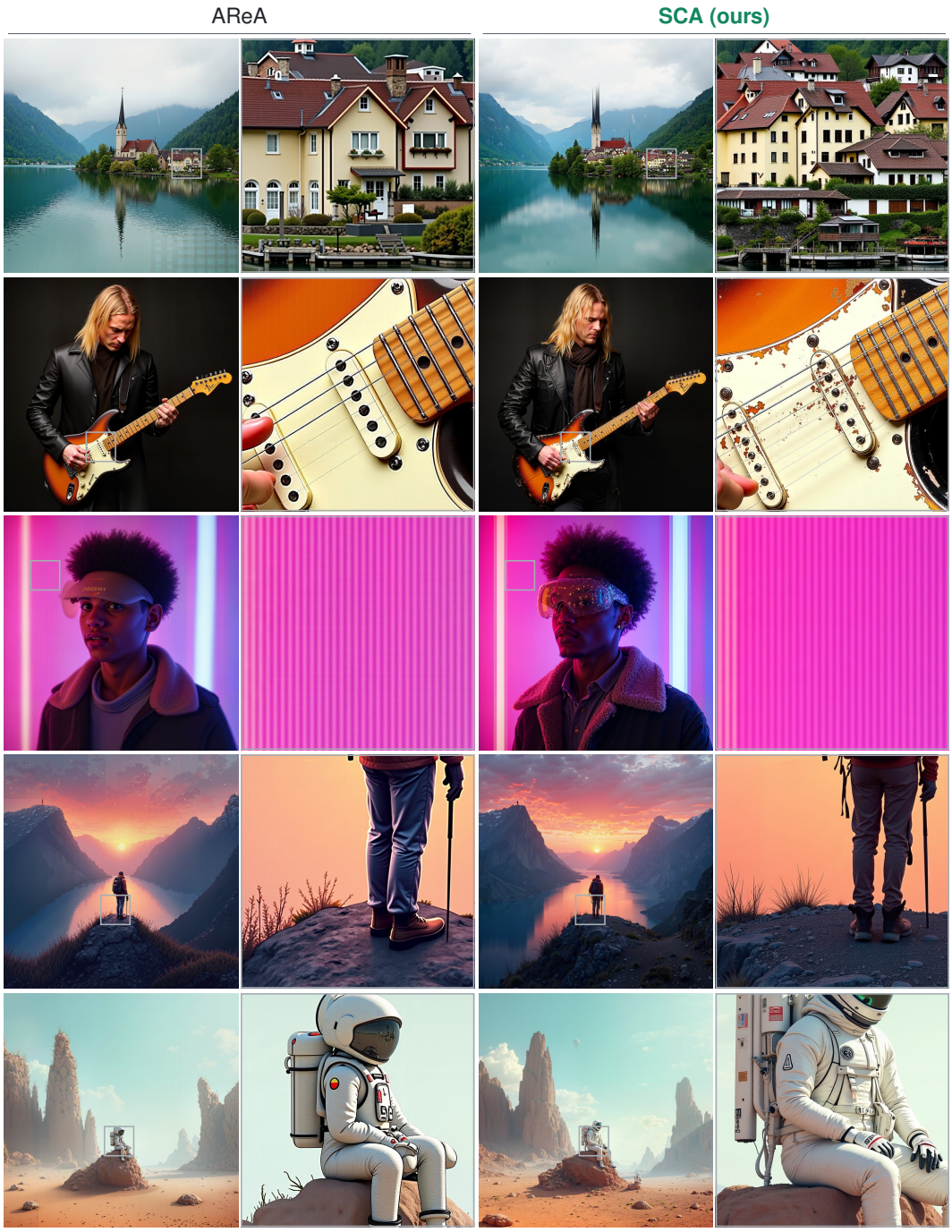

Figure T.3: **Uncurated samples at 4096^2, continued.** Prompts six to ten of the same draw; protocol as in Figure T.2. SCA turns the steeple into a cluster of spires and adds two extra hikers and a second astronaut in rows one, four, and five.

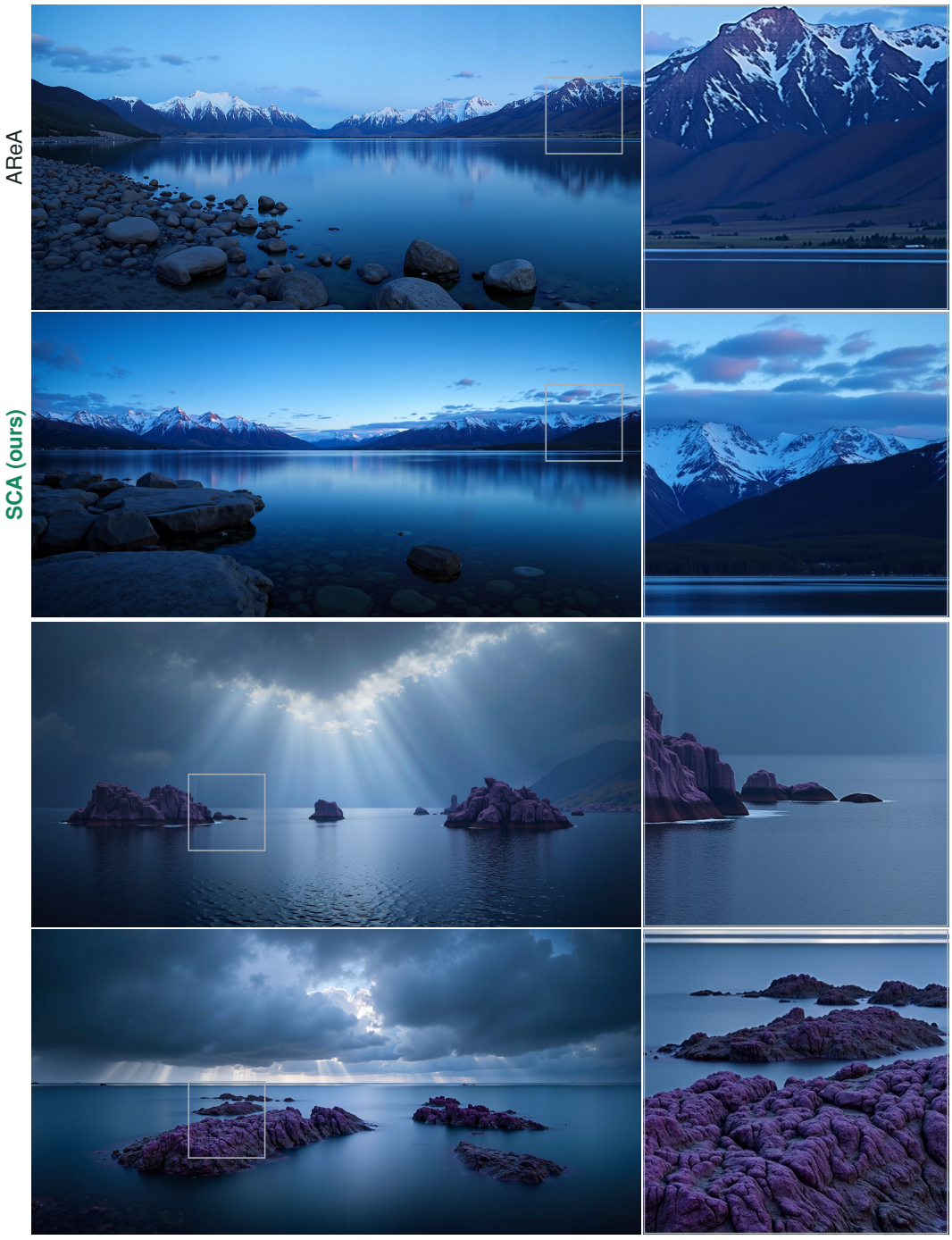

Figure T.4: **At 4096 × 2048 both methods keep the layout; SCA avoids AReA's faint weave pattern and keeps sharper rock texture.** The first two of six prompts drawn at random; AReA above, SCA (with the window schedule) below, each followed by a 1:1 crop of the outlined window.

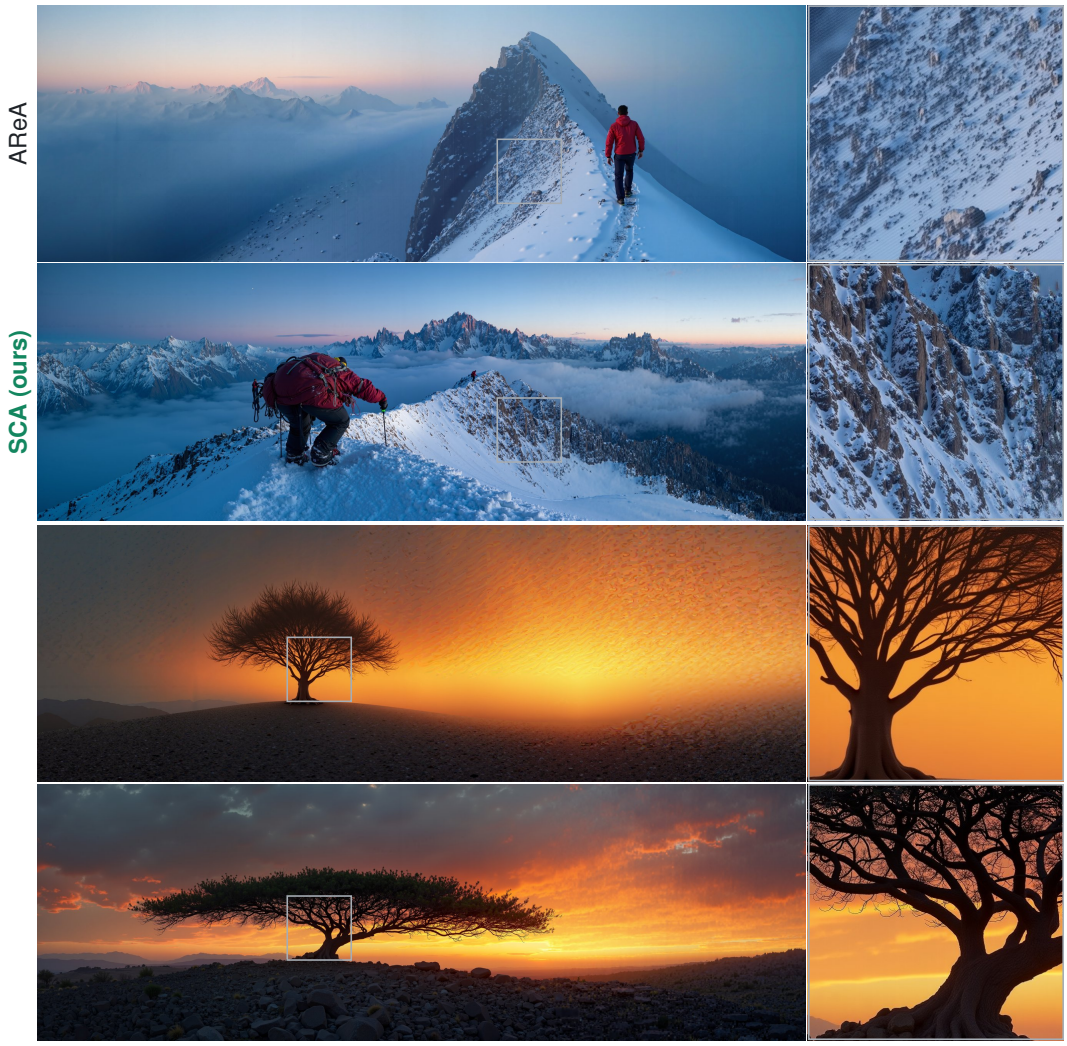

Figure T.5: **At 6144 × 2048 SCA resolves finer texture than AReA but adds two extra climbers.** The first two of six prompts drawn at random; AReA above, SCA (with the window schedule) below, each followed by a 1:1 crop of the outlined window.

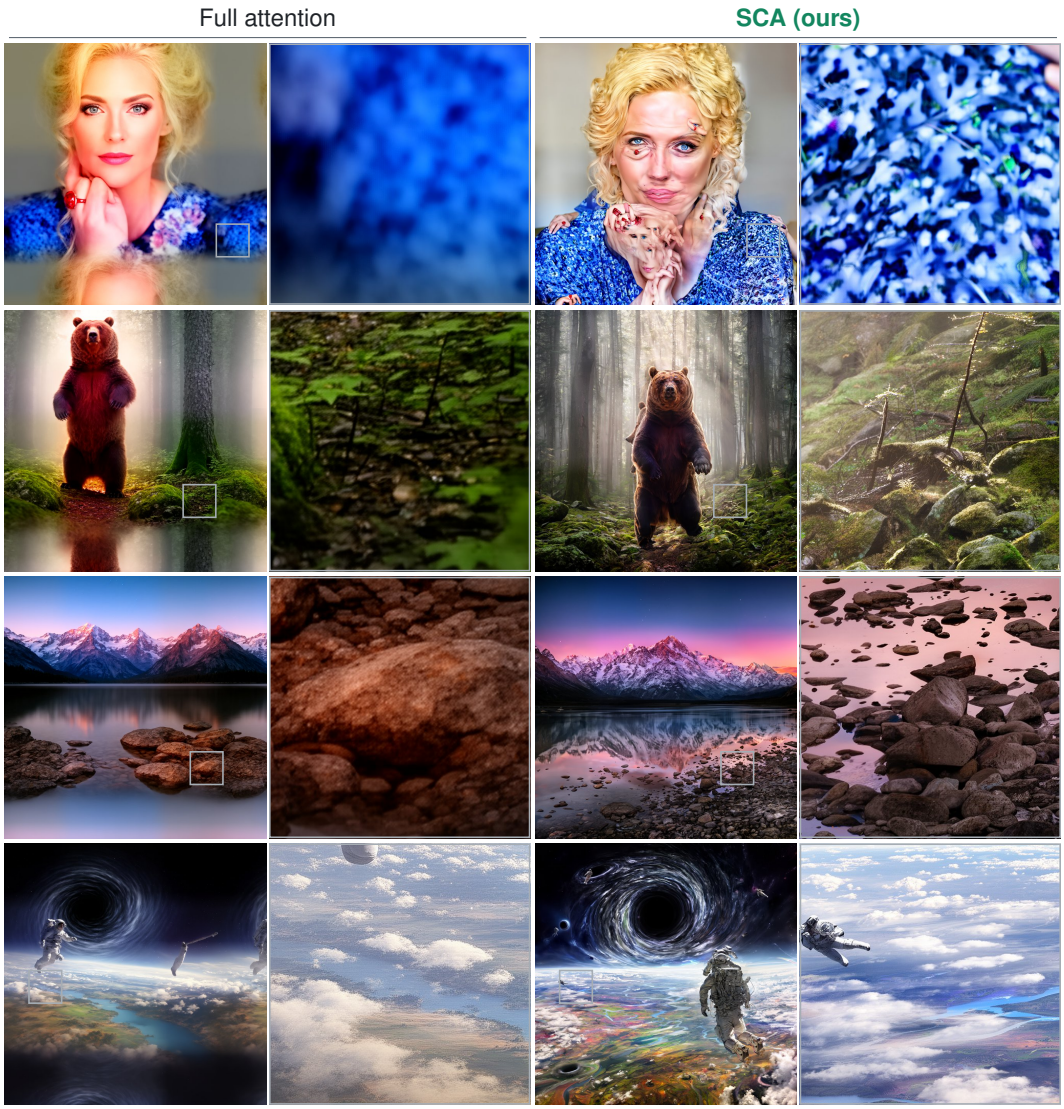

Figure T.6: **FLUX.2-dev at 4096^2: full attention blurs and repeats; SCA is sharp but duplicates subjects in three rows and distorts the portrait.** Four of 50 paired prompts drawn at random (seed 17); 1:1 crops follow each image.

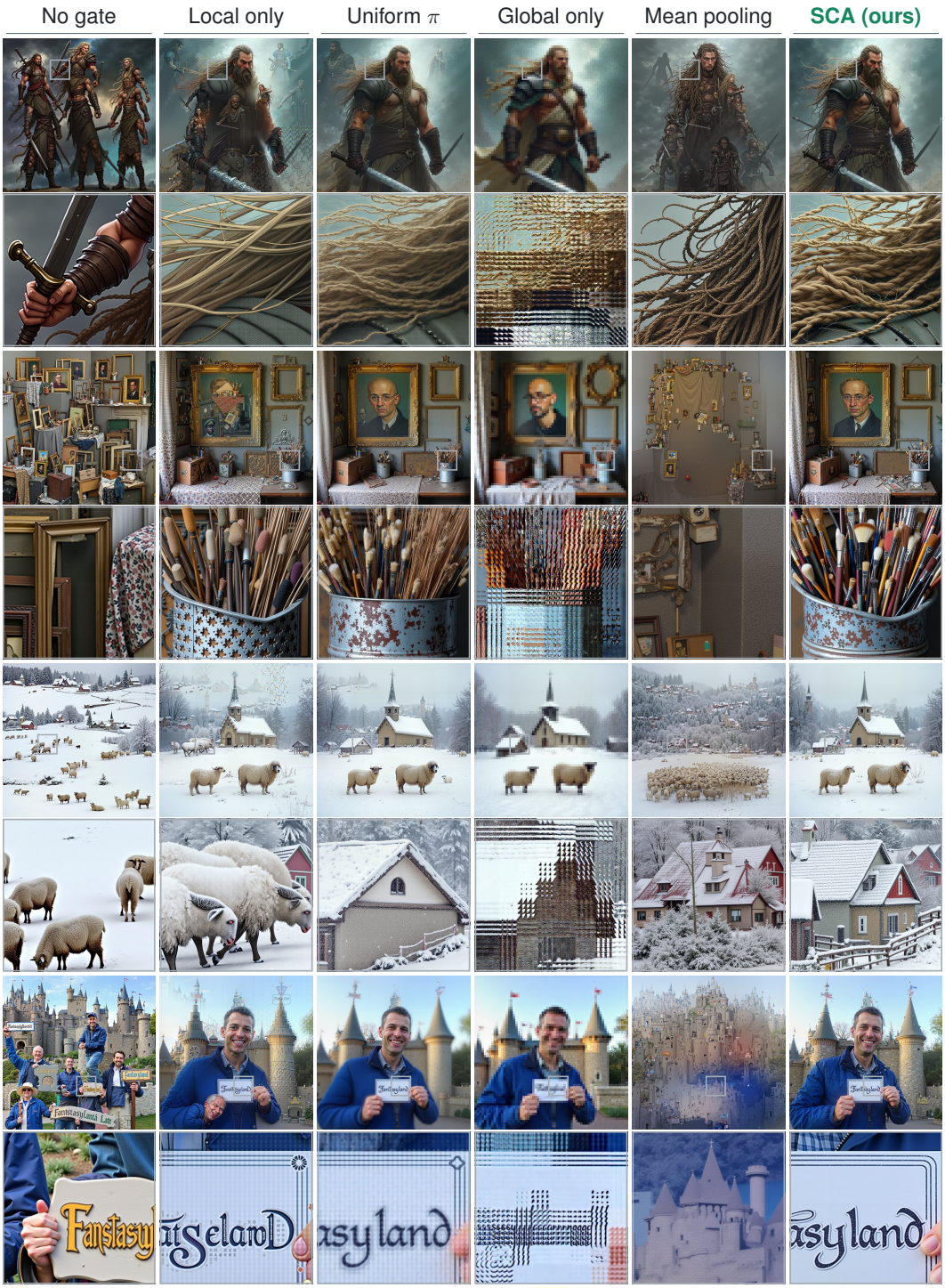

Figure T.7: **Without the gate the subject repeats; global only leaves the pooled branch's coarse grid; mean pooling loses the layout.** Four curated ablation prompts at 4096^2 (Table 3); full images above 1:1 crops of one window.

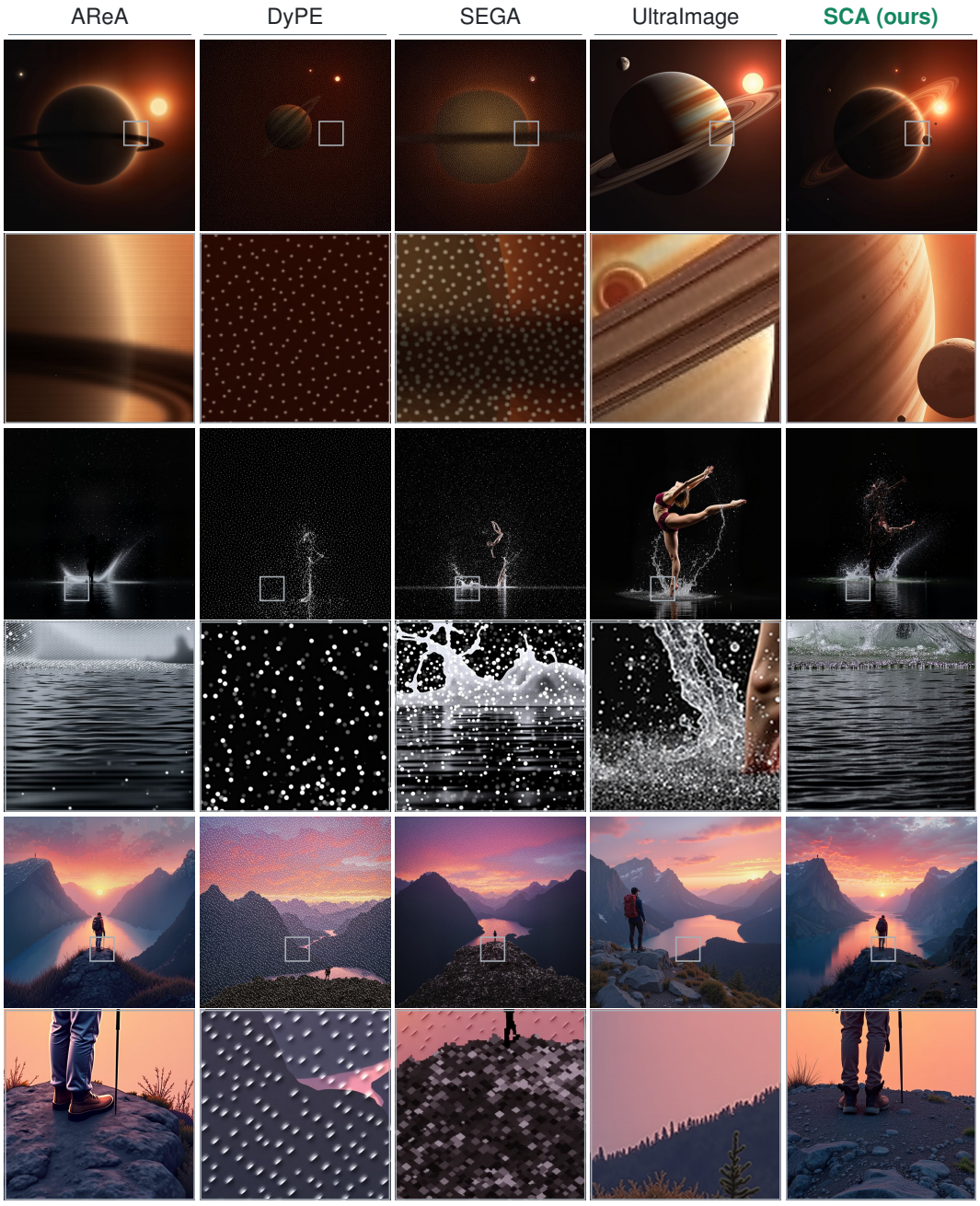

Figure T.8: **At 4096^2 DyPE and SEGA leave speckle, UltraImage softens the planet, and SCA is sharp but adds extra people in two prompts.** Three of 191 non-development prompts drawn at random (seed 17); SCA with the window schedule; full images above 1:1 crops of one window.

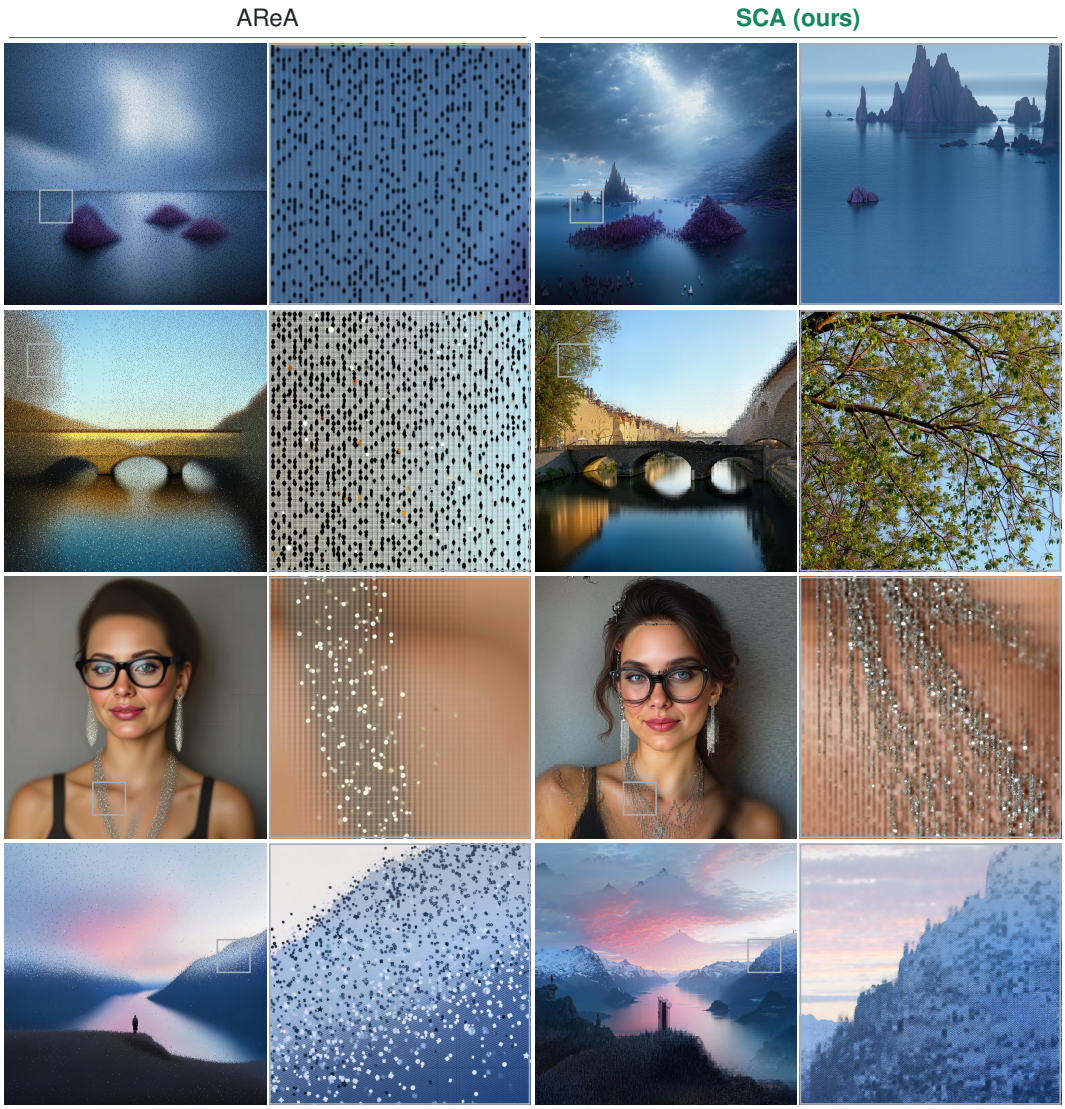

Figure T.9: **At 6144^2 AReA's crops dissolve into speckle or streaks; SCA keeps the layouts but shows stripes, block speckle, or extra figures.** Four of 100 prompts drawn at random; after each image, a 1:1 crop of its window.

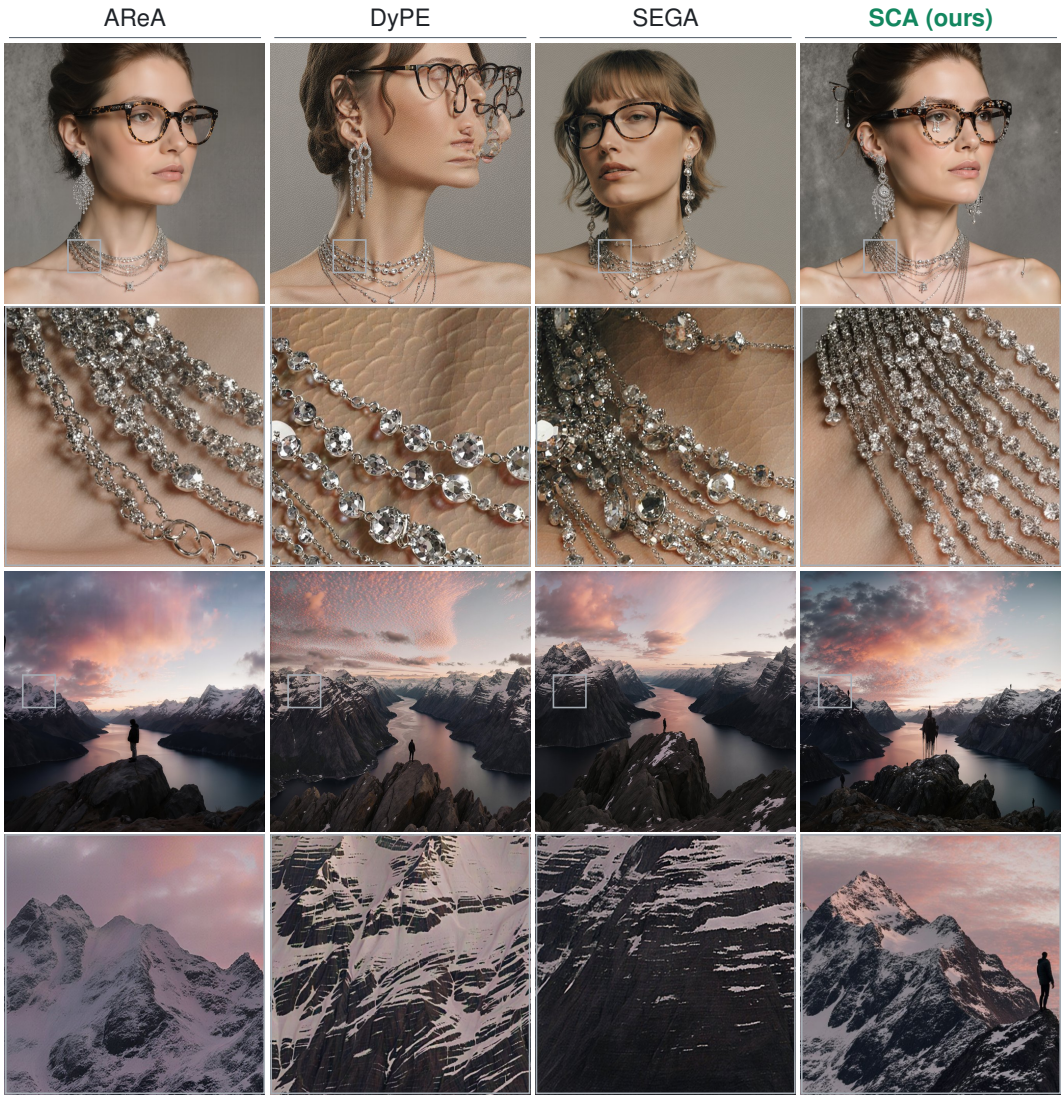

Figure T.10: **On FLUX.1-Krea-dev at 4096^2 the crops look alike, but DyPE ghosts the face and SCA melts the lone figure.** Two of 100 four-method prompts drawn at random; full images above 1:1 crops of one window.

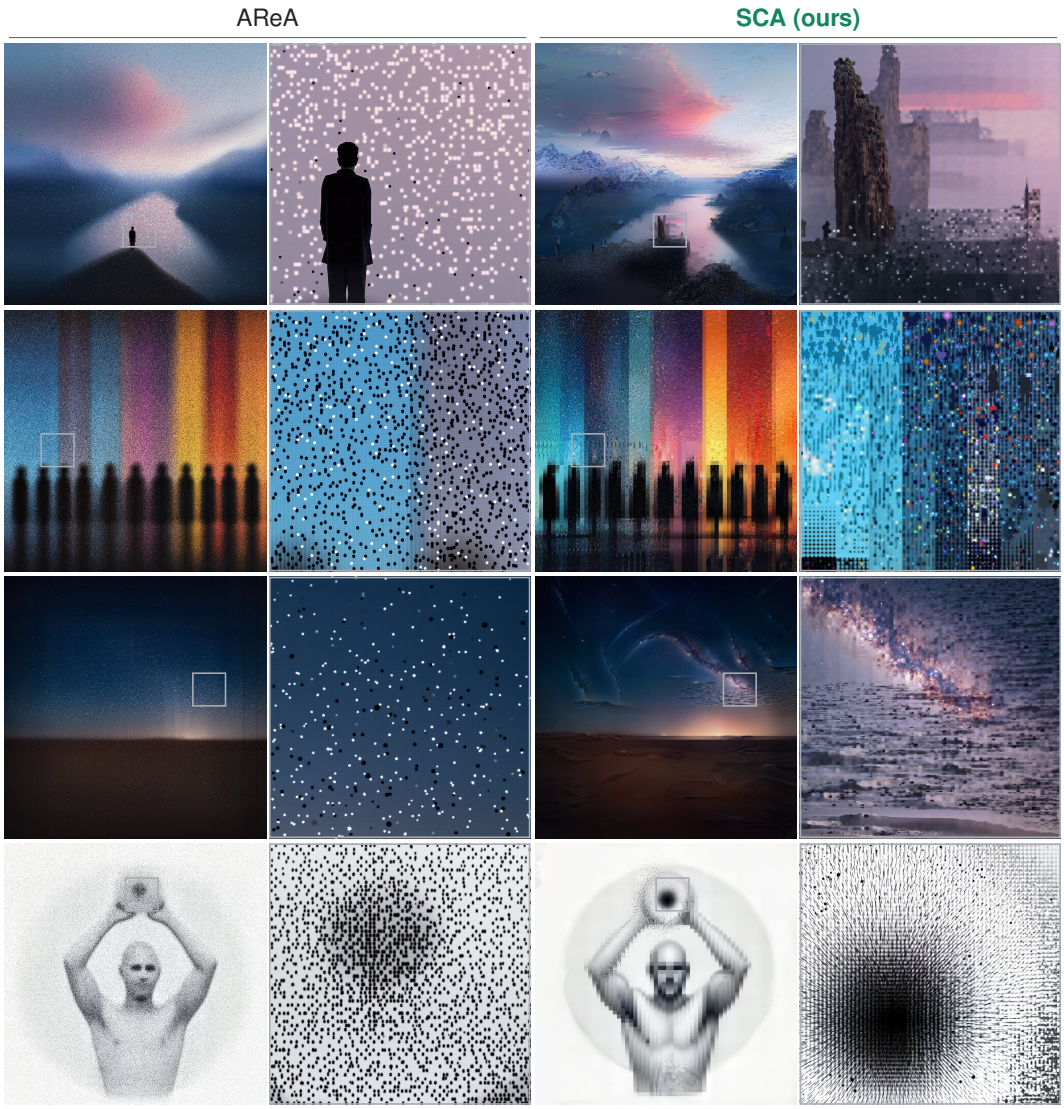

Figure T.11: **At 8192^2 both fail: AReA blurs and speckles, and SCA breaks into blocks and stripes.** Four of 12 paired prompts drawn at random; after each image, a 1:1 crop of its outlined window. Exploratory.

U FAILURE CASES

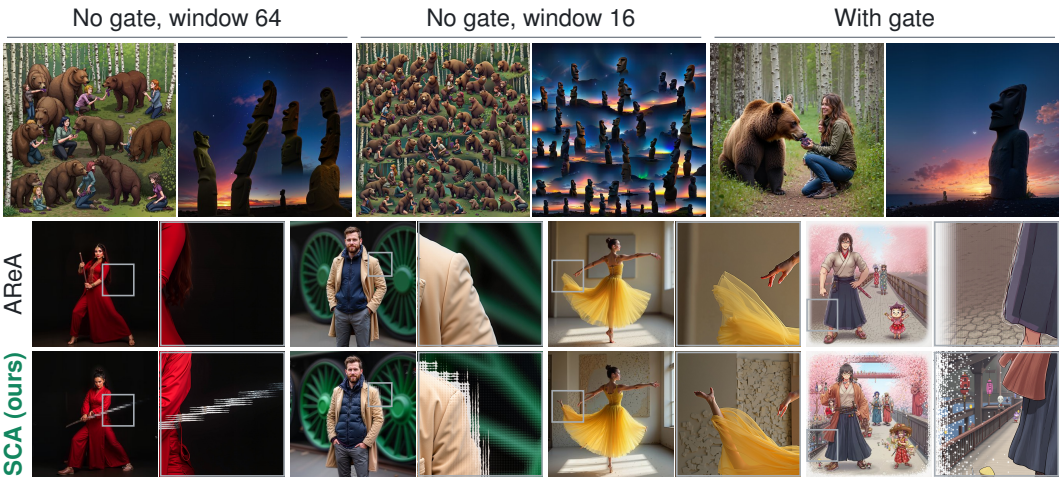

Figure U.1: **Curated failures: without the gate, windows duplicate subjects; SCA without the schedule still makes local errors.** Top: development prompts; the gate leaves small duplicates. Below, 1:1 crops: staircase sword, blocky patch, extra foot, border speckle, none in AReA.

Figure U.1 shows the duplication that windows cause without the gate, and local errors of SCA without the schedule.

Late duplication. Without the schedule, small extra people, animals, or objects appear in the steps right after the gate, when the local branch takes over most heads. In a visual audit of 11 distinct uncurated FLUX.1-dev 4096^2 images of that operator, three add people and two repeat objects (`results/visual_audit_20261020`), and the detector flags 57% of the person or animal prompts (Appendix P); AReA's images of the audited prompts blur or show lattice patterns instead. The schedule halves the flagged share, to 26%, and the copies it leaves are smaller (Figure 4c). The band guard also suppresses some of them but costs preference scores.

Articulated subjects. The underwater-surfer development prompt keeps doubled limbs in every variant; AReA's image shows a large translucent ghost.

Small factors. At 2048×4096 the window covers half the image height, so the local branch alone already sees most of the image along that axis, and the per-head mix gains little over AReA's shrinking windows; aesthetic score, CLIP-IQA, and corrected patch FID and KID are significantly worse.

Larger scales and another backbone. The gallery sampled from the completed 12-prompt 8192^2 run still shows grid and stripe artifacts in both methods (Figure T.11); this exploratory set does not establish general 8K quality. In FLUX.2-dev, improved aggregate metrics coexist with distorted facial geometry and fine texture in the randomly selected portrait example (Figure T.6).

Text-image offsets. The local branch keeps out-of-range text-image offsets (Remark 1).

Metrics. No-reference quality metrics reward sharp repeated subjects (v0 scores higher on CLIP-IQA and ARNIQA), and FID against 195 references at 299^2 resolution sees little of the 4K detail; we therefore report every metric and the qualitative galleries.